Little School on the Prairie: A Push for Structural Transformation

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Abstract

Does education cause structural transformation? Using newly available full count census records from the early 20th century Canadian prairies, I examine the effect of the rollout of schools. I construct new census links and create the finest geolocations available in North American census records. Isolating schooling access at an individual child level, I estimate a difference-in-differences by cohort, relying on variation in school opening dates relative to a child's age. I find that schooling access led children to exit agriculture as adults, move farther away from home, and earn higher incomes. They are significantly more likely to work in a higher skill services occupation, including as teachers, as adults. I interpret these results through the lens of a structural model of education-occupation-migration and find that the expansion of schools led to migration away from the most agriculturally marginal land, towards growing cities. This suggests that education access can act as a push factor for structural transformation out of agriculture in rural areas.

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"And although many farmers would have preferred that their children remain on the land, most knew it would prove impossible. The best they could do was to endow them with education to be mobile."

Goldin and Katz (2000)

1 Introduction

As economies develop they go through a process of structural transformation, where labour is reallocated from agricultural to non-agricultural production. The role of education as a driver of structural transformation has received limited attention (Porzio, Rossi, and Santangelo 2022), despite education being one of the most common policy prescriptions for developing economies. This paper addresses this gap by studying the impact of the expansion of schools in the Canadian prairies at the start of the twentieth century.

The Canadian prairies were a heavily agricultural region that underwent a rapid expansion in school access. By 1906, a new school was being established every day, and the Canadian government increased the frequency of the census to track changes in the population. This provides a unique opportunity to observe how improvements in education affected children and their later-life outcomes during a formative stage of economic development. The prairies are also empirically valuable because they help isolate the role of schooling relative to the traditional drivers of structural transformation: income driven demand shifts and agricultural productivity improvements. Since the primary crop was wheat for export rather than consumption, income driven demand shifts are unlikely to drive changes in agricultural labour. At the same time, agricultural productivity improvements in this period came from new wheat varieties rather than labour-replacing machines, limiting the role for productivity changes to reallocate labour.

The core of the analysis is newly available full-count historical Canadian census records. Rural census records in the prairies include the legal land description from the Dominion Land System as an address, which I use to geolocate households with a one-mile precision. I apply the same procedure to archival records of rural schools, allowing me to compute child-by-year-specific measures of school access. To follow individuals over time, I link the 1906 census of the prairie provinces with the 1911, 1921, and 1931 Canadian censuses. I implement the Multigenerational Longitudinal Panel (MLP) of Helgertz et al. (2022), training the model on variables common to the U.S. and Canadian censuses, allowing U.S.-based training data to be used to link Canadian records. The resulting linked and geolocated panel provides the foundation for the empirical analysis.

With this novel dataset, I then identify the effect of school expansions on individual children, implementing a difference-in-differences design in the spirit of Duflo (2001). I find that children exposed to early schooling were less likely to work in agriculture by 4.3%, migrated farther away from home by 11.2%, and earned higher incomes by 12.6%. These results suggest that education expansions can act as a push factor for structural transformation in rural areas.

I find that children exposed to schooling were more likely to enter high-skill service sector occupations, including becoming teachers themselves. Surprisingly, there is no change in the likelihood of entering manufacturing occupations. I find some evidence that the establishment of a school also drives an increase in average farm sizes of 5.2%. This is nearly identical to the decline in agricultural employment of 4.3%, implying that the effects on children from school construction were not offset by larger population growth or immigration.

I then turn to the aggregate effects. Building on Hsiao (2025), I develop a model of sequential education–occupation–migration choices to capture the effects of education cost decreases. I estimate the amount of education obtained, which is not observable in the data, and calibrate the model to match the estimated education as well as observed occupation and migration flows. I compute counterfactuals where the cost of education acquisition falls. I find that the expansion of schools at the start of the twentieth century was instrumental in both hollowing out agriculturally marginal areas in the southern prairies and encouraging migration to growing cities. I am able to recover these results without any change in underlying productivities or non-homotheticity of preferences, as is typically emphasized in the structural transformation literature. I conclude that education expansions can be an important driver of agglomeration toward more productive areas.

This paper makes four main contributions. First, I provide new evidence on how expanding education access can reallocate labour out of agriculture. This mechanism was first proposed by Caselli and Coleman II (2001) and more recently examined in Porzio, Rossi, and Santangelo (2022). Relative to this work, I provide plausibly exogenous individual-level variation in a context where the typical forces of structural transformation are less likely to be drivers. A long development literature also finds evidence that education causes an exit from agriculture (Duflo 2004; Akresh, Halim, and Kleemans 2023; Cox 2025). This paper is the first to show that the same mechanism, the complementarity of education with non-agricultural production, observed in modern developing economies also occurred historically.

Second, I resolve a key data gap on the role of rural schools in the development of the American education system, and am among the first to isolate exogenous variation in schooling access for children. I show how the development of rural primary schools in North America was an important force before the more widely studied high school movement.

While we have significant evidence on the effects of the high school movement (Goldin and Katz 2008; Nencka, Karger, and Alison 2025), the role of rural primary schools has received little attention due to the difficulty of determining who had access to these schools. The closest work is Schaede and Smith (2025), who focuses on regional effects of school funding instead of individual effects. Relative to the majority of previous literature on rural schooling (Goldin and Katz 2000, 2008; Card, Domnisoru, and Taylor 2022; Lachanski 2024; Althoff, Gray, and Reichardt 2025), I estimate the causal effect of school access. The causal evidence we do have comes from the Rosenwald Schools which targeted Black children in the early twentieth century South (Carruthers and Wanamaker 2013; Eriksson 2020; Mohammed and Mohnen 2025); in comparison, I examine a schooling expansion that was untargeted to any particular group. This paper builds on the literature on Canadian education expansions (MacKinnon and Minns 2009) and provides the first micro-level evidence from Canada.

Third, I assemble new longitudinal linked census records for Canada, building on a long literature in census linking (Abramitzky, Boustan, and Eriksson 2012, 2014; Antonie et al. 2014; Feigenbaum 2016; Bailey et al. 2020; Abramitzky, Mill, and Pérez 2020; Antonie et al. 2020; Price et al. 2021; Helgertz et al. 2022; Buckles et al. 2023; Abramitzky et al. 2024). This is the first paper to do full-count linking of the Canadian prairie census and the 1931 census. J. Feigenbaum, Helgertz, and Price (2023) discusses the role of training data in these methods; I contribute by showing how training data from the United States can be used in Canadian census linking. Relative to the US census, the Canadian census offers several novel features, including enumeration of income starting in 1901 and religion.

Fourth, I contribute new geolocations based on legal land descriptions. These geolocations contribute to a growing literature on homesteading and the development of the American West (Libecap and Hansen 2000; Hansen and Libecap 2004; Libecap and Lueck 2011; Hornbeck 2010, 2012; Lew and Cater 2018; Mattheis and Raz 2019; Smith 2022; Leonard and Kogelmann 2022; French 2022; Hornbeck 2022; Bagagli 2023; Nagy 2023). Relative to these papers, this paper is among the first to examine the role of education in western development. This paper contributes significantly to the geolocation of historical census records. Berkes, Karger, and Nencka (2023) is the closest paper to geolocate rural historical census records in the United States, and only at the county level. The geolocations constructed here are among the finest resolutions available in any historical census records. Finally, this paper is also related to the long literature on the Canadian wheat boom, most recently examined in Pavlik, Geloso, and Pender (2025).

The rest of the paper is organized as follows. Section 2 gives the historical background for the expansion of schools. Section 3 describes the data, linking, and geolocation. Section 4 shows reduced form, individual level results from the expansion of schools. Section 5 shows the model and counterfactual results. Finally, section 6 concludes.

2 Historical Background

2.1 Schools in the West

After the end of the American Civil War and the acquisition of Alaska, the Canadian Government also desired to expand territory westward (Lambrecht 1991). To accomplish this the Canadian government acquired Rupert's Land and the North-West Territories in 1870 from the Hudson's Bay Company. To further expand, British Columbia entered confederation with Canada in 1871, under the condition that a railway be constructed to the west coast from the east coast. The railway would simultaneously link the opposite sides of the dominion, while allowing settlers into the fertile prairies. Upon the completion of the Canadian Pacific Railway in 1885, widespread settlement of the Canadian prairies began. As the prairies were settled and the number of children increased there was increased demand for schools, a need that was filled by rural schools which dotted the prairies.

The system for establishing schools was initially envisioned in the North-West Territories Act of 1875, but was repeatedly refined over the coming decades (Noonan, Hallman, and Scharf 2006). A school would be established when a group of three taxpayers petitioned the territorial, later provincial government, for the formation of a school district with a set of proposed boundaries (Baergen 2005). The districts had a maximum size which varied from 36 square miles in 1881, later to 25 square miles in 1887 (Noonan, Hallman, and Scharf 2006), and later decreased to 16-20 square miles (Baergen 2005), with a minimum number of children and ratepayers in the area (Baergen 2005). The government would approve the proposed boundary, select a location for the school, then organize a meeting and vote of the ratepayers to approve the school district (Baergen 2005). There were often issues with overlapping boundaries of school districts and the need to fill in holes between existing districts (Noonan, Hallman, and Scharf 2006; Saskatchewan Department of Education 1906-1912, 1914). Funding for the school came from a combination of government grants and local property taxes (Baergen 2005; Noonan, Hallman, and Scharf 2006).

The curriculum in rural schools was basic, and consisted of the three R's: Reading, Writing, and Arithmetic (One Room School Project 2023). Teachers were accredited by provincial governments and hired by the school districts. Throughout the early 1900s the expansion of schooling was rapid, giving rise to the motto "A New School Every Day for Twenty Years" in 1900 (One Room School Project 2023). As a result there was often significant teacher shortages and rapid turnover, as a teacher would stay for a year in a rural community before leaving (Noonan, Hallman, and Scharf 2006). Although initially instruction was entirely in English, the Saskatchewan government later encouraged some teaching in other languages to encourage ethnic communities to integrate (Saskatchewan Department of Education 1906-1912, 1914).

Despite being public schools, religious instruction was included in education. Canada's education system was defined on Protestant vs Catholic lines. This conflict resulted in different institutions depending on the province. In Manitoba publicly funded schools allowed for half an hour of religious instruction at the end of each day. In Alberta and Saskatchewan a system of public and separate schools allowed the religious minority in an area to establish their own school if desired (Noonan, Hallman, and Scharf 2006). A detailed discussion of the role of religious education can be found in section A.3. While religious education is an important feature of Canadian education, only 0.6% of schools in Alberta and Saskatchewan are separate in 1911. The handling of separate schools is unlikely to affect the results.

2.2 Surveying

The Dominion Land Survey (DLS) was used to survey land, and created the system of legal land descriptions that I use to geolocate historical census records. The base unit of land is a township, consisting of an area 6 miles by 6 miles. The location of each township is defined by three values: the township number, range number, and meridian. Each 6 mile by 6 mile township is further subdivided into 36 sections. Each section is numbered 1 through 36, starting in the lower right hand corner. Figure 1 shows an example of how sections are laid out within townships, and an example of a single township with geolocated households. Between each section is a road allowance of 90 feet[1]. The road allowances later formed the basis for country roads, and are an important reason why the grid of land parcels remains largely in place today (Lambrecht 1991).

Within a township land was allocated between four parties in a pattern established by the 1881 Dominion Lands Regulations Lambrecht (1991). Odd numbered sections, excluding 11 and 29, were reserved for land grants for railway companies to incentivize railway construction. A small number of colonization companies also purchased railway land and resold it for the purposes of settlement. Even numbered sections, with the exception of 8 and most of 26 were sold as homesteads. Homesteaders were granted entry to a quarter section for a minimal fee, and after three years of cultivation and residence would be granted the title to the land. Sections 11 and 29 were reserved for the purpose of funding local schools. Funds generated were pooled at the federal level and then redistributed to the provincial

1. Decreased to 66 feet and only between every second section horizontally after 1881

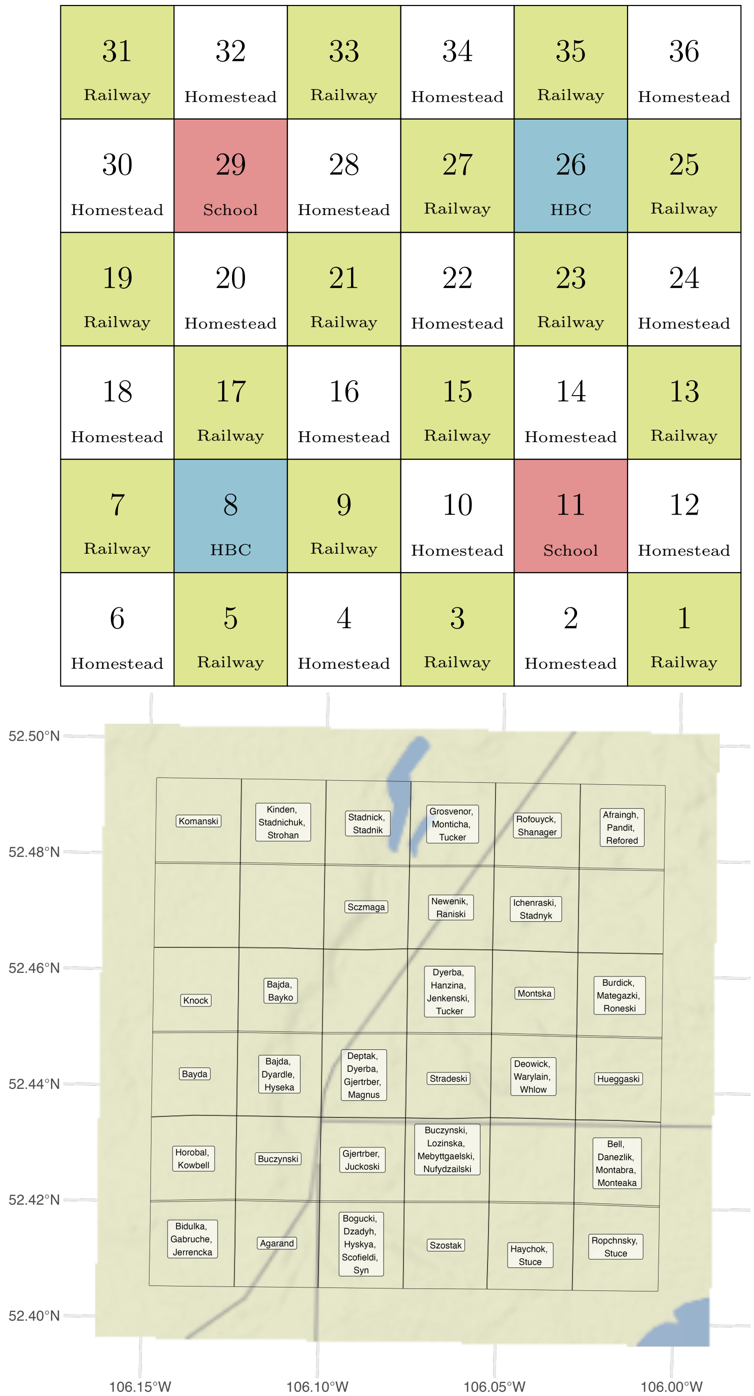


Figure 1: Top: Survey boundaries for sections within each township under the Dominion Land System. Even numbered sections, with the exception of 8 and 26 were sold as homesteads. Odd numbered sections with the exception of 11 and 29 were reserved for railways. Sections 11 and 29 were reserved to fund schools. Section 8, and three-quarters of section 26 (and the entirety of section 26 every fifth township) were given to the Hudson's Bay Company in exchange for Rupert's Land and the North West Territories. Bottom: A single township from the 1921 census: Township 40, Range 1, West of the 3rd Meridian.

and territorial governments[2]. Sections 8 and three quarters of section 26, and the entirety of section 26 in every fifth township in the agricultural parts of the west were granted to the Hudson's Bay Company. The Hudson's Bay Company received these lands in exchange for their surrender of Rupert's Land and the North-West Territories in 1870. Details on the allocation process of land can be found in section A.2 of the appendix.

3 Data

3.1 Census Records

The primary data source for this paper comes from the Canadian complete count census records for 1911, 1921, and 1931, and the census of the prairie provinces in 1906[3]. The population of interest is individuals who lived in Manitoba and the portion of the Northwest Territories which became Alberta and Saskatchewan, starting in 1900. Census records for 1911, 1921, and 1931 can be linked to Census Subdivision (CSD) level polygons which have been digitized (CIEQ, Université Laval and HGIS Lab, University of Saskatchewan 2023). Where needed in 1906 enumeration subdistrict centroids have been constructed from household geolocations. Some variables of interest like months of schooling and income are only available in 1911, 1921, and 1931, while other variables like the section, township, range, and meridian necessary to geolocate households are only available in 1906, 1921, and 1931. A full description of which variables are available in which years can be found in section A.4 of the appendix.

3.2 Linking

To track individuals over time, I need to link census records between waves. I implement the Multigenerational Longitudinal Panel (MLP) method developed at IPUMS to link individuals across waves of the Canadian census (Helgertz et al. 2022). The MLP method is a two stage probabilistic procedure. In the first stage only males are linked across census waves. In the second stage, the first stage links are used to infer links for family members who have been missed in the first stage. The second stage allows for both males and females to be linked. Linking is implemented using the hlink package (Wellington, Harper, and Thompson 2024). A more detailed discussion of this procedure can be found in section A.5 in the appendix.

2. This is the opposite of Schaede and Smith (2025) where school lands were used exclusively for the local school in that township.

3. Data access provided courtesy of The Canadian Peoples (2025)

In both the first and second stage of the linking procedure links are determined probabilistically using a model trained on a set of known links. Unfortunately there is no currently available set of Canadian census links to train the linking model on. Instead I utilize the training data in the replication package of Helgertz et al. (2022) which links US census records. I train a logit model on the set of variables available both in the training data and those that can be assembled from the Canadian census records. This follows the best practice guidelines discussed in J. J. Feigenbaum, Helgertz, and Price (2025). Details on the variables used can be found in section A.5.1 of the appendix.

Table 1 gives the results of the linking procedure. I'm able to link 26.69% of children between the 1906 and 1911 census waves, which declines to 16.27% across the 1906 and 1931 census waves when the difference in time between waves is larger. These results are broadly consistent with the literature, including Antonie et al. (2014) who claim a link rate of 24.22% when linking the 1871 to 1881 Canadian census.

Table 1: Link Rates for Canadian Census

	Total	**Matched**	**Children**	**Adults**	**Male**	**Female**
1906-1911	100.00	22.80	26.69	21.43	26.75	17.76
1906-1921	100.00	19.73	20.59	20.12	25.62	12.02
1906-1931	100.00	16.20	16.27	16.91	23.02	7.18
1911-1921	100.00	19.95	23.06	18.25	24.69	14.67
1911-1931	100.00	13.37	13.92	13.19	18.86	7.21

No automated linking method produces representative samples (Bailey et al. 2020). I therefore want to characterize which populations are more likely to be linked, and what effects the selection into linking is likely to have. Section A.5.3 in the appendix presents some results on the likelihood of being linked versus some characteristics of the individual. Linking rates are larger for the young, people from Ontario and the United Kingdom, wealthier families with larger livestock holdings, sons and heads of households, and Protestants versus Catholics. In general, more prosperous individuals are more likely to be linked across census waves.

The MLP method utilizes geographic distance between the origin and destination as a linking variable. Including geographic distance is a tradeoff between distinguishing between more individuals and improving linking rates, while biasing the linking towards links where individuals don't move. In section A.5.2 of the appendix I compare the results of the linking with the ABE method, which uses only time invariant factors, and find that changes in the link rate by distance are consistent with large improvements in link rate with little cost in increased distance bias. Newer versions of the ABE method also include distance as a linking variable, but only as a tie breaker (Abramitzky et al. 2025).

Based on the characteristics of linked individuals, the linking procedure is likely to bias my estimates towards zero. I'm more likely to link non-immigrants and the wealthy, who are the least likely to benefit from increased school access due to higher levels of parental human capital. Including distance as a linking variable improves the linking, while increasing the likelihood of linking those who stay in rural areas and stay in agriculture. The outcome of interest is leaving agriculture, income, and migration. As a result I'm most likely to link those that remain behind, biasing my estimates towards zero. My estimates should therefore be interpreted as a lower bound. As a secondary measure the likelihood of a link is used as a weight for the observations. Weighting ensures that more weight is put on observations that have the highest likelihood of being correct.

3.3 Schools

Records of school locations[4] and opening dates come from three main sources. Alberta's schools are documented in Baergen 2005, and later digitized by the University of Calgary (University of Calgary, Glenbow Archives School District Database, Courtesy of Libraries and Cultural Resources at the University of Calgary). Saskatchewan's schools are best recorded in Saskatchewan Gen Web 2023, a genealogy website. Manitoba's schools came from Manitoba Historical Society 2019, which are largely based on the extensive archival work of Perfect 1978. In addition I supplement these records with annual reports from the Saskatchewan Department of Education (Saskatchewan Department of Education 1906-1912, 1914).

While school opening dates and locations are nearly exhaustive in Alberta and Manitoba, the Saskatchewan records are inconsistent. I infer missing opening dates from the sequential numbering of schools in Alberta and Saskatchewan. Alberta and Saskatchewan shared education systems prior to 1905 when both provinces were created. Schools constructed prior to 1905 share a numbering system that allows the missing opening dates in Saskatchewan to be inferred from the Alberta school openings (Baergen 2005). Details on the process of inferring opening dates can be found in section A.6. School locations are also known less precisely in Saskatchewan. In cases where the school location cannot be geolocated to a land section, I utilize the township centroid or the centroid of a group of townships. Details on school locations can be found in section A.7.3 of the appendix.

4. I use the term "schools" and "school districts" interchangeably in this paper. Most school districts consisted of a single school, while a small percentage could consist of two schools. In a small number of cases a school district could be formed and then contract with a nearby school district to convey the students elsewhere. In 1913 there were 3,230 school districts in existence with 2,747 having a school in operation (Saskatchewan Department of Education 1906-1912, 1914), although some of these would reflect new districts that had not yet opened their school.

I assume that no schools closed during my period of interest. This is a reasonable assumption as the population of the Canadian prairies was rapidly expanding due to immigration, increasing demand for schools. The movement to consolidate rural schools largely occurred after my study period. Manitoba was the earliest to consolidate schools in 1905, but only had 77 consolidated districts by 1914 out of a total of nearly 1,600 (Saunders, n.d.). Saskatchewan allowed for consolidation with an incentive grant in 1912, but the move was highly unpopular (Baergen 2005). Alberta didn't allow for consolidated districts until 1913 (Sparby 1958), and experienced considerable resistance (Baergen 2005).

3.4 Geolocation

A unique feature of the Canadian prairies is the widespread use of Dominion Land Survey parcels and legal land descriptions as addresses. A legal land description consists of four numbers, which define the location of a one mile square section[5]. In census records, colloquial, and administrative use the legal land description was used as an address. The parcels defined by the Dominion Land Survey continue to be in use today, allowing for precise geolocation over one hundred and fifty years after the survey was established.

Table 2: Geolocation rates by census year for Alberta, Saskatchewan, and Manitoba. Exact geolocations are at the section level or in CSDs that are less than 1.5 miles in area. General geolocations are those that are at the township level or finer. CSD geolocations are at the CSD level. Splitting the geolocation rate by urban and rural households reveals that the majority of geolocations in both census waves are for the rural households.

	Total	**Exact**	**General**	**CSD**	**Exact (Rural)**	**Exact (Urban)**
1906	100.00	54.77	84.41	100.00	70.91	0.00
1921	100.00	65.97	69.61	100.00	83.89	36.35
1931	100.00	42.36	86.55	100.00	70.14	0.00

To geolocate households and schools I link legal land descriptions from census records and school records with GIS datasets of all land sections in the prairies. I merge datasets of land sections in Alberta (Government of Alberta 2020), Saskatchewan (Information Services Corporation (ISC) 2020), and Manitoba (Manitoba Government 2020). For households I locate to the nearest section, and identify potentially erroneous locations based on the location of neighbors and plausible paths that enumerators would have taken. For schools I locate sequentially, starting with the finest possible resolution. I first locate at the nearest quarter section, then section, then township. Details on the geolocation procedure for households and schools can be found in section A.7 of the appendix.

5. Quarter sections, defined by intercardinal directions, improve the resolution to a half mile square but are available inconsistently.

Table 2 gives the rate of geolocation across census waves. The top row gives the geolocation rates in 1906, the most important wave. I'm able to geolocate 54.77% of households in the prairies to a land section. My target population is those that live in rural households, of which I'm able to geolocate 70.91% of households. The results for the 1921 and 1931 census are comparable, with the exception of urban households in 1921 where I am able to precisely geolocate 36.35%. I geolocate an average of 800 thousand individuals per census wave with this method.

Table A13 in the appendix gives the counts of geolocated schools. As of 1911 I'm able to geolocate 73.7% of schools to the exact land parcels, and 97.4% of schools to the township level. Details on school geolocation can be found in section A.7 in the appendix.

3.5 Estimating Farm Size

The only microdata remaining from the Canadian census is from the population schedules[6], with limited agricultural information. To understand some of the effects on farmers, I construct an estimate of farm size by computing a set of Voronoi polygons around each geolocated household. This measure of farm size is possible in the 1906, 1921, and 1931 census. I find that the estimated farm size positively correlates with horse and cattle holdings and negatively correlates with homestead status, consistent with homestead land sections being more fragmented between owners as documented in the US by Smith 2022. Details on the farm size estimation and validation can be found in section A.7.4.

3.6 Other Data

Railways formed the most common mode of transportation in the early 20th century. Railways were also critical for farmers to sell their crops (B. R. Hamilton, n.d.). To account for this I use a dataset of the location and year of construction of Canadian railways (Cartography office, Historical Canadian Railroads).

Agricultural productivity is also a key variable in determining the viability of an area. I make use of the FAO GAEZ v4 database (FAO and IIASA, Global Agro Ecological Zones Version 4 (GAEZ V4))[7]. I include data for expected yields for wheat, oats, barley, and flax, which were the most common crops of the time, and reed canary grass as a proxy for livestock suitability. I note that the use of the crop suitability index has come under scrutiny

6. Other schedules were collected, including agriculture and manufacturing, but these have largely been destroyed.

7. Expected output density with the oldest possible time period (1971-2000), CRUTS32 climate model, historical representative concentration pathway (RCP), rain fed water supply, low input level, and without C02 fertilization.

for historical usage (Rhode 2024). The inclusion or exclusion of the index as a control has no significant impact on the results.

4 Individual Effects of School Openings

To estimate the effect of school openings I follow the methodology of Duflo (2001), estimating the effect of school openings on a younger cohort of children who are able to benefit from openings, relative to an older cohort who are too old to benefit. This cohort difference-in-differences is analogous to a standard difference-in-differences, but with the cohort as the time dimension. Cohorts who are too old to benefit from school construction are equivalent to the pre-period, while cohorts which are young enough to benefit from school construction are the post period. This methodology allows me to estimate the effects of school construction while only observing each person twice: once as a child to determine their school exposure, and a second time as an adult to observe their adult outcomes. The primary outcome variables are whether the individual is working in agriculture, income, and distance from childhood home in 1931.

The sample is boys in 1906 linked to the 1931 census when they are adults. Girls are dropped from the sample as most girls in 1906 are likely to have married by 1931, making linking unreliable due to changing last names. The 1906 census provides precise locations for rural households, while the assembled school data covers school openings in the period from 1871 to 1912. Assuming that migration is limited over short time periods, the geolocations will be most accurate around the year 1906. As a result I focus on changes to schooling exposure occurring between 1900 and 1905, utilizing data on school construction during this time period. This also coincides with the period before rural schools were consolidated.

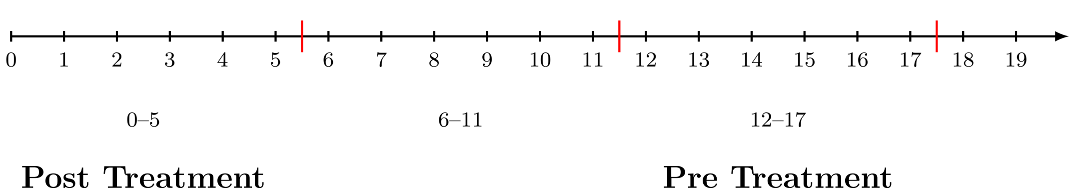


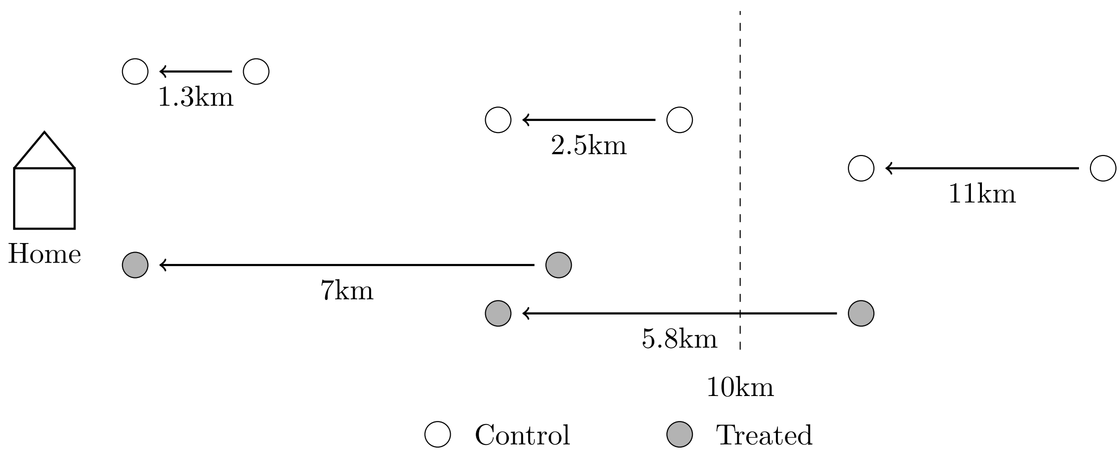
Figure 2: Ages and treatment timing in 1906.

The primary cohort of interest is those between the ages of 0-5 in 1906, who are young enough to benefit from school construction. The pre-period is those 12-17 in 1906, who are too old to benefit from school construction. I consider comparisons to other cohorts including 6-11 and 18-23 as robustness checks.

Unlike Duflo (2001) where treatment intensity is defined at the region level, treatment here is defined at the individual level. A child who sees their distance to school decrease by

5km or more between 1900 and 1905 is considered treated[8], conditional on the final school distance being within 10 kilometers. Control children are those who did not see significant decreases. Control children are therefore split into two groups: those that already had access to school, and those that did not get access. I show comparisons against both of these groups. Note that distance is not included directly in the estimation equation due to possible endogeneity. Instead it is whether there are significant changes in local school availability that determine if a particular individual is treated. Figure 3 gives examples of how treatment and control are assigned to individuals.

Figure 3: Treated and Control Individuals.



Treated individuals (grey) see a decrease of more than 5km in their distance to school based on their home location, conditional on the final distance to school being within 10km. Untreated individuals (white) see a decrease of less than 5km or have a final distance greater than 10km in 1906.

I construct the estimation equation as follows. The comparison is between those who are in the 0-5 cohort ($t_c = \mathbb{1}\{age_{1906} < 6\}$) in 1906 and those who are 12-17 in 1906. An individual is considered treated if the distance to school decreased by 5km or more between 1900 and 1905, conditional on the final school being within 10 kilometers ($s_i = \mathbb{1}\{\Delta dist_{i,1900,1905} > 5km | dist_{i,1905} < 10km\}$). The estimation equation is therefore:

$$y_{icg} = \gamma s_i t_c + s_i + t_c + \beta X_i + \alpha_g + \epsilon_{icg} \tag{1}$$

where y_{icg} is the outcome of interest in 1931, s_i is the treatment, t_c is whether a person is in the 0-5 cohort versus the 12-17 cohort, X_i is a series of controls, and α_g is a geographic

8. I lag the school construction dates by 1 year to reflect the fact that the school data reflects the year the school was established, not the year the school was built. The construction of schools was rapid, however I lag by one year to reflect when the school actually opened.

fixed effect for the enumeration subdistrict.

This design relies on two underlying assumptions. The first is that geographic proximity to school affects the likelihood of obtaining schooling. The second is that children beyond the age of 12 are unlikely to benefit from school construction. I test these assumptions by linking children in 1906, where precise geolocation is possible, to 1911 where schooling is measured, and restrict to children who are likely living in the sample location in both 1906 and 1911. Holding the 1906 locations constant and assuming the child has not moved, I then look at the determinants of school attendance in 1911.

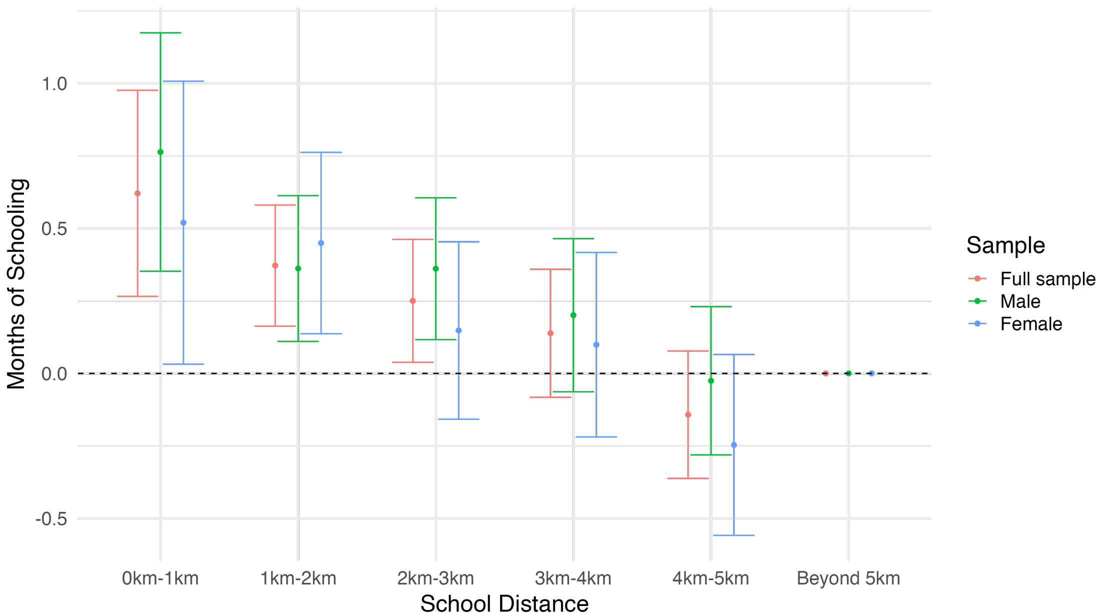
Figure 4 shows the estimated total effect of distance on the number of months of schooling in 1911. All coefficients are relative to children who are five kilometers or farther from the school in 1911. I find that there are significant decreases in the months of school attended in 1911, with children immediately adjacent to the school attending for approximately 0.75 months more than children who are 5km or farther away. The average amount of schooling for the sample is 4.9 months of school, implying a decline of 0.75 months is equivalent to a 15% decline. The distance effects are not statistically different for boys versus girls. Additional results on the intensive margin, extensive margin, and the effects of controls can be found in section A.8.2 of the appendix. Distance was a key factor in determining schooling.

The second assumption is that beyond a certain age, children do not benefit from school construction. To validate this assumption I again turn to the linked 1906-1911 sample. Figure 5 shows the months of school attendance versus age in 1911, split between girls and boys. Monthly school attendance increases between the ages of 5 to 8, is approximately flat between 8 and 12, then decreases beyond the age of 12. Children beyond the age of 12 are unlikely to benefit from school construction near them, as they are already starting to leave school.

Figure 5 also highlights a noticeable gap in schooling between boys and girls after the age of 12. While girls continue to attend school for more than 6 months per year until after the age of 14, boys fall past this threshold at 13. The higher level of education for girls versus boys is attributed to the higher value of farm labour in rural communities, and was noted by school inspectors at the time (see section A.1). The earlier decline in schooling for boys further justifies my identification strategy utilizing older cohorts as a comparison. The difference in schooling between girls and boys has been found in other contexts including the United States, but I am among the first to document it in Canada. A brief discussion of the topic can be found in section A.8.3 in the appendix.

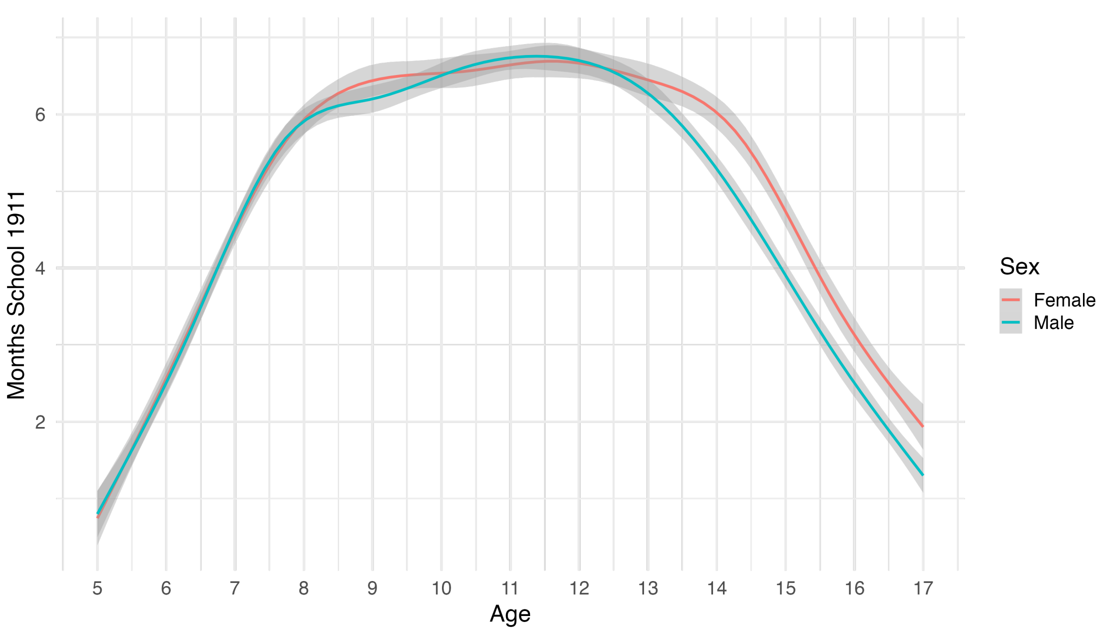
With the design I am taking the stand that although the distance to school is endogenous, the timing of school construction relative to a child's age may be exogenous. This assumption requires that although parents may have moved to be closer to or farther from an individual

Figure 4: Months of Schooling versus Distance



Months of schooling by distance in 1911, in 1km bins. All coefficients are expressed relative to the furthest bin of 5km or more. Includes controls for the interactions of sex, age, and birth order, birthplace, family livestock holdings, estimated agricultural productivity, whether the family lives on a homestead parcel of land, estimated farm size in 1906, and distance to the railway in 1911.

Figure 5: Months of School versus Age



school, they could not manipulate the timing of school exposure for their children. I confirm this by looking at the balance between treated and control individuals. Figure 3 shows these results. I find that the characteristics of treated and control individuals broadly similar.

Table 3: Balancing on Treatment Status

	Treatment Status	
Characteristic	0, N = 11,026	1, N = 3,675
Link Prob.	0.86 (0.19)	0.86 (0.20)
Age	8.17 (5.96)	7.72 (5.86)
Only Child	0.07 (0.25)	0.10 (0.30)
Birth Order	3.15 (2.05)	2.99 (1.98)
Unknown	449	110
RR Distance (1900)	9.12 (1.45)	9.99 (1.23)
School Distance (1900)	8.42 (1.02)	9.90 (0.56)
Immigration Year	1,901.60 (3.66)	1,902.48 (2.53)
Unknown	9,180	2,438

[1] Mean (SD)

Balancing between treated and control groups. A small number of children have an unknown birth order due to missing or identical ages of their siblings. There are a large number of observations with no immigration year, these are native born individuals.

This design also controls for other possible shocks we might worry could confound the results. One possibility is that regions that constructed schools could differ systematically from those that didn't. Enumeration subdistrict controls partly address this concern due to their small size (typically 4 townships, or 12 miles by 12 miles). Including an older pre cohort ensures that other aggregate shocks are also controlled for. Changes in agricultural technology or agricultural prices which affect both young and old cohorts equally and are the typical forces of structural transformation will be controlled for.

A second set of possible shocks we might worry about come from shocks that are cohort specific. This is a particular concern with World War I participation which would affect the 12-17 cohort, who will be 20-25 and therefore of prime military age in 1914 at the start of the war. Thankfully the design accounts for this, so long as participation in the war is not correlated with the likelihood of being treated by school construction. This could still be a potential threat if school construction encouraged more students to enlist in WWI[9].

9. A possible future step would be to link to the 1916 Census of the Prairie Provinces, which asked for military service.

4.1 Agriculture, Migration, and Income

I first examine whether an individual is working on a farm in 1931. Working on a farm is a combination of two categories. The first is those listed as working in an agricultural occupation as defined by their occupation codes. The second group is those listed as labourers who are working on a farm as listed by their industry codes.

Table 4 shows the results. Starting with the simplest specification in column 1, I find that the interaction effect for school construction and cohort is -3.87pp. Combining this with the direct effect of school construction of 1.09pp gives a total effect of -2.78pp or a 4.5% decrease in the likelihood of working on a farm as an adult. The cohort effect is negative in this specification, indicating that the younger "post" cohort is less likely to be in agriculture as adults, but this effect disappears with the inclusion of controls. I weight all observations by the MLP derived probability that a link is correct.

Columns 2-5 of table 4 include additional controls that may influence the results. Column 2 includes controls for birthplace and birthplace of parents, leading to a decrease in the coefficient sizes. Controlling for birthplace is important to capture cultural differences in education. Column 3 adds in controls for homestead status, who are on average poorer, and livestock holdings, who are on average richer. Column 4 includes controls for agricultural productivity from the crop suitability indexes. Although birthplace controls impacted the coefficients, the inclusion of agricultural controls appears to have had minimal effects.

Column 5 of table 4 includes a control for the log distance to railway in 1906, and is the preferred specification. Railways were the primary mode of transportation at the time, providing both manufactured goods and a market to sell grain. I find that the likelihood of remaining in agriculture is positively associated with distance to the railroad in 1906. There is the possibility of interaction effects between railway access and school access. Instead I find that the largest effects from education come from households furthest from the railway as seen in table A18 in the appendix.

The total effect of school construction in column 5 is the combination of the direct school effect and the school and cohort interaction. This gives a total effect of -2.64pp or a 4.3% decline in the likelihood of a child working in agriculture as an adult if exposed to schooling. Note that these estimates reflect a likely underestimate of the true effects due to bias from linking. Children who remain in farming are more likely to be linked than those that leave, so any estimate of the likelihood of leaving is therefore biased towards zero.

The decline in farm work likely coincides with migration for other opportunities. Table 6 repeats the same estimation with migration distance as an outcome. I find that having a school constructed nearby increases the distance from home by 15.3% when using an inverse hyperbolic sine transformation in column 2 and 12.8% when using a log transformation in

Table 4: Farm Work in 1931

Dependent Variable:	Farm Work				
Model:	(1)	(2)	(3)	(4)	(5)
School Constr. × Cohort	-0.0387**	-0.0276	-0.0291*	-0.0295*	-0.0297*
	(0.0170)	(0.0172)	(0.0171)	(0.0171)	(0.0171)
School Constr.	0.0109	0.0045	0.0053	0.0029	0.0033
	(0.0175)	(0.0176)	(0.0176)	(0.0177)	(0.0175)
Cohort	-0.0193**	-0.0162	0.0054	0.0051	0.0043
	(0.0095)	(0.0099)	(0.0099)	(0.0099)	(0.0099)
Log RR Distance (1906)					0.0200***
					(0.0048)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes	Yes
Fam. Bpl.		Yes	Yes	Yes	Yes
Homestead			Yes	Yes	Yes
Livestock			Yes	Yes	Yes
Ag. Productive.				Yes	Yes
Observations	14,701	14,701	14,701	14,701	14,701
R^2	0.05817	0.06562	0.07704	0.07841	0.07989
Dependent variable mean	0.62009	0.62009	0.62009	0.62009	0.62009

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Likelihood of working on a farm as an adult. School construction (treatment) is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905, conditional on it being within 10km by 1906. The post cohort are those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

column 3. These interaction effects are offset by declines from the direct effect of school construction. The total effect of school construction is to increase the distance from home by 11.2% with log, or 8.28% with the inverse hyperbolic sine.

Both the 1906 census records and 1931 census records include the legal land description of individual households. I construct a binary outcome for an adult living on the same land section as they did as a child. Column 4 finds there is an insignificant decline (-1.31pp) in the likelihood of living on the same family farm or a 5.7% decline. Although insignificant the results agree with the previous results on geographic distance.

The same caveat that these migration probabilities are a lower bound applies with one additional feature. The MLP linking method explicitly includes geographic distance as a linking variable. Including distance allows the algorithm to disambiguate more clearly between otherwise identical individuals, as discussed in section A.5. The cost is that any mobility estimates will be biased downwards from this process. These estimates should be taken as a lower bound of the true effect of schooling on migration distance.

One advantage of Canadian census records is that income is recorded starting in 1901, versus the United States which only started reporting income in 1940. I therefore want to examine the income effects of school access. One concern is the selection process into reporting income in the census. Only employees were asked in the 1931 census to report income for the prior 12 months (Canada. Dominion Bureau of Statistics. 1931). The majority of farmers are self-employed and therefore reported no income. The sample size for income is therefore significantly smaller than that for other outcome variables, and reflects income conditional on being an employee, of which the majority are non-agricultural.

Table 5 shows this result. The baseline specification in column 1 reports a 9.67% increase in income for affected cohorts, combined with a 1.49% increase from the direct effect. Including more controls leads to the preferred specification in column 5, which reports a total 12.59% increase in income. As with the results for agriculture where the likelihood of being linked decreases with income, these effects are likely an underestimate of the true income effect.

Enumerators were asked to only record the income of employees, but not all enumerators followed the instructions. Table A22 in the appendix reports income for all farmers, conditional on reporting, in the 1931 census. I find that farmers and farm labourers are amongst the lowest paid occupations. There is a clear hierarchy, with farming employees earning the lowest amount, own account farmers in the middle, and farming employers as the highest earners.

The results presented so far make use of the comparison of children 0-5 in 1906 to those 12-17 in 1906. This estimation can be repeated with other cohorts, creating an event study.

Table 5: Income in 1931

Dependent Variable:			Log Income		
Model:	(1)	(2)	(3)	(4)	(5)
School Constr. × Cohort	0.0967*	0.1057*	0.1110*	0.1170*	0.1173*
	(0.0570)	(0.0587)	(0.0593)	(0.0597)	(0.0598)
School Constr.	0.0149	0.0094	0.0068	0.0095	0.0086
	(0.0545)	(0.0556)	(0.0563)	(0.0556)	(0.0553)
Cohort	-0.3056***	-0.2677***	-0.2612***	-0.2556***	-0.2541***
	(0.0279)	(0.0295)	(0.0306)	(0.0308)	(0.0308)
Log RR Distance (1906)					-0.0330***
					(0.0126)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes	Yes
Fam. Bpl.		Yes	Yes	Yes	Yes
Homestead			Yes	Yes	Yes
Livestock			Yes	Yes	Yes
Ag. Productive.				Yes	Yes
Observations	5,651	5,651	5,651	5,651	5,651
R^2	0.15457	0.16322	0.16709	0.17199	0.17320
Dependent variable mean	6.5234	6.5234	6.5234	6.5234	6.5234

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Income as an adult in 1931. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905, conditional on it being within 10km by 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

Table 6: Migration Distance between 1906 and 1931

Dependent Variables:	Migration Dist. (km)	Asinh Migration. Dist.	Log Migration Dist.	On Family Farm
Model:	(1)	(2)	(3)	(4)
School Constr. × Cohort	29.05***	0.1623*	0.1372*	-0.0131
	(11.20)	(0.0922)	(0.0795)	(0.0180)
School Constr.	-7.393	-0.0504	-0.0544	-0.0007
	(9.664)	(0.0843)	(0.0782)	(0.0161)
Cohort	-0.0974	-0.1423***	0.0322	0.0367***
	(6.861)	(0.0536)	(0.0474)	(0.0098)
Log RR Distance (1906)	-3.566	-0.0296	-0.0064	0.0106***
	(2.779)	(0.0244)	(0.0211)	(0.0041)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes
Fam. Bpl. + Homestead	Yes	Yes	Yes	Yes
Livestock + Ag.	Yes	Yes	Yes	Yes
Observations	14,701	14,701	12,977	14,701
R^2	0.09335	0.11241	0.11932	0.09491
Dependent variable mean	223.17	4.1938	4.0493	0.23624

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Distance from childhood home in 1906 versus adult home in 1931. Column 1 reports the effect on linear distance, which is highly skewed. Column 2 resolves this and reports the estimated effect taking the inverse hyperbolic sine of the distances. To check if the presence of zeros is driving the results, column 3 reports the log migration distance. Finally Column 4 reports a binary outcome for whether an individual is living on the same farm as an adult that they were as a child. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

This allows us to visually confirm that older cohorts do not have any impact from treatment, equivalent to the parallel trends assumption. I split children and young adults between the ages of 0 and 24 in 1906 into four age groups. The excluded age group is those 12-17 as in the difference-in-differences. The specification echoes the earlier difference-in-differences specification:

$$y_{icg} = \sum_{k \neq \text{age}_{12} - \text{age}_{17}} \gamma_k t_k s_i + s_i + t_c + \beta X_i + \alpha_g + \epsilon_{icg} \tag{2}$$

with dummies for each cohort t_k.

Figure 6 shows the results. Income is positive and significant for the youngest cohort, with some evidence of pre-trends. The likelihood of working on a farm is negative only for the youngest cohort, with no evidence of pre-trends confounding the results. Relative to the 12-17 cohort, it is only the youngest cohort of 0-5 who sees significant benefits from school construction.

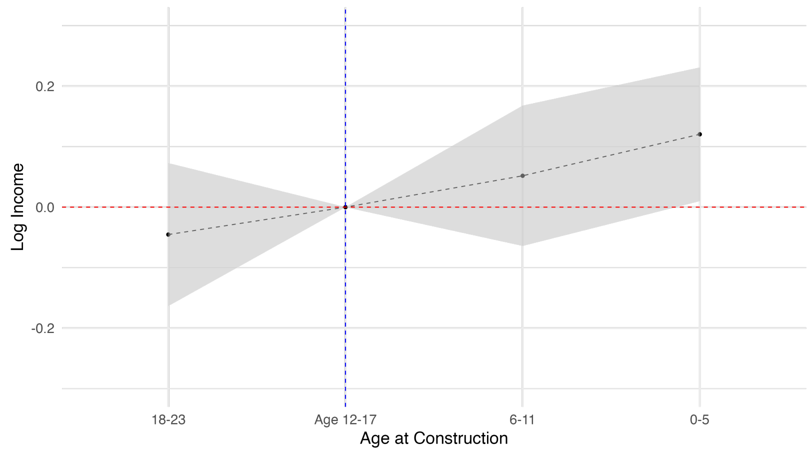
Two further robustness checks can be found in the appendix. I vary the distance threshold used to define treatment and control in tables A19 and A20, finding broadly similar results. As a further robustness check I alter the definition of working in agriculture by including other family members, counting if any family member works in agriculture. This sharpens the result from table 4, and is significant across all specifications with a marginal change in the point estimates.

4.2 Occupation

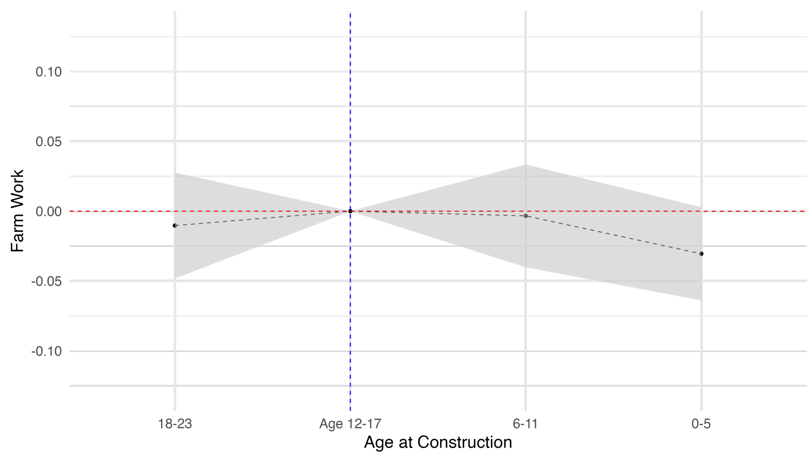
Knowing that people are exiting agriculture, what occupations are they switching into? The 1931 census includes information on the occupation that people held, and this information has been coded to six digit occupation codes using the HISCO definitions. I repeat the same difference-in-differences estimation strategy as above, but use as outcome variables the occupation held by different individuals.

Figure 8 shows a breakdown by the three primary industries, and a separate category for labourer. The estimated effect of school exposure is only significant for the professional and services sector, which is positive. Manufacturing and labour are slightly negative but insignificant. Agriculture is negative but insignificant, reflecting the inclusion of farmers but not labourers working on a farm in the earlier results. The positive result for professional and services is surprising given the typical pathway for structural transformation through manufacturing. These results suggest that education expansions may bypass the traditional route through manufacturing. An alternate explanation could be the relative lack of manu-

Figure 6: Event Study



(a) Income

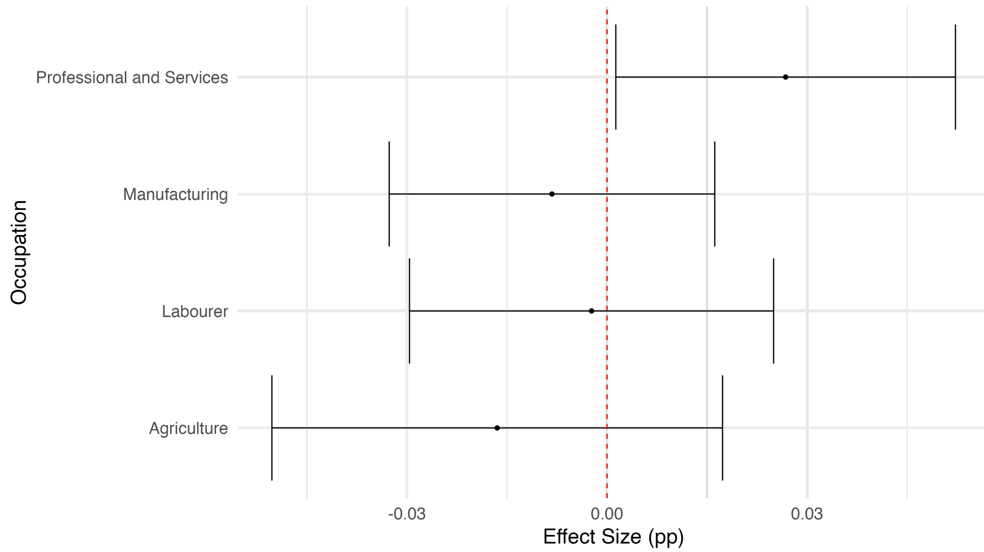


(b) Farm Work

Figure 7: Event study for income and engaging in farm work. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905, conditional on it being within 10km by 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

facturing in the Canadian prairies, a phenomena also mentioned in Goldin and Katz (2000) in Iowa.

Figure 8: Primary Industry



One digit occupation estimation. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905, conditional on it being within 10km by 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

I break this down further into two digit occupations, as seen in figure 9. There is an insignificant decline in the likelihood of becoming a farmer; it is only when combining with labourers working in agriculture that the decline in agricultural work in table 4 can be recovered. There are insignificant declines in the likelihood of working as a store clerk, as a buyer, or in transportation. There is also a significant decline in the likelihood of becoming a housekeeper, although the magnitude of the coefficient is small. In general the occupations that see declines tend to be lower skilled occupations.

In contrast there is a marked increase in the likelihood of working in some higher skill occupations. There are significant increases for the likelihood of being a teacher and manager, both of which are high skill occupations that require education. Teachers were one of the few high skill occupations available in rural areas; it is reassuring that children who had exposure to teachers are more likely to enter this occupation themselves as adults.[10] There

10. This echoes the results of (Andrabi, Das, and Khwaja 2013) who find that education expansions in

are insignificant, but positive, increases for becoming a working proprietor who owns their own business, which would require entrepreneurial ability and is also likely to be higher skilled. There is also a positive but insignificant increase in the likelihood of becoming a plumber. This is somewhat surprising today, but in 1931 indoor plumbing was considered a luxury in the prairies (Dyck 2007), suggesting a premium for plumbers.

There is significant heterogeneity within occupations that reflects childhood background. The two digit category for Agents contains a variety of industries. There is a slight increase in the likelihood of being an agent as an adult, but the occupation of agent is broad and could span a variety of industries. Figure 7 breaks down the industries that agents are working in, and further separates agents by birthplace. The left-hand table shows agents who grew up in rural areas, while the right-hand table shows agents who grew up in urban areas. Agents who grew up in rural areas are most likely to be found working at a grain elevator, while agents who grew up in urban areas are most likely to be found working in insurance. This suggests that although schools may have shifted occupations away from agriculture, children who grew up on a farm may still be using their agricultural knowledge in their later professions and working in agriculture adjacent services. Table A23 in the appendix finds similar effects for managers; those who grew up in rural areas are more likely to be managers of a farm.

Table 7: Agents in 1931

Industry	n
Unknown	130
Grain Elevator	55
Insurance	45
General	12
Real Estate	9
Steam Railway	9
General Farm	6
Lumber Mill	5
Retail Hardware	5
Steam Railroad	5
Other	28

Grew up Rural

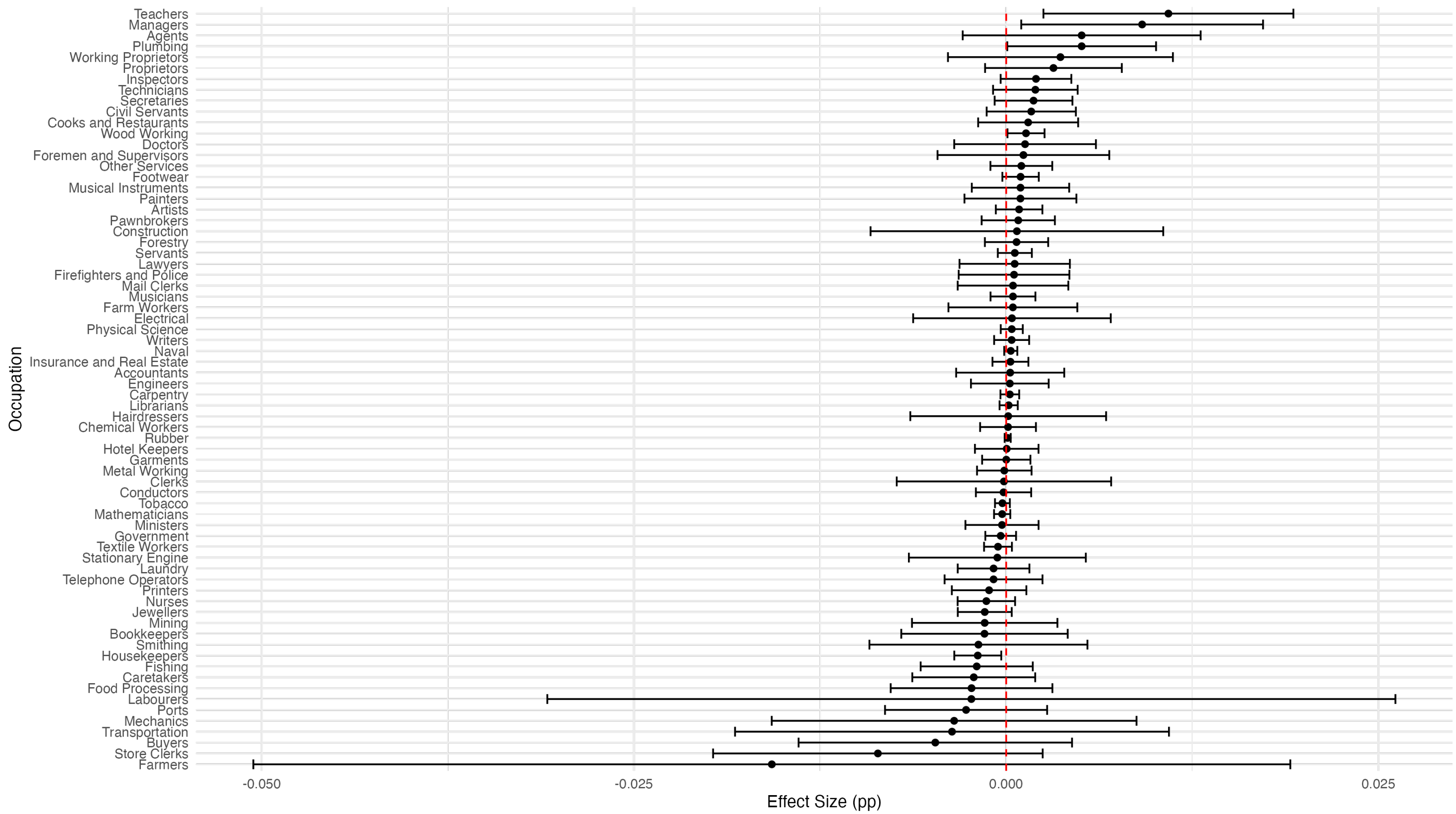
Industry	n
Unknown	96
Insurance	29
Real Estate	10
Grain Elevator	7
Steam Railway	5
Mill Farm	4
Wholesale Grocery	3
Financial	2
Automobile	1
Barber Shop	1
Other	15

Grew up Urban

Taken together I argue that the expansion of schools in western Canada did successfully endow children with useful human capital, which they used in non-agricultural occupations.

Sub-Saharan Africa in turn lead to more teachers.

Figure 9: 2 Digit Occupations



Two digit occupation estimation. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905, conditional on it being within 10km by 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

The types of occupations being switched into are disproportionately higher skill, while the occupations being left are lower skill. At the same time the persistence of working in agriculture adjacent industries is also clear. Children pick up additional skills from school, but they retain human capital from growing up in rural areas that makes them more suitable to work in agriculture adjacent industries.

One possible competing mechanism that could be driving the results is that exposure to a teacher, who was often more educated than the rest of the population and largely from other regions of Canada, may inspire children to leave their homes as adults. Only a quarter of teachers were born on the prairies in 1911[11]. Exposure to teachers from other provinces or countries may have inspired children to leave the farm, without necessarily providing them with human capital.

I examine the role of the teacher, and link children in 1911 to their likely teacher. The 1911 census does not include precise geolocations, so I cannot assign teachers to their nearest schools. Instead, I assign each child to the nearest teacher by page distance in the census, taking inspiration from Card, Domnisoru, and Taylor (2022). As discussed in Card, Domnisoru, and Taylor (2022) this strategy is only reliable in rural areas, which is my sample of interest anyway. Details for this procedure can be found in section A.12 of the appendix. To test the validity of this linking method I first replicate the main result of Card, Domnisoru, and Taylor (2022) that female students perform better when taught by a female teacher in table A25 in the appendix. This suggests that the linking of children to teachers is performing correctly.

I then examine outcomes for students by teacher birthplace. If teachers had a strong inspirational effect, then we would expect that students taught by teachers from farther provinces or countries would have different outcomes than students taught by local teachers. I find no obvious effects from having a teacher from different provinces on either contemporaneous schooling or later life outcomes. The results can be found in tables A26 and A27 in the appendix.

4.3 Farm Size

Large distances in the prairies required farms be occupied by the owner or renter of the land. If access to education increased the likelihood of leaving agriculture as an adult and moving away from home, then all else being equal the average farm size should increase. This could not be the case however if school construction encouraged further immigration (Schaede and Smith 2025) or increased birth rates.

11. See table A24 in the appendix for a breakdown.

I test this by estimating the effect of school openings on farm size. I utilize the set of geolocated farms in 1906, 1921, and 1931 in their entirety, allowing for new farmers to take over existing farms. Using the geolocation for each farm combined with the panel of school opening dates gives different exposures for each farm to access to school. A farm is defined as being close to a school if it had a school constructed within 5 kilometers by 1912 at the end of the school opening data $p_i = \mathbb{1}\{dist_i < 5km\}$. Farms are considered to have had a school early relative to other farms if that school was constructed before 1906 $s_j = \mathbb{1}\{date_j < 1906\}$. Finally there is a time dimension consisting of the years 1906, 1921, and 1931 (T_t).

The following specification is then estimated in a triple difference setup:

$$y_{ijt} = \beta p_i s_j T_t + p_i s_j + s_j T_t + p_i T_t + p_i + T_t + \alpha_j + \epsilon_{i,j,t} \tag{3}$$

where β is the coefficient of interest. Instead of including s_j directly, a school fixed effect α_j is included to control for any geographic characteristics.

Table 8 shows the results. The coefficient in the third row shows the interaction between having a school constructed early and nearby in 1906, the second row the interaction with 1921, and the third row with 1931. It is this top coefficient, which gives the estimated effect on farm sizes in 1931 from school construction. Column 1 gives the most basic specification, while columns 2 through 5 add controls for farm size. I choose the farm size controls based on the farm size validation results from table A14 in the appendix, controlling for family characteristics which are available in all census waves and correlate with farm size. Column 5 gives the preferred specification.

I find that farm sizes increase by 5.24% around where schools are constructed in 1931. Schools are initially constructed in areas with 5.14% smaller farms (as indicated in the "Close" row), consistent with schools being built in more populated areas. The 5.24% increase in farm sizes around schools closely aligns with the 4.3% decline in farm work estimated in section 4.1. If anything the direct effects of school construction appear to understate the amount of farm consolidation occurring.

5 Model

I now show how we can recover structural transformation from a model of education choice, industry choice, and migration. In counterfactuals, I want to show how decreasing costs of education can both rationalize the results and explore spatial heterogeneity.

I draw from the model of Hsiao (2025), and extend it by adding a separate agricultural and non-agricultural sector. This allows my model to nest Lagakos and Waugh (2013) and allow

Table 8: School Construction and Farm Size

Dependent Variable:	Ln Farm Area				
Model:	(1)	(2)	(3)	(4)	(5)
Early × Close × Year = 1931	0.0677*	0.0467	0.0500	0.0522*	0.0524*
	(0.0383)	(0.0316)	(0.0316)	(0.0315)	(0.0314)
Early × Close × Year = 1921	-0.0062	-0.0161	-0.0134	-0.0152	-0.0159
	(0.0383)	(0.0319)	(0.0319)	(0.0317)	(0.0316)
Early × Close	-0.0174	-0.0040	-0.0087	-0.0068	-0.0063
	(0.0313)	(0.0269)	(0.0269)	(0.0268)	(0.0267)
Constant	6.358***				
	(0.0292)				
Early	-0.0099				
	(0.0323)				
Close	-0.1090***	-0.0521***	-0.0533***	-0.0516***	-0.0514**
	(0.0242)	(0.0200)	(0.0200)	(0.0200)	(0.0200)
Year = 1921	-0.1588***	-0.1835***	-0.2336***	-0.1981***	-0.1982***
	(0.0309)	(0.0254)	(0.0254)	(0.0255)	(0.0254)
Year = 1931	-0.0735**	-0.0786***	-0.1347***	-0.0821***	-0.0822***
	(0.0309)	(0.0256)	(0.0256)	(0.0258)	(0.0257)
Early × Year = 1921	-0.0166	0.0384	0.0481	0.0358	0.0347
	(0.0387)	(0.0321)	(0.0321)	(0.0322)	(0.0321)
Early × Year = 1931	-0.0557	-0.0074	0.0019	-0.0209	-0.0211
	(0.0381)	(0.0318)	(0.0318)	(0.0319)	(0.0319)
Close × Year = 1921	0.0505*	0.0463**	0.0476**	0.0503**	0.0500**
	(0.0267)	(0.0218)	(0.0217)	(0.0217)	(0.0217)
Close × Year = 1931	0.0381	0.0349	0.0345	0.0314	0.0311
	(0.0273)	(0.0221)	(0.0221)	(0.0221)	(0.0221)
Homestead			-0.2102***	-0.2117***	-0.2118***
			(0.0052)	(0.0052)	(0.0052)
RR Dist in Year				0.0381***	0.0381***
				(0.0036)	(0.0036)
Family Size					-0.0005
					(0.0006)
Fam. Adult Males					0.0044***
					(0.0016)
School District		Yes	Yes	Yes	Yes
Observations	412,918	412,918	412,918	412,918	412,918
R^2	0.00667	0.13645	0.15332	0.15527	0.15577

Clustered (Enum. Sub.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

for sorting between agricultural and non-agricultural sectors on the basis of comparative advantage. I model the agricultural sector in a similar way to Porzio, Rossi, and Santangelo (2022), with a smaller share for human capital in agriculture than non-agriculture, but also include diminishing returns in the non-agricultural sector.

Note that I exclude the traditional demand factors of structural transformation coming from improvements in agricultural productivity and non-hometheticity of preferences are absent. I do this for the sake of clarity, and do not claim that the traditional forces of structural transformation are absent from this context.

5.1 Education, Sector, and Migration

Each individual i makes three sequential choices. In the first stage they receive an education cost shock ϵ and choose education e. In the second stage they receive a sector shock z_g, and choose between the agricultural sector F and the non-agricultural sector N. In the final stage they receive a productivity shock by location ε_ℓ and choose a location ℓ. The assumption is that parents are forward looking and altruistic towards their children.

I solve the model in reverse order. In the third choice over locations, the individual makes a choice over locations ℓ given their birthplace j, cohort k, sector choice g and education level e. Utility from a location is:

$$v_{jkg}(e, \varepsilon) = \max_{\ell}\{v_{jkg\ell}(e, \varepsilon_\ell)\}$$

The utility in each location ℓ can be split into utility from amenities a_ℓ, wages $w_{jkg\ell}$, the migration cost to get there $\tau^m_{jkg\ell}$, and an idiosyncratic utility term ε_ℓ which is distributed Fréchet with dispersion parameter θ.

$$v_{jkg\ell}(e, \varepsilon_\ell) = \frac{a_\ell w_{jkg\ell}(e)}{\tau^m_{jkg\ell}}\varepsilon_\ell$$

Wages in a location ℓ in sector g for education level e depend on the wage rate per unit of human capital and the number of units of human capital.

$$w_{jkg\ell}(e) = r_{g\ell} h_{jkg\ell}(e)$$

Units of human capital can be decomposed into human capital from education e with some elasticity η, a sector specific skill z_g, and a general skill $s_{jk\ell}$. Human capital can be expressed as:

$$h_{jkg\ell}(e) = e^\eta s_{jkg\ell} z_g$$

The utility from choosing location ℓ conditional on being from origin j, cohort k, having education e, and having chosen sector g is therefore:

$$v_{jkg\ell}(e,\varepsilon_\ell) = \frac{a_\ell r_{g\ell} e^\eta s_{jkg\ell} z_g}{\tau^m_{jkg\ell}} \varepsilon_\ell \tag{4}$$

Following Hsiao (2025) I define the location utility $\tilde{v}_{jkg\ell}$ as the total of all components which do not vary at the individual level. This definition is crucial because it acts as an input for both the market access to locations MA_{jkg} and the choice probabilities for locations $\bar{m}_{jkg\ell}$. These components are:

$$\tilde{v}_{jkg\ell} = \frac{a_\ell r_{g\ell} s_{jkg\ell}}{\tau^m_{jkg\ell}} \tag{5}$$

$$\text{MA}_{jkg} = \sum_\ell \tilde{v}^\theta_{jkg\ell} \tag{6}$$

$$\bar{m}_{jkg\ell} = \frac{\tilde{v}^\theta_{jkg\ell}}{\text{MA}_{jkg}} \tag{7}$$

The derivations of equations 6 and 7 follow from standard Fréchet logic and can be found in the appendix. The expected utility and wage before realizing the productivity shock can also be computed as:

$$\bar{v}_{jkg}(e) = \mathbb{E}[\max_\ell v_{jkg\ell}(e,\varepsilon_\ell)|e] \tag{8}$$

$$= z_g e^\eta \gamma_\ell \text{MA}^{\frac{1}{\theta}}_{jkg} \tag{9}$$

$$\begin{aligned}\bar{w}_{jkg}(e) &= \mathbb{E}[\max_\ell w_{jkg\ell}(e,\varepsilon_\ell)|e] \\ &= e^\eta z_g \gamma_\ell \text{MA}^{\frac{1}{\theta}}_{jkg} \frac{\sum_\ell (\frac{\tau^m_{jkg\ell}}{a_\ell}) \tilde{v}^\theta_{jkg\ell}}{\text{MA}_{jkg}}\end{aligned}$$

The derivation for both equations can be found in section B.2 in the appendix.

In the second stage, before the location choice is made, individuals make a choice over which sector to work in. Starting from a location j and cohort k the individual must choose over sectors $g \in \{F, N\}$. The sectoral utility is therefore:

$$u_{jk} = \max_g \{\bar{v}_{jkg}(e)\}$$

where $\bar{v}_{jkg}(e)$ is the utility from each sector as defined in equation 8. They receive a Fréchet sector specific skill shock with dispersion parameter ϕ. Using the same Fréchet logic as before, I recover the choice probabilities of choosing industry g as:

$$\bar{n}_{jkg} = \frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}}}{MA_{jk}} \tag{10}$$

where $\mathrm{MA}_{jk} = \sum_g (\mathrm{MA}_{jkg}^{\frac{1}{\theta}})^{\phi} = \sum_g (\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}})$. The expected utility before the choice of sector and expected productivity in each sector are therefore:

$$\bar{u}_{jk}(e) = \gamma_g \gamma_\ell e^{\eta} \mathrm{MA}_{jk}^{\frac{1}{\phi}}. \tag{11}$$

$$\bar{z}_{jkg}(e) = \mathbb{E}[\max_g z_g | e] = \gamma_g \bar{n}_{jkg}^{-\frac{1}{\phi}} \tag{12}$$

Derivations can be found in section B.3 in the appendix. With the mean utility after the sector choice, we can now compute the expected level of education. Education is chosen as a continuous quantity e, given the expected utility from the sector choice from having that level of education $\bar{u}_{jk}(e)$ and a cost of education $c_{jk}(e)$.

$$t_{jk}(\epsilon) = \max_e \{\bar{u}_{jk}(e) - c_{jk}(e, \epsilon)\}$$

The education cost can be decomposed into a common cost for education τ_{jk}^e and an idiosyncratic component ϵ giving $c_{jk}(e, \epsilon) = e\tau_{jk}^e \epsilon$. Utility from labour and education is therefore:

$$t_{jk}(\epsilon) = \max_e \{\gamma_s \gamma_\ell e^{\eta} \mathrm{MA}_{jk}^{\frac{1}{\phi}} - e\tau_{jk}^e \epsilon\}$$

Solving for the utility maximizing e gives:

$$e_{jk} = \left(\frac{\gamma_s \gamma_\ell \eta \mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\tau_{jk}^e \epsilon} \right)^{\frac{1}{1-\eta}}. \tag{13}$$

5.2 Production

Production occurs in two sectors, agriculture and non-agriculture, which have differing rates of human capital usage. Non-agricultural production is given by:

$$Y_{N\ell} = A_{N\ell} H_{N\ell}^{\kappa} \tag{14}$$

where Y is non-agricultural output, $A_{N\ell}$ is non-agricultural productivity, H_ℓ is human capital in location ℓ, and κ is the production elasticity. Agricultural production is given by:

$$Y_{F\ell} = A_{F\ell} T_\ell^\delta H_{F\ell}^{\kappa-\delta} \tag{15}$$

where $Y_{F\ell}$ is agricultural output, $A_{F\ell}$ is agricultural productivity, T_ℓ is land, $H_{F\ell}$ is human capital in agriculture, and δ is the production elasticity of land. δ serves an important role in the model, diminishing the returns to human capital more quickly in the agricultural sector than the non-agricultural sector. I follow Hsiao (2025) and interpret these diminishing returns as being a net congestion force. In this model with two sectors however, the diminishing returns are larger in the agricultural sector reflecting congestion on a finite amount of land.

Following Hsiao (2025), output in each sector and each location is independent of each other, and firms are perfectly competitive. For each sector this gives:

$$Y_g = \sum_\ell Y_{g\ell}.$$

Taking the first order condition of equation 14 with respect to $H_{N\ell}$ and setting it equal to the wage rate per unit of human capital $r_{N\ell}$ gives:

$$r_{N\ell} = \kappa A_{N\ell} H_{N\ell}^{\kappa-1} = \frac{\partial Y_{N\ell}}{\partial H_{N\ell}}. \tag{16}$$

The stock of human capital $H_{N\ell}$ is not directly observable. We know that the wage rate paid is equal to the stock of human capital times the wage rate per unit of human capital $w_{jkg\ell} = r_{g\ell} h_{jkg\ell}$. In aggregate for a given location sector this gives $W_{jkg\ell} = r_{g\ell} H_{jkg\ell}$. We can recover the aggregate wages paid to a given location by:

$$W_{g\ell} = \sum_{jk} N_{jk} \bar{n}_{jkg} \bar{m}_{jkg\ell} \bar{w}_{jkg\ell}. \tag{17}$$

Combining equations 14, 16, and the equation for the relationship between wages and human capital gives:

$$Y_{N\ell} = \frac{r_{N\ell} H_{N\ell}}{\kappa} = \frac{W_{N\ell}}{\kappa}, \tag{18}$$

which gives a relationship between wages and output. The same logic applies to agricultural production, yielding a first order condition of:

$$r_{F\ell} = (\kappa - \delta) A_{F\ell} T_\ell^\delta H_{F\ell}^{\kappa-\delta-1} \tag{19}$$

and an equivalent relationship between output and wages:

$$Y_{F\ell} = \frac{W_{F\ell}}{\kappa - \delta}. \tag{20}$$

Comparing the non-agricultural to agricultural sectors, for the same level of total wage payments $W_{g\ell}$ the amount of output will be lower due to δ.

5.3 Equilibrium

An equilibrium consists of a set of wage rates in each sector location $r_{g\ell}$ such that human capital markets clear in each sector location. That is:

$$H_{g\ell}^D(r_{g\ell}) = H_{g\ell}^S(r_{g\ell}) \quad \forall g, \ell.$$

The demand for human capital in each sector location depends on the representative firm. Inverting equation 16 I recover for the non-agricultural and agricultural sectors:

$$H_{N\ell}^D(r_{N\ell}) = \left(\frac{\kappa A_{N\ell}}{r_{N\ell}}\right)^{\frac{1}{1-\kappa}}$$
$$H_{F\ell}^D(r_{F\ell}) = \left(\frac{(\kappa - \delta) A_{F\ell} T_\ell^\delta}{r_{F\ell}}\right)^{\frac{1}{1-\kappa+\delta}}$$

The amount of human capital supplied to each location depends on the aggregate wage rate and the wage rate per unit of human capital. This gives:

$$H_{N\ell}^S(r_{N\ell}) = \frac{W_{N\ell}}{r_{N\ell}}$$
$$H_{F\ell}^S(r_{F\ell}) = \frac{W_{F\ell}}{r_{F\ell}}$$

5.4 Estimation

From the model I derive two estimating equations, taking account the additional sector relative to Hsiao (2025). Derivations for these equations can be found in section B.4.

$$\underbrace{log(\bar{w}_{jkg\ell})}_{\text{Data}} - \underbrace{log(\bar{e}_{jk})}_{\text{Estimate}} = \underbrace{log(\frac{\tilde{\epsilon}}{\bar{\epsilon}\eta})}_{\text{Constant}} - \underbrace{log(a_\ell)}_{\text{FE}} + \underbrace{log(\tau^e_{jk})}_{\text{FE}} + \underbrace{log(\tau^m_{jk\ell g})}_{\text{Residual}} \tag{21}$$

Estimating equation 21 requires several pieces of data. I first aggregate the data to the 1931 census division level (Gaffield et al., CCRI 1931 Census of Canada: Aggregate Data and Geography = CCRI 1931 Recensement Du Canada: Données Agrégées et Cadre Géographique), using geolocations for children in 1906 and adults in 1931[12]. These census divisions serve as the origins and destinations in the model. $\bar{w}_{jkg\ell}$ comes directly from wages at the origin, cohort, industry, destination level. This comes directly from the data, with the exception of agricultural wages which are imputed as few farmers reported income as described in section A.10. $\bar{e}_{jk}$ is the average education level at the origin cohort level, using the estimates described in section C.3. The results of this data assembly process can be found in section C of the appendix.

I then estimate equation 21. a_ℓ comes from origin fixed effects. τ^e_{jk} comes from destination-cohort fixed effects. $\tau^m_{jk\ell g}$ is a residual. Estimating this equation gives τ^e_{jk} and $\tau^m_{jk\ell g}$ giving the cost of education access and migration, and amenity a_ℓ.

$$\underbrace{log(\bar{w}_{jkg\ell})}_{\text{Data}} = \underbrace{log(\frac{\tilde{\epsilon}\gamma_g\gamma_\ell}{\bar{\epsilon}^\eta})}_{\text{Constant}} + \underbrace{\eta log(\bar{e}_{jk})}_{\text{Estimate}} + \underbrace{log(r_{\ell g})}_{\text{FE}} - \underbrace{\frac{1}{\theta}log(m_{jkg\ell})}_{\text{Data}} - \underbrace{\frac{1}{\phi}log(n_{jkg})}_{\text{Data}} + \underbrace{log(s_{jkg\ell})}_{\text{Residual}} \tag{22}$$

The second estimating equation 22, requires both the original data on wages and education. In addition, it requires data on the probability of migrating from any origin and cohort to any industry and destination $m_{jkg\ell}$, and from any origin cohort to any industry n_{jkg}. I take parameters θ, and ϕ from the literature. The elasticity of human capital with respect to education η, I take as 0.12, informed by OLS estimates of the return to human capital in table C30. Finally parameter δ I compute from the 1931 aggregate census tables, the details of which can be found in section C.4. After controlling for these values, I estimate equation 22 via OLS to recover $r_{\ell g}$ and $s_{jkg\ell}$. Table 9 gives the values for each parameter.

12. This is non-trivial as there are two sets of census geographies in 1931 and some other waves. One set, used for collection, consists of census districts and subdistricts. Each line on the census is identified in one of these subdistricts. The second set of geographies is for census divisions and subdivisions. These units are used to disseminate the results from the census, and are what is used in census publications. Thank you to Marc St-Hilaire who made me fully aware of this distinction.

Table 9: Select Elasticities

Parameter	Value	Source
θ Frechét Shape Parameter	20	Hsiao (2025)
ϕ Frechét Shape Parameter	4 (midpoint)	Lagakos and Waugh (2013)
κ Production Parameter	0.925	Bryan and Morten (2019)
η Human Capital Elasticity	0.12	OLS
δ	0.11	Estimated from 1931 Census Tables

I follow Hsiao (2025) and compute counterfactuals using hat algebra. To compute changes I rely on the standard definition of the percentage change of a variable $\hat{x} = \frac{x'}{x}$. After calculating labour supply and labour demand in changes, I search over the set of $\hat{r}_{g\ell}$ until $\hat{H}^D_{g\ell} = \hat{H}^S_{g\ell} \quad \forall g, \ell$. Details for the computation of all terms in changes can be found in section B.5.

5.5 Decreases in Education Costs

The key counterfactual I compute is to examine how a decrease in education costs affects aggregate outcomes. I proceed by considering a counterfactual where the education cost falls by 50% uniformly across all locations.

Table 10 shows the result of a uniform decline in education costs. As the aggregate amount of human capital increases, production increases in both the agricultural and non-agricultural sectors, leading to a 16% increase in production in the non-agricultural sector and a 10% increase in the agricultural sector. This leads to downward pressure in wages per unit of human capital in both sectors. The fact that human capital is more complementary with non-agricultural production leads wages per unit of human capital to fall more slowly in the non-agricultural sector. Finally we see a reallocation of labour between the two sectors, with the workforce increasing by 3.3% in the non-agricultural sector and decreasing 1.5% in the agricultural sector.

Adding a sector choice results in a dramatically different outcome than Hsiao (2025). The amount of education does not factor directly into the location choice in equation 7. In Hsiao (2025), this means that changes in education costs will not affect the location choice decision. In this model however, increases in the aggregate amount of education change the decision of which sector to work in. After making a sector choice individuals face different migration costs between locations, therefore shifting their location choice. In this model uniform education costs changes can therefore affect location choice probabilities.

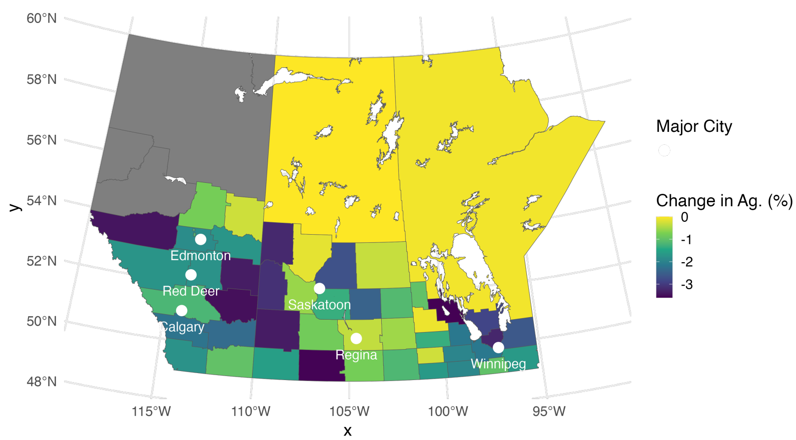
Figure 10 shows the effects of a 50% decline in education costs. As education costs fall all locations experience a decline in the likelihood of choosing agriculture versus non-

Table 10: Changes in the Non-Agricultural and Agricultural sectors in response to 50% Decline in Education Cost

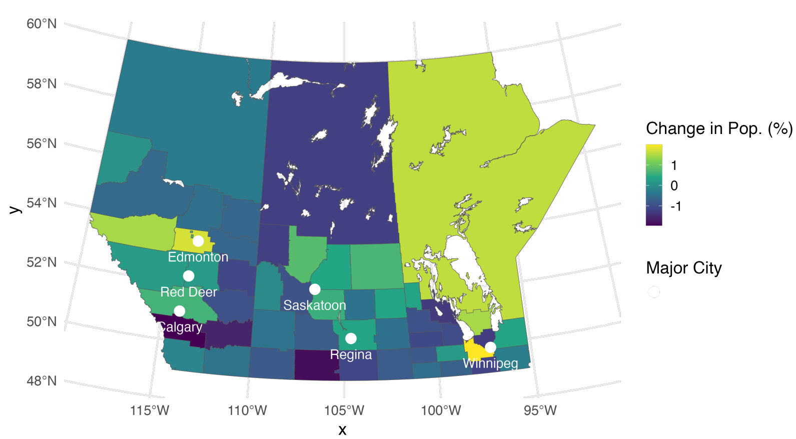
Sector	Production Change	Wage Change	Population Change
Non-Agriculture	1.1622	0.9880	1.0330
Agriculture	1.1092	0.9763	0.9851

Production Change gives the percentage change in output given in equations 18 and 20. Wage Change gives the changes in the wage rate per unit of human capital. Population change gives the percentage change in the population working in each sector. Note that the population changes do not equal each other as there is a larger initial share in agriculture than non-agriculture.

Figure 10: Regional Effects of a 50% Decline in Education Costs



(a) Change in Probability of Working in Agriculture



(b) Change in Population

Figure 11: Top: Percentage change in the probability of choosing agriculture as a result of a 50% decline in education costs. Bottom: Percentage change in population as a result of a 50% decline in education costs.

agriculture, as can be seen in the upper panel. There is considerable spatial heterogeneity, with regions close to Winnipeg seeing declines in the likelihood of agricultural employment, and the arid regions in southern Alberta and Saskatchewan also seeing declines in agricultural employment.

The lower panel shows the effects on population. Population declines are seen in the areas surrounding Winnipeg and in southern Alberta and Saskatchewan. Taking these facts together, we see that improved access to schooling drove people away from marginal agricultural land in the arid south, and towards growing cities.

These results show that the spatial role of increased education access was to exacerbate existing regional inequalities. The counterfactually lower education costs shift more population towards already growing cities, notably Winnipeg. Winnipeg was already the largest population center in 1931, and further improvement in education access would have only increased the size of the city further. The role of education in decreasing population in southern Alberta and Saskatchewan further highlights this. The late 1910s and early 1920s saw failed harvests due to drought in the southern dry belt, leading farms and towns to be abandoned (Jones 1987). These results suggest that education would not have acted as a hedge against climate risk, but rather incentivized children to leave more than they already did.

6 Conclusion

This paper examines the role of education in structural transformation by studying the rollout of schools in the Canadian prairies at the start of the twentieth century. Utilizing variation in individual level exposure to new schools, I find that rural children who had access to education are more likely to be found in non-agricultural occupations as adults, migrate farther from home, and earn higher incomes. I interpret these results through the lens of a model, and find that these patterns are consistent with human capital being complementary with non-agricultural production. Expansions in education access do not benefit everyone equally, and the most marginal agricultural regions see population declines as education access improves.

Despite differences in time, technology, and institutions, there are lessons from the expansion of schools in the Canadian prairies that can inform policy today. Early schooling in the Canadian prairies was not sophisticated, and the schools were not known for high quality education, yet they still resulted in significant gains for children. Having a school, even if poor quality, is significantly better than having no school at all. Students themselves can become teachers, further driving education expansion, as has been observed in Sub-Saharan

Africa (Andrabi, Das, and Khwaja 2013). Finally, the impacts of education expansions may lead the population to enter the services sector and bypass the manufacturing sector. This experience echoes that of many countries in Asia today (IMF 2024).

Two branches of questions remain. On the macroeconomic side it is still unclear to what extent increased access to schooling benefitted those in the agricultural sector. While rural schools clearly benefitted those able to switch to non-agricultural occupations through higher incomes, did the additional human capital also confer productivity benefits to farmers? Farmers should also benefit from additional human capital, although likely at a lower rate. If agricultural productivity increased, what role did this play in Canada's increased dominance in wheat exports? What are the welfare effects on those left behind after the education expansion?

Separate from the economic effects are the social spillovers from school expansions. Schooling was one of the first interactions between the Canadian government and recent immigrants who settled the prairies. A key motivation behind the rapid establishment of rural schools was the desire to assimilate these immigrants (Saskatchewan Department of Education 1906-1912, 1914). While I find no differences by teacher background, it is worth asking if these schools affected long-run assimilation. Similar questions apply to indigenous children, who might have been better able to preserve their culture if allowed to access non-reserve education. Finally, teaching was one of the few occupations easily accessible to women, and female teachers had a disproportionate impact on their communities. Two of the Famous Five, a group of suffragists from western Canada, were teachers. While this paper discusses the economic impacts of the rural schooling, the social impacts may be just as important.

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A Appendix

A.1 Selected Quotes

There is significant evidence from the department of education reports that school attendance in rural areas was significantly lower that desired, and that boys in particular were leaving school at a faster rate than girls. The following quotes provide some accounts of the challenges in rural schools from the Saskatchewan Department of Education (Saskatchewan Department of Education 1906-1912, 1914).

"Parents who keep children out of school because of distance or danger, appear to have no hesitation in sending the boys to market" - A. Kennedy, Inspector of Schools, 1910

"The attendance in rural districts is not so regular as one would expect or could hope for. Undoubtedly, it is in part due to the scarcity of farm help and the long distances some children have to go; but I fear it is also caused by a lack of interest on the part of many parents" - H.H. Smith, Inspector of Schools, Saskatoon, December 31, 1909

"When a boy is old enough to work an outfit on a farm, his school days are over" - John S. Huff, Davidson, Saskatchewan., March 1st, 1912

A.2 Land Allocation

Dominion land policy was marked by frequent changes between the acquisition of Rupert's Land and the North-West Territories and later settlement. I rely on Lambrecht (1991) as the primary source of both narrative and legislative background.

To enable settlement existing claims on prairie land has to be resolved. The Canadian government signed seven major treaties between 1871 and 1877 across the prairies to extinguish indigenous land title (Lambrecht 1991). The exact contents of these "Numbered Treaties" varied, but throughout they established reserves for indigenous people, administered by the Indian Act. The second group of claimants was the Métis, people who were the result of intermarriage between indigenous and european parents, who were granted 1.4 million acres under the Manitoba Act of 1870 and later lands in the North West Territories in 1885. The settling of Métis land claims was a source of continuous error, with repeated issuance of scrip which could be redeemed for land as late as 1925 (Lambrecht 1991). The overwhelming motivation behind the settling of indigenous and Métis land claims was to extinguish any claims that would impede settlement and risk the establishment of Canadian sovereignty [13].

Non reserve land would then be administered by the Dominion Lands Act in 1872 (Lambrecht 1991). The act laid out a survey for lands in Manitoba, the North-West Territories (to largely become Alberta and Saskatchewan in 1905), and later the railway belt and peace river block in British Columbia. Based on the Public Land Survey System in the United States, the Dominion Land Survey would both measure and allocate land across western Canada, and remains largely in place as of writing as the dominant way to describe parcels of land.

The Dominion Land Survey uses a regular hierarchical pattern to survey land. The base unit of land is a township, consisting of an area 6 miles by 6 miles. The location of each township is defined by three values, the township number, range number, and meridian. Township numbers give the north-south position, start at the southern border of Canada on the 49th parallel, and count up as one proceeds north. Range numbers start at the Prime Meridian (approximately 97° 27'), and increase as one moves west. Range numbers reset

13. Prime Minister John A. MacDonald would defend his role as minister of the interior with regards to Métis land claims stating "Whether they had any right to those lands or not was not so much the question as it was a question of policy to make an arrangement with the inhabitants of that Province ... in order to introduce law and order there and assert the sovereignty of the Dominion" (Lambrecht 1991)

as after crossing one of six meridians. Any position in the east-west direction is therefore defined by a range number, and the position of the range relative to the meridians. For example, a township could be located by 5-23-W2, or the 5th township, 23rd range, west of the second meridian. Figure A1 shows the extent of the Dominion Land System in western Canada.

Figure A1: Survey boundaries for townships under the Dominion Land System. Each square consists of a 6 mile by 6 mile township. The red lines give the meridians. Each township can be uniquely identified by a township number, giving the north-south direction, and a range number plus position relative to a meridian, giving the east-west direction.

Map: Index to townships in Manitoba, Saskatchewan, Alberta and British Columbia showing the townships for which official and preliminary plans have been issued to January 1st, 1929. Topographical Survey of Canada, Department of the Interior. Compiled, drawn and printed by the Topographical Survey of Canada, Ottawa. [cartographic material]
Source: Library and Archives Canada/Maps, plans and charts/NMC43265

As described in section 2.1 the primary landholders after settlement were railways, homesteaders, the Hudson's Bay Company, and land reserved for the funding of schools. Despite the relative simplicity of this formula, there was significant variation and modification of land policy that is not captured by it.

To fund the construction of railways, approximately 32 million acres was given away as grants to encourage construction between 1881 and 1894. The system was initiated to subsidize the construction of the Canadian Pacific Railway (CPR), but also applied to a multitude of smaller railways as well. Like Canadian land policy in general, there was significant variation in policy over time before the pattern seen in figure A1 became firmly established.

The 1870's featured a variety of attempts at establishing land grants and highlighted conflicting goals in Canadian policy. The first grant system in 1871 allowed three full townships (approximately 18 miles) on either side of the railway to be withdrawn as a grant, superseded in 1872 by the Dominion Lands Act which allowed land to be withdrawn up to 20 miles on either side. Reserving these lands for the railway had the unintended consequence of limiting settlement, another main goal of the government. To overcome this the land grant policy was modified again in 1879 to allow belts of land at different distances to have different checkerboard patterns with homesteads and railway land. Finally in 1881 the Canadian Pacific Railway Act authorized the current system, with alternate sections within 24 miles of the railway given to the CPR for a total of 25 million acres.

Other railways constructed across the prairies had similar arrangements with the government. Individual contracts typically specified 6,400 acres of land for mile of line, the equivalent of every other section within a 20 mile band around the railway. In all of these contracts, including the CPR, was an indemnity provision which allowed the railway to not take a particular parcel of land if it was not "fairly fit for settlement". This provision meant that railways would often select lands that were far from their railway line, especially as rail lines started to cross. Although land grants were removed in 1894, the indemnity provision led to a delay in railways selecting their parcels, which was not resolve until 1908.

After the railway land grant system ended there were still significant supports for railway construction through subsidized corporate debt. The Railway Act allowed a rail company to pass through the subsidy lands of another, encouraging more railway construction. Railways began to consolidate as well, with Canadian Northern forming from absorbing smaller railways with land grants, and later merging with Grand Trunk Pacific to form Canadian National.

Railway construction was primarily devised to support settlement, which was enabled through homesteads. The homestead system was designed to compete with the United States, including advertisements at which referred to Canada as the "Last Best West" (Sharp 1947). As a result the homestead system in Canada resembles that of the United States, but with additional features that would make it more appealing to would be settlers. As with railway land there was some variance in policy over time.

At the core of the system was a 160 acre quarter section, which would occupy one quarter of a section of land. The majority of homesteads would be on even numbered sections, except for 8 and 26, with some railway sections also being sold as homesteads if the railway used the indemnity provision to not take possession of a particular parcel. Obtaining a homestead first required "entry" by filing for a homestead at a Dominion Lands office, followed by a three year period of homestead duties which required living on and cultivating that parcel. After fulfilling homestead duties, the homesteader could apply for letters patent, or "perfecting" the homestead, and become the owner of the parcel. Some homesteaders purchased their land in less than three years.

While applying for entry, homesteaders were able to apply for pre-emption, or a second parcel of land which was contiguous with the homestead. The homesteader could cultivate this land immediately, and could be purchased after fulfilling homestead duties. The combination of pre-emption and railway indemnity slowed the rate of settlement by restricting the amount of land available. As a result the right of pre-emption was removed in 1890, then reintroduced in 1908 after the railway indemnities were resolved. More than half of homestead entries would include a pre-emption.

The eligibility for a homestead was intentionally broad. In 1874 the required age was reduced from 21 to 18, and in 1876 women were excluded unless they were the head of the household. There were no nationality requirements until 1908, when it was required that a homesteader had to be or intend to be a British subject. After 1916 at the start of World War I, entry was restricted to British subjects.

There were some other mechanisms that could be used to obtain homestead land. From 1883 to 1886 it was possible to have a second homestead, and this was again possible after 1908. Widespread crop failures after WWI prompted the government to allow a second homestead if the first had failed. In 1881 there were also colonization companies, which would purchase land and receive rebates depending on the success of their sales. Approximately 3 million acres were allocated to these companies, but they failed after the completion of the initial boom of construction with the CPR in 1885.

The final method for land allocation was as a reward for military service. Grants and scrip were given for service in the militia and North-West Mounted Police starting in 1871, with repeated extensions for entry until 1900. A similar program was established for soldier and nurses who participated in the Boer War, and involved the transfer of approximately 2 million acres in Saskatchewan and 1 million acres in Alberta. This transfer was criticized for not imposing any domicile conditions unlike regular homesteads. Finally soldiers returning from WWI were able to make an entry for an additional 160 acres in addition to a regular homestead, with subsided loans to strengthen agricultural life and "make men settlers".

The Hudson's Bay Company (HBC) was granted 5% of all available land in the fertile belt in exchange for Rupert's Land and the North-West Territories as part of the deed of surrender. This land amounted to section 8 in every township, the entirety of section 26 in every fifth township, and three quarter of section 26 in four fifths of the townships [14]. The fertile belt was defined as the land bounded by the United States border to the south,

14. $\frac{1}{36} + \frac{1}{36} * \frac{1}{5} + \frac{3/4}{36} * \frac{4}{5} = 1/20$

the North Saskatchewan river to the north, the Rocky Mountains to the west, and the Laek Winnipeg-Lake of the Woods water system to the east. In addition the HBC was given title to any land that it had fur trading posts on and adjacent parcels not to exceed 50,000 acres. There were some negotiations to allow the government to fufil other objectives if section 8 or 26 were needed, but by 1924 a total of 6,639,059 acres had been given to the HBC.

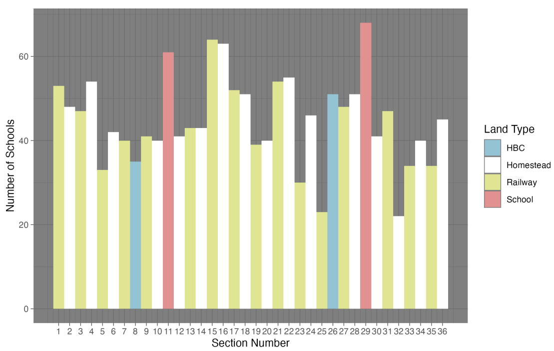
The HBC, recognizing the value of the land that it had been granted, held onto it's portfolio of land strategically and only sold parcels when the surrounding area had been settled to maximize the value (B. Hamilton, n.d.). This strategy was achieved by first allowing settlement to occur in the surrounding land, including the development of railways. Land closer to railways would sell faster (B. Hamilton, n.d.). The HBC continued to sell land throughout the twentieth century, and by 1961 formed the Rupert's Land Trading Company to administer the remaining parcles. By 1984 the final 5,100 acres in Saskatchewan were donated to the Saskatchewan Wild Life Federation (B. Hamilton, n.d.).

Sections 11 and 29 were explicitly reserved to fund school boards. Unlike the United States (Schaede and Smith 2025) these "school sections" were not sold or rented to fund local school boards. Instead revenue from sales and rentals was collection by the federal government and placed in the School Endowment Fund. These funds were invested in Dominion securities, with the revenues distributed to the provinces.

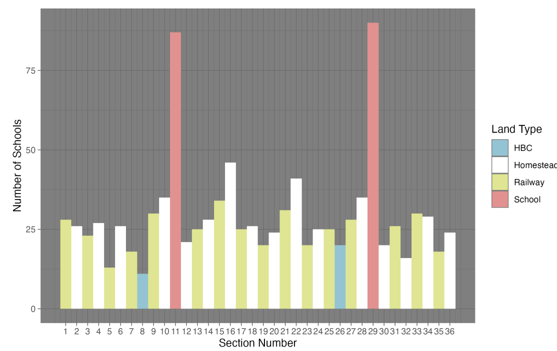
As with the HBC land the government held off on selling land until the surrounding land was settled to maximize the value. There was a bias towards leasing land rather than selling, to maintain a steady flow of income.

There is some evidence that schools were more likely to locate on school sections relative to other parcels. This appears largely true in Saskatchewan, less so in Alberta, and not at all in Manitoba. This could reflect the lower transaction costs to construct schools on land explicitly designated for schools. Figure A8 shows the distribution of schools across sections.

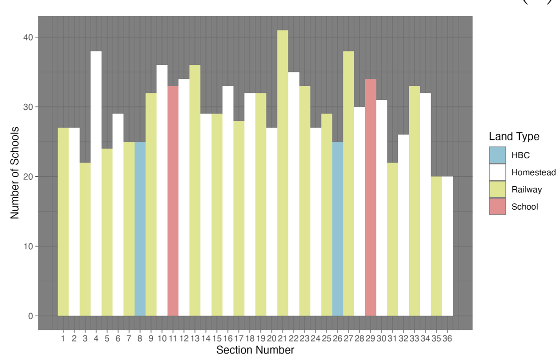
Figure A3 shows the distribution of population by section in 1906 and 1921, two of the census years where the section number is available. School and HBC sections were the least



(a) Alberta



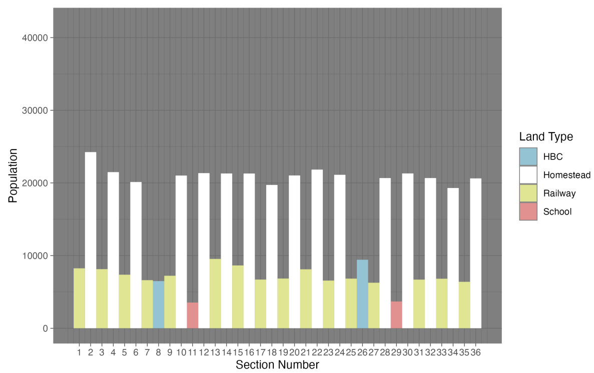
(b) Saskatchewan



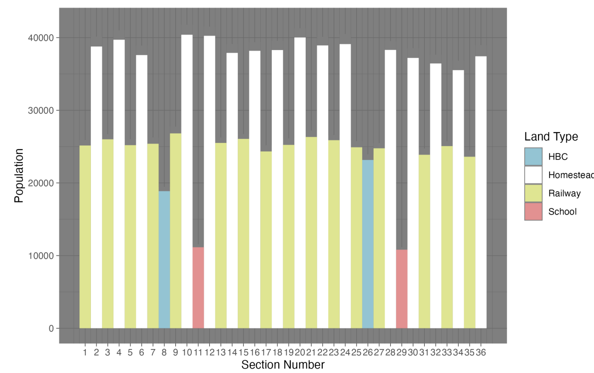
(c) Manitoba

Figure A2: Distribution of schools by section type by province.

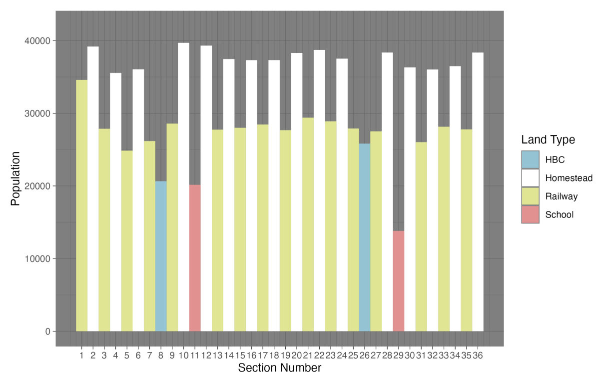
populated in both years, consistent with those sections being sold off strategically once the surrounding areas had been settled. The highest population sections are the homesteads, although this gap narrows in 1921. This is consistent with railway lands being slower to settle due to the indemnity provision in railway contracts.



(a) 1906



(b) 1921



(c) 1921

Figure A3: Population distribution between land parcel types in 1906, 1921 and 1931. Early in the settlment process the majority of settlers lived on even numbered sections which were sold as homesteads. Over time more settlement occurred on railway, school, and Hudson's Bay Company lands, as evidenced by the figure from 1931.

A.3 Separate Schools

The Northwest Territories Act of 1875 gave the Territorial Council the right to pass ordinances regarding education, on the condition that the religious minority of either Protestants or Catholics could establish their own school and ratepayers would only be liable for the school of their religion (Moir 1934; Noonan, Hallman, and Scharf 2006). The first major ordinance, The North-West Territory School Ordinance of 1884, created a system of school districts, with every district designated as either Protestant or Catholic. The original intention was to create a "Dual" system similar to Quebec and Manitoba, with separate school systems for Protestants and Catholics (Noonan, Hallman, and Scharf 2006). Protestant and Catholic schools were overseen by a Catholic and Protestant sections, overseen by the Board of Education (Noonan, Hallman, and Scharf 2006). Each section separately determined subjects, textbooks, and teachers for the schools of their own faith (Moir 1934).

The "Dual" system didn't last, and was changed to a "minority" system similar to that of Ontario in 1885 (Noonan, Hallman, and Scharf 2006). The minority system allowed two schools to operate simultaneously in a given area. The first school, either Protestant or Catholic, would be designated a "public" school. The minority in an area could then vote and create a "separate" school district, either Protestant or Catholic, for people of that faith. In several instances changing population of Catholics and Protestants led to a separate school, which was supported by the majority of tax payers (Noonan, Hallman, and Scharf 2006). Initially the boundaries of a separate school district did not need to overlap with the public school district, but this was changed so that both school districts had the same boundaries in 1892 (Noonan, Hallman, and Scharf 2006; Moir 1934).

Although the minority system remained in place in the Northwest Territories, the differences between the Protestant and Catholic sections, and their influence, decreased over time. In 1885 the Board of Education was reduced to five members, two Catholic, two Protestant, and the lieutenant governor (Noonan, Hallman, and Scharf 2006). In 1887 the Board of Education took over the power to license teachers, meaning that Protestants and Catholics both jointly determine the teachers for each other's schools, despite differences in the materials being taught (Moir 1934). In 1891 and 1892 most of the power of the sections was removed with the lieutenant-governor-in-council taking over licensing of teachers, appointment of inspectors, the use of uniform textbooks, and uniform examination standards (Moir 1934). Finally in 1901 a commissioner of education was created to administer schools, to be advised by an Educational Council consisting of three Protestants and two Catholics.

The differences in instruction across schools were also reduced over time. Religious education was limited to the last half hour of the day after 1892, with the religion determined by the local school board (Noonan, Hallman, and Scharf 2006). Following the Riel Rebellion in 1885 and a subsequent rise in anti-French sentiment, English was mandated as the language of instruction in 1901 (Noonan, Hallman, and Scharf 2006), with provisions for teaching some primary courses in French (Moir 1934).

In practice, the majority of separate schools were Catholic, although a small number of separate Protestant schools were established in Catholic majority areas. To fund schools, Protestants and Catholics each paid taxes to the school of their respective faith, and after 1886 no owner of property was able to escape taxation all together (Moir 1934). Prior to 1909 residents could determine which school board they would like to fund, but after a legal case it was determined that the assessment of taxes would be on the basis of religious faith, not personal preference (Noonan, Hallman, and Scharf 2006). Further legal opinions clarified that for the purposes of establishing a separate school, a Catholic was someone who believed that religious authority rests with the pope[15] (Noonan, Hallman, and Scharf 2006).

Details on how separate schools are handled in the data are discussed in section A.6.1.

A.4 Census Records

The Canadian full count census records are available courtesy of The Canadian People's project (The Canadian Peoples 2025) and a partnership with Ancestry.com. The records used in this project consist of waves starting in 1901 and ending 1931, the most recent wave released[16]. I utilize both census waves that cover the entirety of Canada in 1901, 1911, 1921, and 1931, and census waves that cover the prairie provinces only in 1906, 1916, and 1926. These additional waves, taken to track the growth of the west after the creation of Alberta

15. As of 2018 this definition is still used in Alberta. This confusingly leads people who are Eastern Orthodox or Ukrainian Orthodox to be considered Protestants for the purposes of establishing a separate school board (Government of Alberta 2018).

16. The Canadian government operates on a 92 year lag for releasing census records (Canada 2023b)

and Saskatchewan in 1905 (Canada 2023a), mean that the relevant areas have census waves every five years.

Although the census was being conducted frequently, different years of the census contain different information. The availability of different variables is recorded in table A1. One important advantage that the Canadian census has is that income is available every ten years starting in 1901, nearly forty years prior to the United States first asking about income in 1940.

A.5 Linking

In the first stage, when linking males, blocking is done on sex, age within a three-year window, the first letter of the surname, and the recorded birthplace (aggregated to the province, United States, United Kingdom, Ireland, or continent). Following Helgertz et al. (2022), the set of potential links is restricted to those with a levenstein distance of 0.79 in the first name, and 0.84 in the surname. The list of potential links is then scored with trained model to predict the likelihood of two records being linked. In order for a link to be accepted, the probability of a link being correct must be greater than a threshold $\alpha = 0.25$, and the score for a proposed link must better that the next best link by at least $\beta = \frac{\alpha_1}{\alpha_2} = 1.1$.

In the second stage the links in the first stage are used to infer family links across census waves. A second round of linking is then completed after blocking at the family level. A second model is used to score links within the family, with the same thresholds for α and β applied. Linking within family also allows blocking variables to be ignored, allowing for links where some characteristics may have changed dramatically either due to enumerator error or deliberate choice on the part of the person being recorded. This is especially important to correct for transcription errors.

Table A1: Variables available in the Canadian census by census wave. Rural locations include the Section, Township, Range, and Meridian allowing for precise geolocation of households. 1911 excludes this information so location to the nearest township is the best possible. Some years include the exact address for urban households, although 1916 only includes the street. Income is consistently available in full census years, but missing in prairie province census years.

Year	**Name**	**Months in School**	**Location (Rural)**	**Location (Urban)**	**Income**	**Other**
1901	Yes	Yes	Not Digitized	Not Digitized	Yes	Locations contained in a separate schedule, which has not been digitized.
1906	Yes	No	Yes	No	No	Contains information on livestock holdings by each individual.
1911	Yes	Yes	No	No	Yes	Contains information on insurance holdings by individual.
1916	Yes	No	Township	Street	No	Rural addresses at the township level only. Contains information on insurance holdings by individual.
1921	Yes	Yes	Yes	Yes	Yes	Contains information on house materials.
1926	Yes	Not Digitized	Not Digitized	Not Digitized	No	
1931	Yes	Yes	Yes	Yes	Yes	Contains estimated value of home.

A.5.1 Linking Variables

In order to score potential links I use the training data from Helgertz et al. (2022) which links US census records. The variables used in stage 1 of linking between individuals can be found in table A2. The names of variables created in the Canadian data are identical to those in the training data for easy reproducibility.

To compute the distances between census records I utilize the link between the census records from The Canadian Peoples 2025 and the census subdivision (CSD) files computed by CIEQ, Université Laval and HGIS Lab, University of Saskatchewan 2023. I take the centroid of each CSD for the purpose of computing the distance, and compute pairwise distances between all CSDs as a lookup table. CSD polygons are only available in complete census years (e.g. 1901, 1911, 1921, etc.), and not for years where a census of the prairie provinces was taken (e.g. 1906, 1916, 1926, etc.). For years where I do not observe the CSD, I create a synthetic CSD based on the exact locations I can observe combined with the enumeration subdistricts or enumeration districts. For details on geolocation, see section A.7 of the appendix.

The variables used in stage 2 of linking can be found in table A3. Note that the number of variables and type is lower than in stage 1. This reflects the fact that the stage 2 matching occurs within households based on the stage 1 link, thus requiring fewer variables to reach a similar level of precision. Unlike stage 1 no blocking is required, instead variables are included for sex, age, name initials, and birthplace codes that were previously used for blocking. This allows second stage links between entries that might have differences across these variables. For example and individual whose sex was misrecorded in one year would not be linked in stage 1 due to the blocking on sex, but could be linked in stage 2 if another family member was linked and enough other characteristics are similar between the two census waves.

For details on the Hlink package used the implement the linking, please see the Github page for Hlink (Wellington, Harper, and Thompson 2024).

Table A2: Linking variables in Stage 1. Code refers to the name of the variable in both the Canadian data and Helgertz et al. (2022). Type gives the data type of the variable. Description offers a brief description of the variable and how it is created.

Variable	**Code**	**Type**	**Description**
Last Name JW	namelast_jw	0-1	Jaro-Winkler string distance between last names.
First Name JW	namefrst_jw	0-1	Jaro-Winkler string distance between first names.
Second Generation	sgen	Binary	Is the individual a second generation immigrant?
County Distance	distance	Continuous (m)	Distance between counties in each census wave, in meters.
Distance Squared	distance2	Continuous (m^2)	County Distance squared.
Birth Year	byrdif	Continuous	Difference between recorded birth years.
Middle Initial	mi	0, 1, or 2	0 if middle initial is empty, 1 if middle initial links any possible middle initial, 2 if there is no possible match.
First Soundex	fsoundex	Boolean	1 if soundex links, 0 if otherwise.
Last Soundex	lsoundex	Boolean	1 if soundex links, 0 if otherwise.
Multiple Exact links	exact_mult	Boolean	1 if there are multiple exact links with identical first and last name, 0 if otherwise
Immigration Year	immyear_caution	0, 1, 2	0 if there is no difference in immigration year or <= 5 years, 1 if the difference is <= 10 years, 2 if the difference is > 10 years. 0 if either record is missing the immigration year
Hits	hits	Integer	The number of potential links for a given individual, looking from the first dataset
Hits Squared	hits2	Integer	Hits squared.

Table A3: Linking variables in Stage 2. Code refers to the name of the variable in both the Canadian data and Helgertz et al. (2022). Type gives the data type of the variable. Description offers a brief description of the variable and how it is created.

Variable	**Code**	**Type**	**Description**
Last Name JW	namelast_jw	0-1	Jaro-Winkler string distance between last names.
First Name JW	namefrst_jw	0-1	Jaro-Winkler string distance between first names.
Sex	sexmatch	Boolean	Indicates if two records have matching sex.
Relation Type	idtype	1, 2, or 3	1 if both entries are biologically related family, 2 if both entries are non biologically related famiy, 3 otherwise.
Birthplace	bplmatch	Boolean	1 if birthplace code links, 0 if otherwise.
Birth Year	byrdifcat	Continuous	Difference between recorded birth years. No restriction on maximum possible difference like in stage 1.
First Initial Match	f1_match	1 or 2	1 if the first initial of the first first name links any of the potential first initials, 2 otherwise.
Middle Initial Match	f2_match	0, 1, or 2	0 if middle initial is empty, 1 if middle initial links any possible middle initial, 2 if there is no possible match. Analogous to middle initial match in stage 1.

A.5.2 Linking Comparison

The linking method used in this paper from Helgertz et al. (2022) (MLP), is a relative newcomer to the linking literature. The more established method of Abramitzky et al. (2021) (ABE) offers greater transparency in the linking process, at the cost of excluding some potentially useful information that might allow for larger link rates and more precise links.

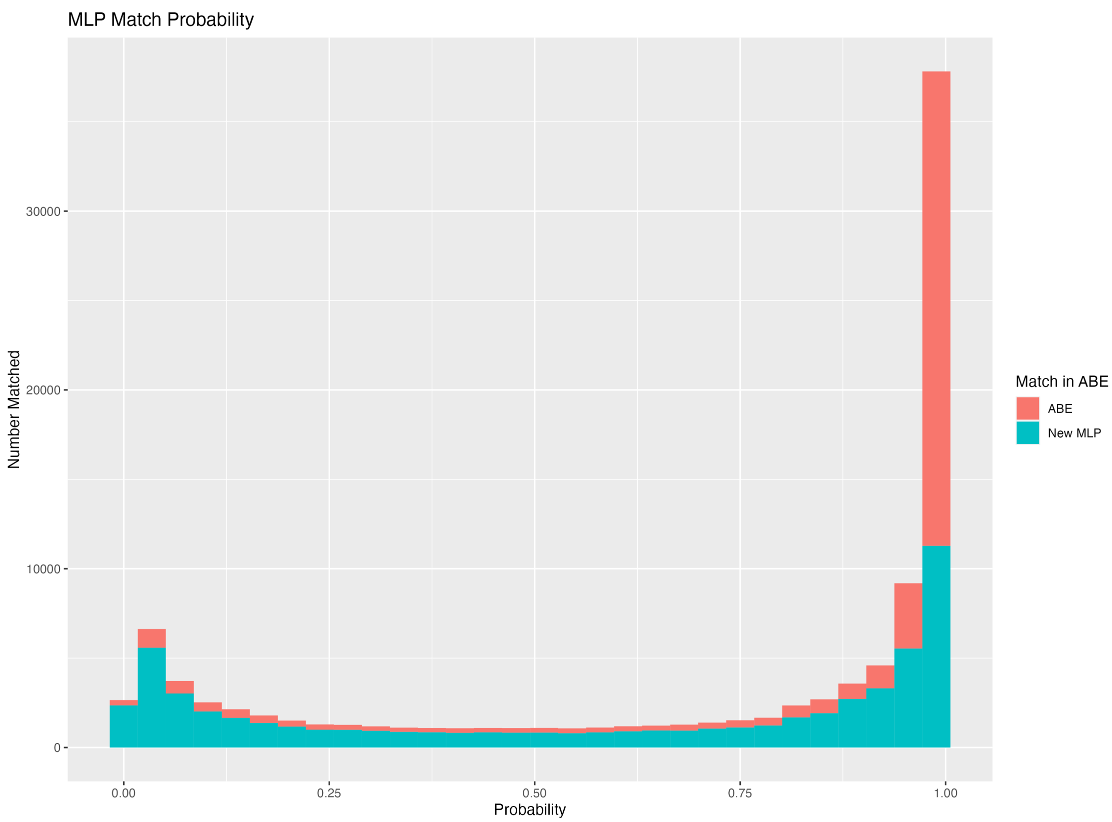
To compare the performance of these two methods I focus on links between 1906 and 1911 and compute links using both the ABE and MLP methods. Table A4 shows the number of links using both the MLP and ABE methods for the target sample of children. Both methods have a significant overlap in the linked population of 41359, but MLP recognizes a much larger set off additional links of 64664, relative to ABE at 14602. This is consistent with MLP using more information to disambiguate links, leading to a higher link rate.

Table A4: Links by ABE vs MLP between the 1906 and 1911 Canadian Census. Links are generated for children who are between the ages of 0 and 17 in 1906 and who live in Alberta, Saskatchewan, or Manitoba. MLP links are generated using the procedure document in section A.5.1. ABE links are generated by first blocking on sex, place of birth, and fathers place of birth. No NYSIIS name cleaning is applied, and the loosest requirements for an exact link are used, requiring uniqueness only on the exact age to be considered linked.

Method	In MLP	Not in MLP	Total
In ABE	32867	23094	55961
Not in ABE	47303	197091	244394
Total	80170	220185	300355

To understand where additional links are coming from, I plot the likelihood scores for each link in figure A4, and decompose the links into whether they are an ABE link or are a new MLP link. These likelihoods can only be computed for links that are completed by MLP, thus the links are the equivalent of those in column 1 of table A4. I find that the ABE links have a large mass near 1, with a tail that extends all the way to zero. This suggests that ABE is picking up the highest quality links, but that by failing to take into account additional information some very low quality links are created. MLP in contrast picks up the high quality links that ABE does, but also many others whose likelihood is lower. Importantly by creating an explicit score, the researcher is able to select only those links which have a high probability of being a correct link. Note that this method does not score any ABE links which are not captured at all by MLP, the "In ABE" and "Not in MLP" group in table A4. If these links were to be included, there would be a heavier tail of low probability links attributed to ABE. The results suggest that MLP is able to tell apart links that may be present, but are not as obvious.

Figure A4: MLP link Probability vs ABE Links. This figure includes all links that are captured by MLP. Links that are not captured by MLP do not have a score, and cannot be included.

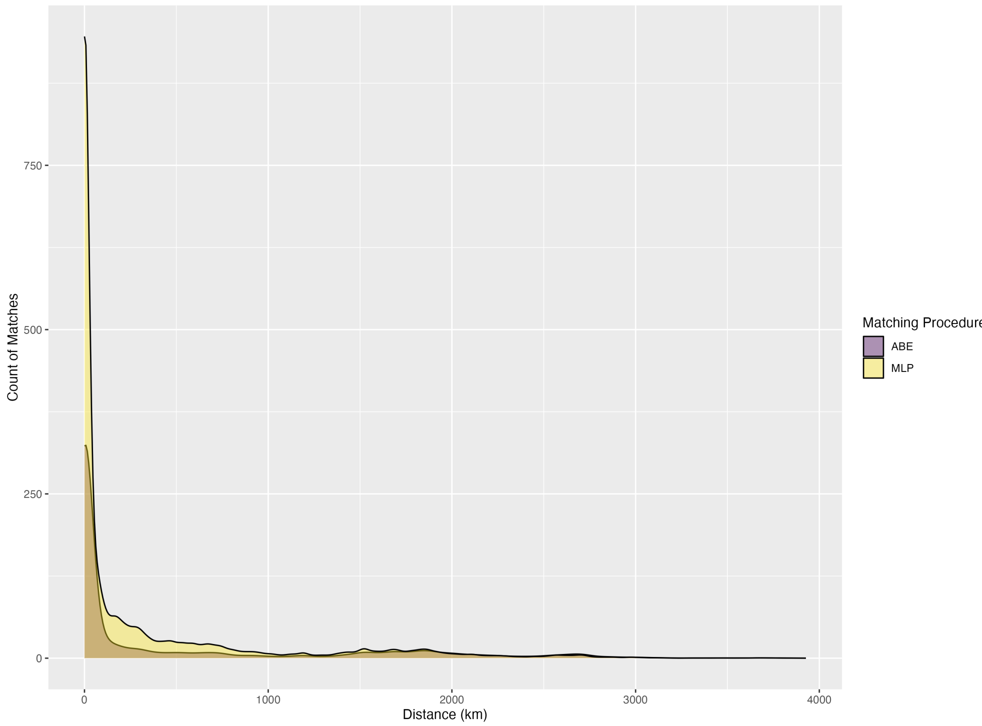


The ability to distinguish between two otherwise identical records in MLP comes at a cost of potentially introducing bias in the linking procedure. One of the most likely sources of bias is the inclusion of distance as a linking variable. If the locations of the individual between two census waves are closer, then a higher probability of a link being true will be predicted. The consequence of this is that any predicted mobility estimates will be biased towards zero, as the link probability decreases with distance. To get a sense of the magnitude of this problem, we can compare the density distribution between links made by ABE versus those made by MLP in figure A4. I find that MLP dramatically increases the number of links for individuals in close proximity, but that there are gains in the count of links across the entire distribution of distances. The distribution of counts can be seen in figure A5a. When examining the density in figure A5b I find that the densities are remarkably similar, although there is slightly more mass at lower distances. Taken together this suggests that although including distance as a linking variable may introduce some bias, it is unlikely to dramatically change any results.

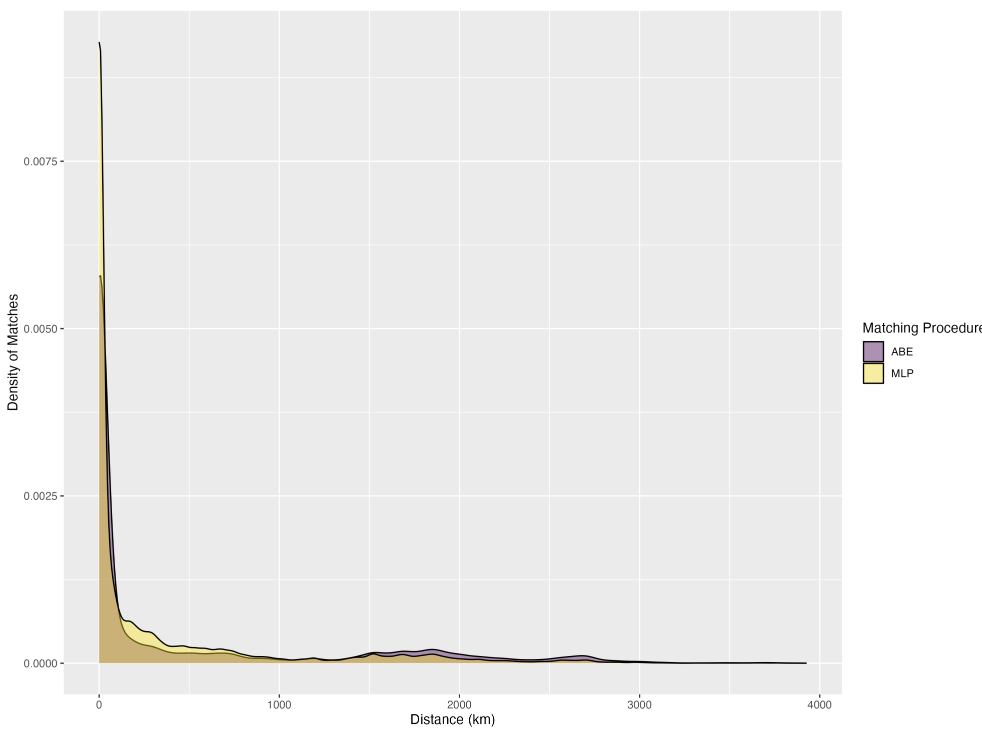
A.5.3 Linking Charateristics

All linking proceedures are biased and are likely to be unrepresentative of the population that we want to link (Helgertz et al. 2022). It is therefore important to understand which populations are likely to be linked and if this may bias any of the results. To do this, I estimate the likelihood of being linked conditional on being in the first of a pair of census waves. This choice is made so that I can utilize the 1906 sample as the first wave, which only include the prairie provinces, at the cost of being an understatement of the likelihood of being linked as some people will have emmigrated or died and will be impossible to link.

To understand why I take the the likelihood of being linked conditional on appearing in the first census wave, consider the opposite. To evaluate the effect of the link between 1906 and 1911, we would take the later wave, and remove anyone born after 1906 and who immigrated after 1906. The fraction of people remaining who are linked would be the estimated linkage rate. The 1911 census contains the entirety of Canada, so the total link rate would be too low, as only those in 1906 who were in Alberta, Saskatchewan, and Manitoba would be linked, relative to the entire population of Canada in 1911. This would not be an accurate measure of the likelihood of being linked. An alternate approach would be to restrict the sample to those living in Alberta, Saskatchewan, and Manitoba in 1911 to



(a) Count of links.



(b) Density of links.

Figure A5: Distribution of distance between links in 1906-1911, for children in Alberta, Saskatchewan, and Manitoba who are between the ages of 0 and 17.

get the denominator correct, but this would remove the possibility of people moving out of the prairies. Instead I take the earlier year, being conscious that some individual will not be linked.

For each pair of census waves I regress the likelihood of being linked of a set of characteristics of the individual including age, birthplace, relation to the head of household, enumeration subdistrict, religion, whether the household lives on a homestead, family holdings of cows and horses, and agricultural productivity of wheat. All effect of estimated jointly. Tables A5, A6, A7, A8, A9, and A10 give the results.

A.6 Schools

After collecting the original school data from each of the three sources, several cleaning steps are required to obtain the opening date and location of each school.

For the Saskatchewan data there can be multiple locations reported for each school district number. This can reflect different original sources for a school's location, or can reflect if a school district spans a geographic area with the exact location of the school being unknown. Table A11 shows an example of the data. To assign unique identifiers I use the fastLink R package (Enamorado, Fifield, and Imai 2023) to deduplicate records and assign unique identifiers.

To remove duplicates I take the most precise location as the location of the school. In the case where two schools have the same level of precision (i.e. both have locations at the section level), I keep both and assign them separate identifiers. This ensures that I am likely to be over counting schools rather than under counting. I keep a single entry for any schools that don't have a location, to keep the total number of schools as accurate as possible.

Many entries in the Saskatchewan data are missing the school opening dates. In cases where the opening date is missing I infer it from the school district number. Within the Northwest Territories (later split to form Alberta and Saskatchewan in 1905) school districts were numbered sequentially. As a result if the school district numbers are known for each

Table A5: Linking rates by Age

Dependent Variable:	Matched				
Model:	(1)	(2)	(3)	(4)	(5)
	1906-1911	1906-1921	1906-1931	1911-1921	1911-1931
Age = 1	0.0730***	0.0786***	0.0673***	0.0686***	0.0367***
Age = 2	0.0074*	0.0155***	0.0583***	-0.0178***	0.0132***
Age = 3	-0.0355***	0.0100***	0.0521***	-0.0302***	0.0088***
Age = 4	-0.0083**	0.0209***	0.0443***	0.0052***	0.0201***
Age = 6	-0.0268***	-0.0569***	-0.0872***	-0.0325***	-0.0582***
Age = 7	-0.0458***	-0.1299***	-0.1473***	-0.0806***	-0.1143***
Age = 8	-0.0642***	-0.1663***	-0.1502***	-0.1197***	-0.1272***
Age = 9	-0.1267***	-0.1751***	-0.1359***	-0.1434***	-0.1211***
Age = 10	-0.1999***	-0.1869***	-0.1474***	-0.1499***	-0.1134***
Age = 11				-0.1073***	-0.0706***
Age = 12				-0.1558***	-0.1033***
Sex	0.1307***	0.1465***	0.1467***	0.1197***	0.1084***
Family Size	0.0041***	0.0010***	0.0001	-5.8×10^{-5}	-7.95×10^{-5}***
Birthplace	Yes	Yes	Yes	Yes	Yes
Relation	Yes	Yes	Yes	Yes	Yes
Enum. Subdist.	Yes	Yes	Yes	Yes	Yes
Homestead	Yes	Yes	Yes		
Milk Cows	Yes	Yes	Yes		
Horses	Yes	Yes	Yes		
Wheat Prod.	Yes	Yes	Yes		
Religion				Yes	Yes
Observations	430,101	430,101	430,101	1,191,247	1,191,247
R^2	0.08367	0.08690	0.10244	0.05913	0.06569

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates by age for different decades of life. The reference group is those in the 5th age group, those between the ages of 40-49. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A6: Linking rates by Birthplace

Dependent Variable:	Matched				
Model:	(1)	(2)	(3)	(4)	(5)
	1906-1911	1906-1921	1906-1931	1911-1921	1911-1931
Sex	0.1307***	0.1465***	0.1467***	0.1197***	0.1084***
Family Size	0.0041***	0.0010***	0.0001	-5.8×10^{-5}	-7.95×10^{-5}***
Birthplace = UnitedStates	-0.0347***	-0.0533***	-0.0462***	-0.0260***	-0.0283***
Birthplace = Canada+BC	-0.0491***	-0.0425***	-0.0391***	-0.0131***	-0.0101***
Birthplace = MB	-0.0378***	-0.0389***	-0.0417***	0.0113***	0.0062***
Birthplace = Maritimes	-0.0322***	-0.0213***	-0.0155**	-0.0111***	-0.0132***
Birthplace = QC	-0.0563***	-0.0268***	-0.0199***	0.0067	0.0032
Birthplace = Caribbean	-0.1479***	-0.0886***	-0.1132***	-0.0441**	-0.0571***
Birthplace = Nordic	-0.0760***	-0.0547***	-0.0334***	-0.0327***	-0.0133***
Birthplace = UnitedKingdom	0.0319***	0.0322***	0.0358***	0.0003	0.0090***
Birthplace = Ireland	-0.0157*	-0.0178**	-0.0030	-0.0009	0.0065*
Birthplace = RestofEurope	-0.1094***	-0.0683***	-0.0469***	-0.0306***	-0.0115***
Birthplace = ROW	-0.0722***	-0.1175***	-0.0977***	-0.0556***	-0.0498***
Age	Yes	Yes	Yes	Yes	Yes
Relation	Yes	Yes	Yes	Yes	Yes
Enum. Subdist.	Yes	Yes	Yes	Yes	Yes
Homestead	Yes	Yes	Yes		
Milk Cows	Yes	Yes	Yes		
Horses	Yes	Yes	Yes		
Wheat Prod.	Yes	Yes	Yes		
Religion				Yes	Yes
Observations	430,101	430,101	430,101	1,191,247	1,191,247
R^2	0.08367	0.08690	0.10244	0.05913	0.06569

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates for different birthplaces. The reference group is those born in Ontario. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A7: Linking rates by Land Type

Dependent Variable:	Matched		
Model:	(1)	(2)	(3)
	1906-1911	1906-1921	1906-1931
Sex	0.1313***	0.1467***	0.1466***
Family Size	0.0041***	0.0011***	9.13×10^{-5}
Homestead = HBC	0.0059	-0.0019	-0.0015
Homestead = Railway	0.0004	0.0018	0.0012
Homestead = School	0.0079	-0.0006	0.0009
Age	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes
Relation	Yes	Yes	Yes
Province	Yes	Yes	Yes
Milk Cows	Yes	Yes	Yes
Horses	Yes	Yes	Yes
Wheat Prod.	Yes	Yes	Yes
Observations	430,101	430,101	430,101
R^2	0.05674	0.06955	0.08882

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates for different land types. The reference group is those living on a homestead section. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A8: Linking rates by Livestock Holdings

Dependent Variable:	Matched		
Model:	(1)	(2)	(3)
	1906-1911	1906-1921	1906-1931
Sex	0.1313***	0.1467***	0.1466***
Family Size	0.0041***	0.0011***	9.13×10^{-5}
Milk Cows = 1-4	0.0266***	0.0266***	0.0199***
Milk Cows = 5-9	0.0537***	0.0475***	0.0372***
Milk Cows = 10-49	0.0618***	0.0495***	0.0403***
Milk Cows = 50+	0.0173	0.0080	-0.0086
Horses = 1-4	0.0311***	0.0280***	0.0205***
Horses = 5-9	0.0547***	0.0499***	0.0385***
Horses = 10-49	0.0634***	0.0559***	0.0436***
Horses = 50+	0.0406**	0.0231*	0.0205**
Age	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes
Relation	Yes	Yes	Yes
Province	Yes	Yes	Yes
Homestead	Yes	Yes	Yes
Wheat Prod.	Yes	Yes	Yes
Observations	430,101	430,101	430,101
R^2	0.05674	0.06955	0.08882

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates by livestock holdings. The reference group is those holding no cows or horses. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A9: Linking rates by Relation to Head

Dependent Variable:	Matched				
Model:	(1)	(2)	(3)	(4)	(5)
	1906-1911	1906-1921	1906-1931	1911-1921	1911-1931
Family Size	0.0043***	0.0013***	0.0004	-4.14×10^{-5}	-6.45×10^{-5}***
Relation = Boarder	-0.1487***	-0.1326***	-0.1166***	-0.1034***	-0.0951***
Relation = Daughter	-0.1340***	-0.2163***	-0.2192***	-0.1301***	-0.1627***
Relation = Lodger	-0.1398***	-0.1533***	-0.1187***	-0.1049***	-0.0944***
Relation = Other	-0.1284***	-0.1156***	-0.0941***	-0.0972***	-0.0842***
Relation = Sister	-0.1942***	-0.2063***	-0.1794***	-0.1756***	-0.1637***
Relation = Son	0.0110***	0.0118***	0.0425***	0.0319***	0.0435***
Relation = Wife	-0.0379***	-0.0605***	-0.0819***	-0.0285***	-0.0458***
Age	Yes	Yes	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes	Yes	Yes
Enum. Subdist.	Yes	Yes	Yes	Yes	Yes
Homestead	Yes	Yes	Yes		
Milk Cows	Yes	Yes	Yes		
Horses	Yes	Yes	Yes		
Wheat Prod.	Yes	Yes	Yes		
Religion				Yes	Yes
Observations	430,101	430,101	430,101	1,191,247	1,191,247
R^2	0.08044	0.08238	0.09728	0.05538	0.06190

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates by relation to the head of household. The reference group is the head of household. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A10: Linking rates by Religion

Dependent Variable:	Matched	
Model:	(1)	(2)
	1911-1921	1911-1931
Sex	0.1197***	0.1084***
Family Size	-5.8×10^{-5}	-7.95×10^{-5}***
Religion = Orthodox	-0.0393***	-0.0289***
Religion = Anglican	0.0653***	0.0438***
Religion = Presbyterian	0.0681***	0.0441***
Religion = Methodist	0.0670***	0.0425***
Religion = Unitarian	0.0625***	0.0405***
Religion = Lutheran	0.0402***	0.0292***
Religion = Baptist	0.0630***	0.0425***
Religion = Mennonite	0.0105	0.0008
Religion = Pentecostal	0.0389	0.0661
Religion = Evangelical	0.0583***	0.0583***
Religion = Fundamentalist	0.0321	0.0247
Religion = Adventist	0.0461***	0.0233***
Religion = Mormon	0.0671***	0.0310***
Religion = Holiness	0.0512***	0.0319***
Religion = Spiritualist	0.0476**	0.0152
Religion = Christian	0.0283***	0.0144***
Religion = Muslim	-0.0350	-0.0278
Religion = Jewish	0.0203***	-0.0094*
Religion = Dharmic	0.0365*	0.0319
Religion = Polytheist	-0.0700***	-0.0650***
Religion = Bahai'i	-0.1391***	0.9297***
Religion = Ethical	0.0169	0.0188
Religion = Other	-0.0984***	-0.0716***
Age	Yes	Yes
Birthplace	Yes	Yes
Relation	Yes	Yes
Enum. Subdist.	Yes	Yes
Observations	1,191,247	1,191,247
R^2	0.05913	0.06569

Clustered (Enum. Subdist.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Linking rates by religion. The reference group is Catholics. Each column gives the estimated effect for different pairs of linked years. Columns (1), (2) and (3) focus on the rural population who are able to be geolocated and are the population of interest. Columns (4) and (5) are census waves which include all of Canada, but are restricted to people in Alberta, Manitoba, and Saskatchewan.

Table A11: An example of assigning unique identifiers in the Saskatchewan data. I use the fastLink package (Enamorado, Fifield, and Imai 2023) to deduplicate records, taking into account the school name, the school district number, and the township, range, and meridian. To resolve duplicates I first take the most precise locations with section numbers (or in some cases quarter section numbers), then the less precise locations with township range and meridian locations only.

Unique ID	School Name	School District Number	Section	Township	Range	Meridian
186	Alexander	1029	NA	20	29	W3
186	Alexander Plain	1029	1	20	29	W3
200	Alexandra	1908	NA	27	19	W2
200	Alexandra	1908	35	27	20	W2
200	Alexandria	1908	NA	27	19	W2
200	Alexandria	1908	NA	27	20	W2
200	Alexandria	1908	NA	28	19	W2
200	Alexandria	1908	NA	28	20	W2
200	Alexandria	1908	NA	27	19	W2

school, and school opening dates are known for Alberta, school opening dates can be inferred for Saskatchewan. Table A12 shows an example of how this process works in practice. When a school could have two possible opening dates, I take the earlier of the two.

After the formation of Alberta and Saskatchewan in 1905 school districts in Alberta continued to be numbered following the same order, while Saskatchewan continued with the same numbering scheme until 1911, when the government went back and filled in numbers occupied by Alberta schools (Baergen 2005; Saskatchewan Department of Education 1906-1912, 1914). I utilize reports from the Saskatchewan department of education to get the year of school opening until the end of 1912.

I utilize the school district numbers to infer the year of opening for any missing year. For Saskatchewan, no information is available on the date of school opening, but Alberta and Saskatchewan shared the same school numbering system until 1905 when the provinces were created (Baergen 2005). Using the known dates for Alberta, I then create a joint list of school districts, and interpolate between the known Alberta dates to determine an estimate of the opening date for each Saskatchewan school.

One possible concern with this approach is that some schools could be missing in the

Table A12: Inference procedure for schools with missing establishment years. If an entry for Saskatchewan has a corresponding entry in Alberta, then it remains NA, as the school number was originally allocated to a school in Alberta when both provinces were part of the Northwest Territories. If there is no corresponding Alberta school then the date of opening can be inferred from the earlier of the two dates that surround it. Instead I turn to the annual reports of the Saskatchewan Department of Education for the overlapped values, giving a year of 1911 for Clunie.

Province	School District Name	School District Number	Year (Data)	Year (Inferred)	Year (Archive)
Alberta	Anderson	434	1896	1896	1896
Saskatchewan	Clunie	434	NA	NA	1911
Saskatchewan	New Finland	435	NA	1896	1896
Saskatchewan	Glenmorris	436	NA	1896	1896
Alberta	Bon Accord	438	1897	1897	1897

records for Alberta or Saskatchewan. After Alberta and Saskatchewan were formed Alberta continued with the same numbering scheme (Baergen 2005), while Saskatchewan went back and filled in new school districts with the numbers previously allocated to schools now in Alberta (Saskatchewan Department of Education 1906-1912, 1914). As a result any schools missing from the Alberta records and present in the Saskatchewan records before the numbers previously used in Alberta were filled up would be falsely attributed to have been constructed earlier than they actually were. To resolve this I manually record which school districts were opened from the Saskatchewan Department of Education annual reports from 1906 through 1912, which record which school district numbers were reassigned (Saskatchewan Department of Education 1906-1912, 1914).

For Manitoba I scrape Manitoba Historical Society 2019. Each school has a single page describing the location of the school and some brief history. To get the opening dates of each school I utilize the ChatGPT API to read the body of each page and return the year of construction of the school. The total school openings in each year from 1870 to 1911 can be seen in figure A6.

There are two caveats for utilizing the school data. The first is that I do not observe closing dates for any schools, and therefore assume that any school that opens remains open

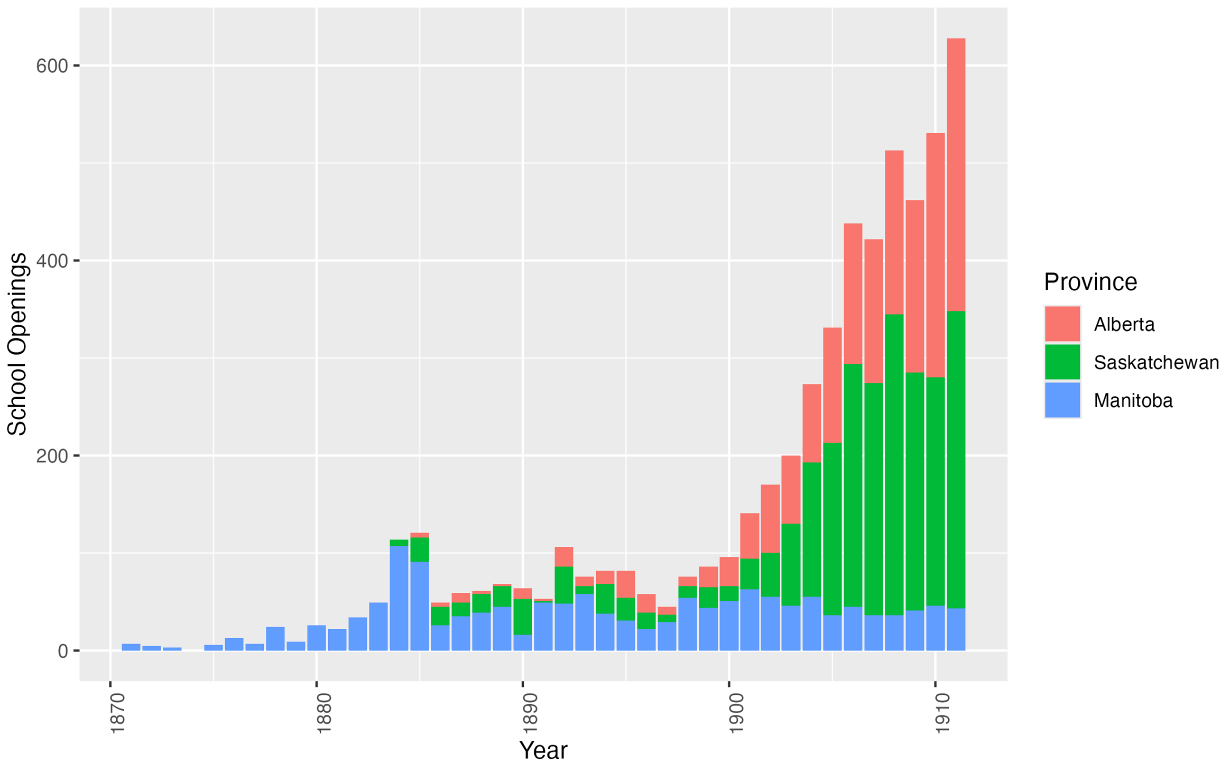


Figure A6: School openings for Alberta, Saskatchewan, and Manitoba from 1870 to 1911. Prior to 1870 only a handful of school existed in Manitoba. The last year that I have data of school openings for is 1912.

until 1911. This is a reasonable assumption as settlement ensured that there would always be demand for schools. The second caveat is that I do not observe if schools moved locations, and in some cases will have a second location in the dataset with way to tell when a move occurred. In these cases I take both locations as school locations, with separate school identifiers.

Table A13 shows the totals split by province of the number of schools in the database and compares this to the number of schools listed in each province's department of education reports. I find very comparable numbers for Alberta with 1784 schools listed as being organized in 1911, versus 1750 in the data.

Table A13: Summary statistics by province for the number of schools in 1911.

Type	**Total**	**Alberta**	**Saskatchewan**	**Manitoba**
Actual	5955	1784	2573	1598
Observed	5618	1750	2370	1498
Geolocated	5472	1750	2227	1495
Geolocated (Section)	4141	1619	1027	1495

I also compare the rate of school geolocations. "Actual" lists the number of schools reports in the 1911 report from the department of education for Alberta (Government of Alberta 1912), Saskatchewan (Saskatchewan Department of Education 1906-1912, 1914), and Manitoba (Government of Manitoba 1911, 1912). This figure includes both public and separate schools in Alberta and Saskatchewan while the data contain only public schools, and is a slight overestimate. "Observed" lists the number of schools I find in the data in 1911, which requires that they have an opening date before 1911. Any schools with an unknown opening date are dropped. "Geolocated" reports the number of schools, of those which are observed, which I am able to geolocate. "Geolocated (Section)" reports the number of schools where the geolocation is at the section level (1 mile square) or finer. Details on the geolocation procedure can be found in section A.7 of the appendix.

A.6.1 Separate Schools

As discussed in section A.3, the question of separate schools for Catholics and Protestants was an important feature of the late nineteenth and early twentieth century prairies. In the data I therefore need to be careful how different school boards are handled.

In Manitoba separate schools were eliminated in 1890 (Noonan, Hallman, and Scharf 2006), and the data reflect the entire set of possible schools.

In Alberta the data only contain public schools, either Protestant or Catholic, and do not contain any information on separate schools established by the minority faith. Checking the reports from the Department of Education I find that of the 1,784 schools in existence at the end of 1911, only 14 were separate schools and 5 were private schools (Government of Alberta 1912). In addition 9 of these schools were located in cities or towns, and are likely to be located near other schools as they are separate school districts. It is therefore unlikely that they are affecting my distance calculations.

In Saskatchewan there are 13 of 2,516 total school districts which were separate in 1911 (Saskatchewan Department of Education 1906-1912, 1914). Although I can observe separate schools in the data, the numbering system for separate schools is separate for that of public schools, making it impossible to infer the opening dates from school numbers alone. Although the numbering system is shared with Alberta prior to 1905, I also don't know the school numbers for separate schools in Alberta as described the paragraph above. In addition some of the numbers of separate schools were back filled as with public schools in section A.6, so that even a small number of opening dates is insufficient to determine the others. As a result archival research is therefore required to determine the opening date for these schools. I therefore drop all separate schools from the Saskatchewan data to be consistent with Alberta.

A.7 Geolocation

After merging shapefiles for parcels in Alberta (Government of Alberta 2020), Saskatchewan (Information Services Corporation (ISC) 2020), and Manitoba (Manitoba Government 2020), I drop parcels which do not conform to the Section-Township-Range-Meridian system. This consists primarily of river lots located in Manitoba, near Winnipeg, with 13,258 dropped out of a total of 196,864 in Manitoba. For each parcel I take the centroid as the location of the parcel. I create three different levels of aggregation at the quarter section, section, and township level.

A.7.1 Geolocating Households

To geolocate households I first split households into rural and urban households. Rural households are located via the section, township, range, and meridian, while urban households are located using CSD polygons from CIEQ, Université Laval and HGIS Lab, University of Saskatchewan (2023). In most cases the rural geolocation is more precise than the urban geolocation, as each section is only a single mile square.

To locate rural households in 1906 and 1921 I first link each household using the section, township, range and meridian to the known coordinates for that section. This gives an initial set of geolocations for each household in rural areas.

One unique feature of census records is that a single enumerator would have to travel between households in order enumerate each household. In addition the instructions to enumerators consistently advise across census years to start at one end of a district, then to move systematically to the other end[17]. As a result the path of an enumerator is informative if the geolocation of a household is correct, and significant deviations from a regular path are likely to be errors in geolocation.

17. For example the enumerators instructions in 1906 advise *In a township, parish or other country district, where the houses are scattered, it is advisable to start on a road or highway at the border line of the district and visit in succession every occupied house until the other side of the district is reached, when the next road may be taken in the same way, and so on until the whole area assigned to the enumerator is covered, taking care to finish the census of one farm or lot before proceeding to the next.* (Government of Canada 1906)

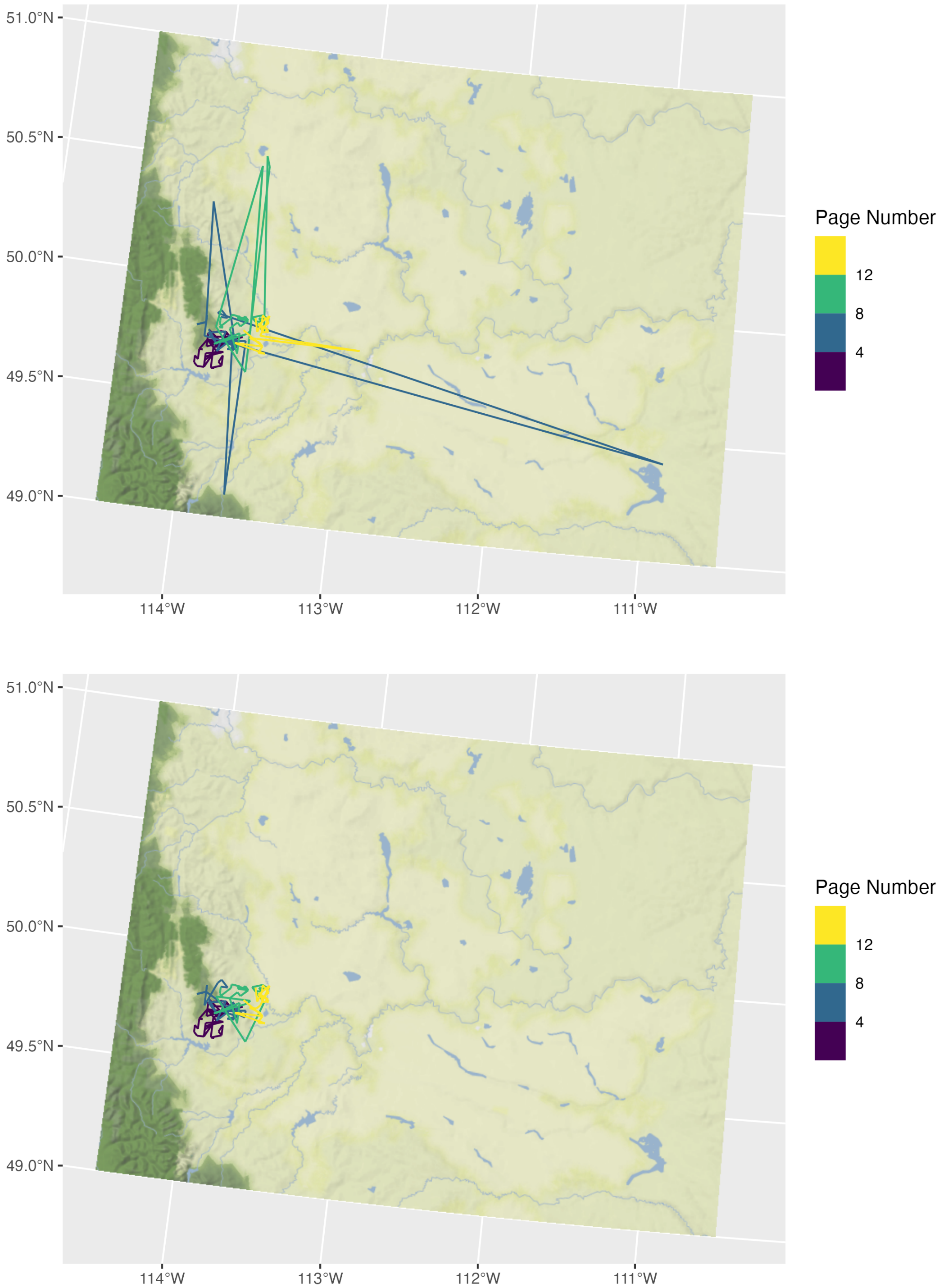


Figure A7: Enumeration path for a single enumeration subdistrict, before and after the cleaning proceedure is applied. Any location which is farther than four standard deviations in either longitude or latitude from the centre of the enumeration subdistrict is removed.

To utilize this feature I first take the raw locations in 1906, and group them by enumeration subdistrict. Any observations which are greater than four standard deviations away from the mean longitude and latitude of the enumeration subdistrict are removed. Figure A7 shows an example of this process. In 1921, where I have CSD polygons, I remove any geolocations which are greater than 3 kilometers from the edge of their CSD polygon. The remaining addresses I consider to be the exact location, as they correspond to a single section of land.

For any addresses that remain unlocated I do a second round of locating to the nearest township, then a third round of locating to the nearest city. In 1921 there a several small towns which have an area close to that of a section. If the area of the CSD is less than 1.5 square miles (relative to a section which has an area of 1 square mile), then I consider the location to also be exact.

A.7.2 Creating Synthetic CSDs

In complete census years (1901, 1911, and 1921) CSD boundaries and centroids have been digitized by the HGIS lab at the University of Saskatchewan (CIEQ, Université Laval and HGIS Lab, University of Saskatchewan 2023). For the census of the prairie provinces and the most recent complete census year (1906, 1916, 1926, and 1931) CSDs have not been digitized. To overcome this challenge I create synthetic CSDs using the enumeration subdistricts and the location of known households.

I first take the known locations from geolocated households, and then compute the centroid of those locations. I use this centroid if there are at least (number of households) located in that enumeration subdistrict. This gives the centroid of the synthetic CSD when the CSD is unknown for (A number) of enumeration districts out of a total of (Another number). For the remaining (A number) enumeration subdistricts I use the centroid of the enumeration district as the centroid of the enumeration subdistrict. This leaves only two enumeration subdistricts in northern Alberta and Saskatchewan which I geocode to an approximate location.

A.7.3 Geolocating Schools

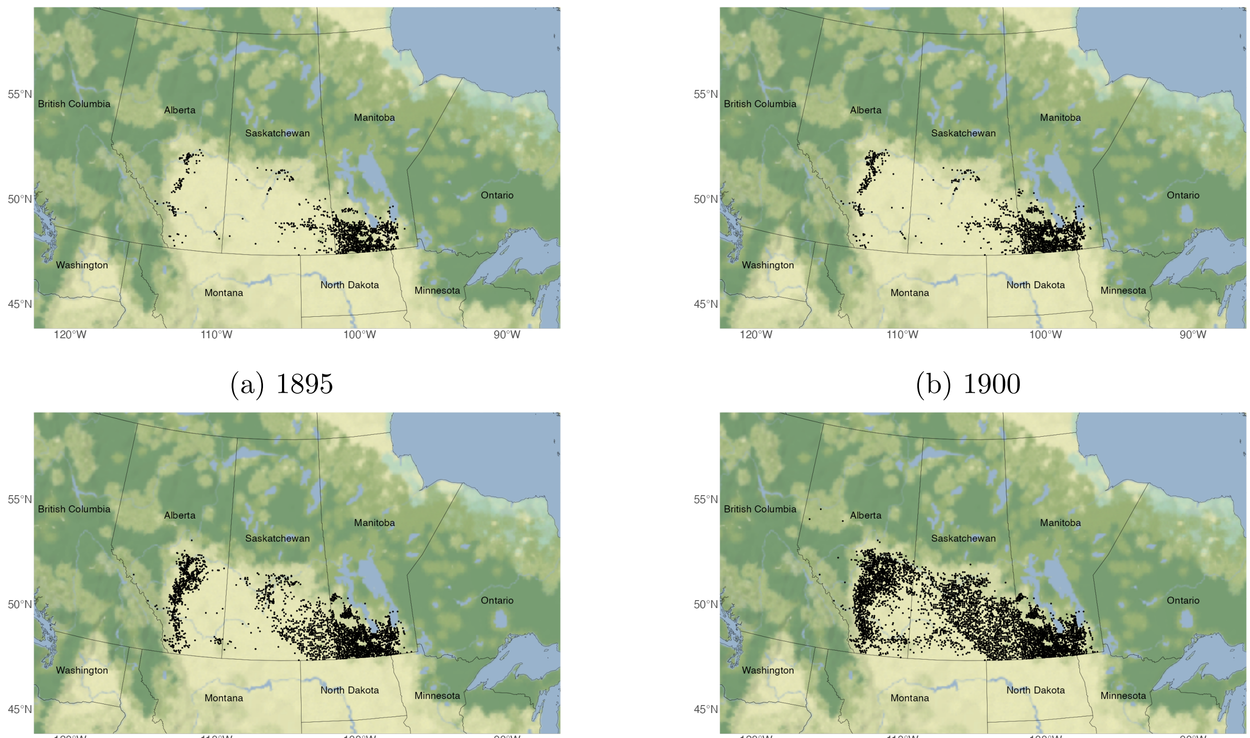
I implement two procedures to geolocate schools.

For Alberta and Saskatchewan, I know the legal land description for each school, and use this to get the parcel of land that corresponds to that description. I start with the finest possible geography, the Quarter Section, and try to link each school. The remaining schools I attempt to link at the section level, and any remaining after that I link at the township level. As shown in table A11 there are cases where a school only has location information at the township level, will multiple possible townships reflecting the entire area of the school division. To resolve this I take the centroid of a group of townships as the location of the school.

For Manitoba, where I already have longitude and latitude, I use the existing coordinates to retrieve the legal land description of the location of the school. I include this information in my school database for possible use in later regressions.

The bottom rows of table A13 show the number of schools I am able to geolocate in 1911. For both Alberta and Manitoba I am able to geolocate nearly all schools reported, while in Saskatchewan I am only able to geolocate 93% of schools. When I focus on the schools I am able to geolocate most precisely, at the section level, I find I'm able to geolocate 93% of schools in Alberta, 99% in Manitoba, but only 43% in Saskatchewan. As a result I show that the results are robust regardless of the school locations used in Saskatchewan.

Figure A8 shows the expansion of school across western Canada from 1895 to 1910. Although schools were initially formed around Winnipeg in Manitoba, westward expansion led settlers to move farther west, and construct schools. As a result we see the expansion of schools in Saskatchewan and Alberta (also seen in figure A6).



(a) 1895

(b) 1900

(c) 1905

(d) 1910

Figure A8: Expansion of schools in western Canada from 1895 to 1910.

A.7.4 Farm Size Estimation

For the census waves in 1906, 1921, and 1931, I can observe the exact geolocations but have no information on land holdings. Instead I can estimate the farm size by taking the set of geolocations and estimating the set of Voronoi polygons around the geolocations. Voronoi polygons are computed from a set of points on a plane, where the plane is separated into regions based on which point is closest. Doing this for the set of households, gives an estimate of the amount of land surrounding an individual geolocated household. A small number of farms will be at the edge of the computed Voronoi polygons and will have a size defined by an arbitrary computational boundary. Farms at the edge are dropped from the sample.

The geolocation proceedure geolocates households to the nearest land section. In practice there can be up to four occupied quarter sections within a given land section, leading to multiple houses sharing the same geolocation. In the case where multiple households share the same geolocation, the area of the farm is divided by the number of households sharing that location.

To validate the estimated farm size I regress estimated farm size on the characteristics of the head of household recorded in the census. Table A14 presents these results for 1906, 1921, and 1931. 1906 is the most informative year as the census also recorded livestock holdings. I find that the estimated size of a farm positively correlates with the number of horses and cattle that the household has, and negatively correlates with the number of hogs. This is expected as the space requirements for hogs are significantly lower than for cattle or horses.

Expanding to 1921 and 1931 in columns 2-5 shows some common patterns with 1906. I find a consistently positive correlation with the number of adult males in a household, consistent with male farm labour being valuable. There is a strongly negative correlation with the household being located on a homestead section of land. Homestead sections were typically split between more owners and had a higher population density as seen in figure 1, as well as in the population values by section in figure A3. There is no consistent pattern for different levels of agricultural productivity.

The 1921 and 1931 census also include information on occupations and income. Column's 2-5 show that being an employer is associated with having a larger farm, consistent with being a landowner with hired help. Individuals listed as employees are shown to have 20% smaller farms. In each case the omitted category is being on Own Account, the typical category for independant farmers. Columns 3 and 5 include income as a covariate, showing that higher incomes are generally associated with smaller farms.

A.8 Additional Results

A.8.1 Distance to School

I compute distance to school in 1911 taking the set of geolocations from 1906 and computing the distance to the nearest school open in 1910. Using the 1910 distances instead of 1911 reflects that the 1911 census asked for the months at school in 1910 (Canada. Census and Statistics Office. 1911). Lagging the set of schools by a year also reflects that fact that school opening dates measured reflect the establishment dates of the district, not necessarily the construction date of the school. The median school distance in 1910 is approximately 2.9km, with a significant spread as seen in figure A9.

The sample is restricted to children whose Census Subdivisions (CSD) are within 100km of each other between 1906 and 1911, to capture children who have likely not moved in the intervening five year period. CSD boundaries changed significantly between census waves due to population changes, making a high threshold necessary. The sample is further restricted to those that are at least five years old in 1911 (so they are guaranteed to be alive in 1906) and are less than eighteen years old in 1911, or the time that modern secondary schooling ends. The sample is further restricted to those that are listed as a descendant of the head of household in 1911. This restricts the sample to children who are living with their parents, and drops those who may have already moved away for work or higher schooling in 1911.

Table A14: Farm Size Validation

Dependent Variable:	Ln Farm Area				
Model:	(1)	(2)	(3)	(4)	(5)
	1906	1921		1931	
Asinh Horses	0.0606***				
	(0.0052)				
Asinh Milk Cows	0.0015				
	(0.0055)				
Asinh Hogs	-0.0124***				
	(0.0040)				
Asinh Cattle	0.0229***				
	(0.0034)				
Asinh Sheep	-0.0027				
	(0.0063)				
Family Size	-0.0007	-0.0036***	-0.0031	0.0023***	0.0017
	(0.0019)	(0.0008)	(0.0021)	(0.0008)	(0.0022)
Fam. Adult Males	0.0234***	0.0131***	0.0142**	0.0252***	0.0162***
	(0.0043)	(0.0024)	(0.0058)	(0.0027)	(0.0051)
Homestead	-0.3010***	-0.1785***	-0.1636***	-0.1529***	-0.1324***
	(0.0185)	(0.0067)	(0.0180)	(0.0062)	(0.0137)
Asinh Wheat	-0.1594	-0.3176***	-0.3648*	-0.1152	-0.0621
	(0.1284)	(0.0699)	(0.2144)	(0.0808)	(0.1945)
Asinh Oats	0.3309*	0.0435	0.5829	-0.2737	-0.3143
	(0.1804)	(0.1850)	(0.5406)	(0.1727)	(0.3072)
Asinh Grass	0.0190	-0.0326	-0.0202	0.1199	0.1430*
	(0.0218)	(0.0199)	(0.0300)	(0.0763)	(0.0755)
Asinh Flax	-0.0159	-0.0064	0.0144	-0.1160	-0.0575
	(0.0559)	(0.0306)	(0.0291)	(0.0872)	(0.0924)
Asinh Barley	-0.2379	0.2689	-0.2563	0.3547**	0.3551
	(0.1613)	(0.1678)	(0.4808)	(0.1439)	(0.2614)
Employer		0.0544***	0.0249	0.0243***	0.0122
		(0.0105)	(0.0254)	(0.0048)	(0.0134)
Employee		-0.2150***	-0.2009***	-0.2013***	-0.2306***
		(0.0120)	(0.0267)	(0.0107)	(0.0166)
Log Income			-0.0180***		-0.0248***
			(0.0065)		(0.0082)
Enum. Sub.	Yes	Yes	Yes	Yes	Yes
Observations	100,562	217,269	29,211	195,664	32,303
R^2	0.21181	0.16214	0.29698	0.18517	0.29221

Clustered (Enum. Sub.) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Regressing estimated farm size from Voronoi polygons on covariates at the individual level. Column 1 gives the results for 1906, columns 2-3 for 1921, and columns 4-5 for 1931. Columns 3 and 5 restrict to household heads who earned some income.

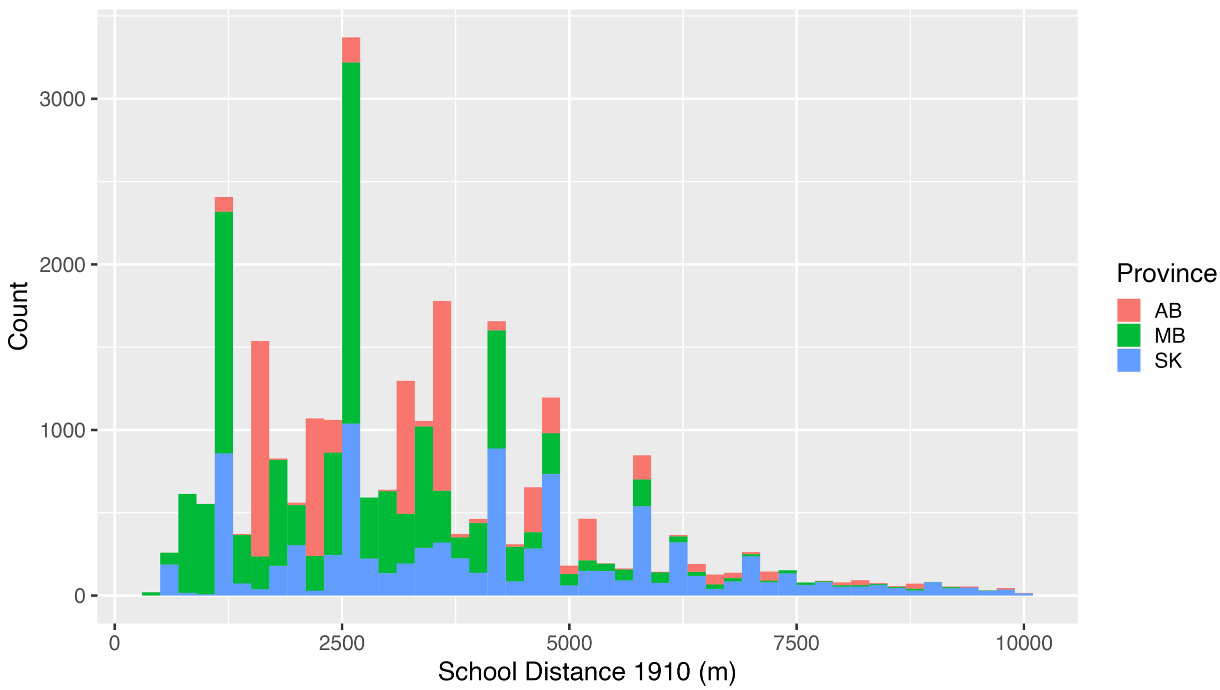


Figure A9: Distance to school for children in 1910, based on 1906 locations and restricting the sample to those that have not moved. Note the distribution has distinctive peaks including those at approximately 1250m and 2500m. This reflects the precision of the geolocation, with 0.75 miles being 1207m, and 1.5 miles being 2.41km.

A.8.2 Intensive and Extensive Margin and Distance

Despite the significant anecdotal evidence that geographic distance reduced school attendance (including quotes in section A.1 in the appendix), there is no quantitative evidence that the distance to school affected attendance in this context. I observe the number of months of school attended(the flow of education) in full Canadian census waves (1901, 1911, 1921, 1931)[18]. I link children in 1906, where precise geolocation is possible, to 1911 where schooling is measured, and restrict to children who are likely living in the sample location in both 1906 and 1911.

Holding the 1906 locations constant and assuming the child has not moved, I then compute the distance to school in 1911. I then estimate the following equation:

$$y_{ig} = \sum_{k=1}^{5} \beta_k \mathbb{1}\{d \in k_{1km}\} + X_i + \alpha_g + \epsilon_{ig} \tag{23}$$

where y_{ig} is the number of months of school that the child attended in 1911, or whether or not a child attended school in 1911. β_k is the distance to the nearest school in 1911 in $1km$ rings, with distances beyond $5km$ as the reference group. α_g is an enumeration subdistrict fixed effect. X_i is a series of controls including the full set of interactions of sex, age, and birth order, birthplace, family livestock holdings, estimated agricultural productivity, whether the family lives on a homestead parcel of land, estimated farm size in 1906, and distance to the railway in 1911.

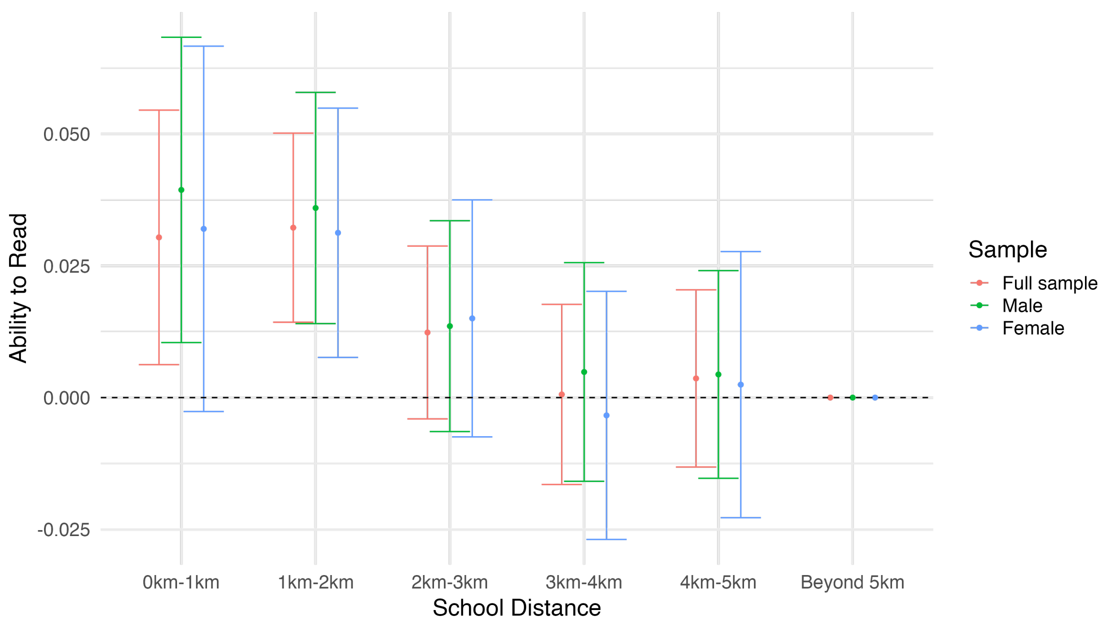
There is a monotonic decline in the estimated effect size with distance, with the effect being statistically insignificant from zero for the full sample when children are in the 3-4km bin. These estimates align with the recommended school district size, which was for 16-20 square miles in Alberta (Noonan, Hallman, and Scharf 2006). Assuming a circular district, this gives a radius of approximately 2.5 miles or 4km. This suggests policy makers of the

18. I do not observe the amount of education obtained (the stock) as adults in any census wave. This information is available in the 1936 Census of Prairie Provinces, which will be released in 2028 in accordance with a 92 year rule.

time accurately judged the largest possible distance that a child would be able to travel to school.

School attendance may have declined due to distance, but that this may have had no effect on learning. To guard against this possibility I repeat the estimation in equation 23, but use the ability to read and write as an outcome variable instead of the months of schooling. The estimates show a similar decline in reading and writing ability with distance, with children adjacent to a school being approximately 3pp more likely to read than children farther than 5km from the nearest school. The likelihood of a child reading is 86% in the sample, so a decline of 3pp is roughly a 3.5% decline in the likelihood of reading as a child.

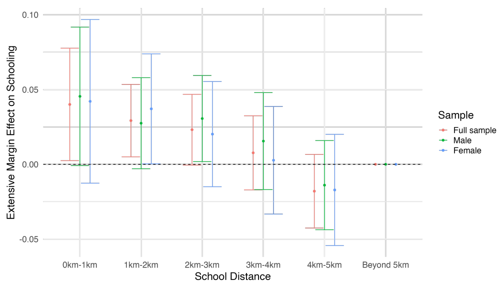
Figure A10: Reading and Writing versus Distance



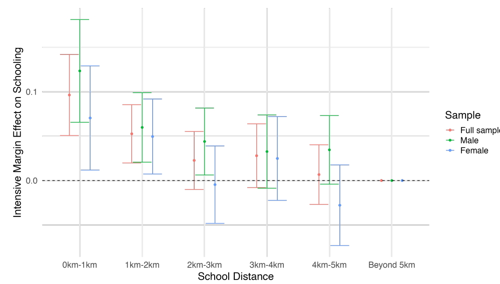
Reading and Writing Ability by distance in 1911, in 1km bins. All coefficients are expressed relative to the furthest bin of 5km or more. Includes controls for the interactions of sex, age, and birth order, birthplace, family livestock holdings, estimated agricultural productivity, whether the family lives on a homestead parcel of land, estimated farm size in 1906, and distance to the railway in 1911.

Figure A11a gives the estimation results repeating the estimation in figure 4, but examining the extensive margin of school attendance only. Figure A11b also repeats the same estimation strategy, but dropping observations with zero attendance to examine intensive margin effects of school attendance. In both cases there is a clear decay pattern over distance, although the estimates are less precise than the main effect.

Figure A11: Intensive and Extensive Margin of School Attendance Over Distance



(a) School Attendance (Extensive Margin)



(b) Months of Schooling (Intensive Margin)

Extensive and intensive margin effects of distance on schooling in 1911, in 1km bins. All coefficients are expressed relative to the furthest bin of 5km or more. Includes controls for the interactions of sex, age, and birth order, birthplace, family livestock holdings, estimated agricultural productivity, whether the family lives on a homestead parcel of land, estimated farm size in 1906, and distance to the railway in 1911.

The average child who is within 1km of a school is approximately 5 percentage points more likely to have attended school in 1911 relative to a child who is farther than 5km from a school. The mean school attendance for all children is 61.5%, so a decline of 6 percentage points is a significant determinant of school attendance. The intensive margin gives a difference of approximately 0.55 months of schooling for a child who is within 1km of a school compared to a child who is more than 5km away. The intensive margin appears to decay faster in space relative to the main effect, with the effects rapidly approaching zero in the 2-3km bin.

Comparing the intensive margin effects to the total effects show in figure 4, I find that the main effect of distance is a 15% decline in schooling versus a 9% decline from the intensive margin. The extensive margin effect is therefore 6% of the decline in schooling.

Table A15: Attendance vs Distance

Dependent Variable:	Attends School 1911					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Log School Distance	-0.0260***	-0.0251***	-0.0262***	-0.0258***	-0.0254***	-0.0253***
	(0.0062)	(0.0062)	(0.0062)	(0.0062)	(0.0061)	(0.0061)
Link Prob.	0.0482**	0.0472**	0.0463**	0.0362*	0.0321	0.0320
	(0.0207)	(0.0206)	(0.0207)	(0.0205)	(0.0204)	(0.0204)
Log RR Distance		-0.0076**	-0.0080***	-0.0077***	-0.0083***	-0.0080***
		(0.0030)	(0.0030)	(0.0030)	(0.0029)	(0.0030)
Log Farm Area			0.0089**	0.0080**	0.0051	0.0050
			(0.0038)	(0.0038)	(0.0039)	(0.0039)
Sex/Age/BO Interactions	Yes	Yes	Yes	Yes	Yes	Yes
Enumeration Subdistrict	Yes	Yes	Yes	Yes	Yes	Yes
Birthplace				Yes	Yes	Yes
Homestead					Yes	Yes
Livestock					Yes	Yes
Ag. Productive.						Yes
Observations	26,878	26,878	26,878	26,878	26,878	26,878
R^2	0.36958	0.36981	0.37001	0.37277	0.37438	0.37518

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Estimates of the effect of distance to school on school attendance in percentage points. Includes controls for the interactions of sex, age, and birth order, enumeration subdistrict, birthplace, whether the family lives on a homestead parcel of land, family livestock holdings, estimated agricultural productivity, estimated farm size in 1906, distance to the railway in 1911, and the likelihood of a link being correct. All observations weighted by the link probability.

Table A16: Months of School vs Distance

Dependent Variable:	Log Months School 1911					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Log School Distance	-0.0391***	-0.0390***	-0.0390***	-0.0395***	-0.0378***	-0.0362***
	(0.0075)	(0.0075)	(0.0075)	(0.0075)	(0.0075)	(0.0075)
Link Prob.	0.0561*	0.0556*	0.0556*	0.0541*	0.0490*	0.0466
	(0.0293)	(0.0293)	(0.0293)	(0.0291)	(0.0292)	(0.0291)
Log RR Distance		-0.0032	-0.0032	-0.0033	-0.0039	-0.0047
		(0.0034)	(0.0034)	(0.0034)	(0.0033)	(0.0033)
Log Farm Area		0.0027	0.0027	0.0024	-0.0003	-0.0006
		(0.0044)	(0.0044)	(0.0044)	(0.0045)	(0.0045)
Sex/Age/BO Interactions	Yes	Yes	Yes	Yes	Yes	Yes
Enumeration Subdistrict	Yes	Yes	Yes	Yes	Yes	Yes
Birthplace				Yes	Yes	Yes
Homestead					Yes	Yes
Livestock					Yes	Yes
Ag. Productive.						Yes
Observations	16,816	16,816	16,816	16,816	16,816	16,816
R^2	0.31302	0.31309	0.31309	0.31498	0.31885	0.32006

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Estimates of the effect of distance to school on months in school in 1911, expressed in logs. Includes controls for the interactions of sex, age, and birth order, enumeration subdistrict, birthplace, whether the family lives on a homestead parcel of land, family livestock holdings, estimated agricultural productivity, estimated farm size in 1906, distance to the railway in 1911, and the likelihood of a link being correct. All observations weighted by the link probability.

A.8.3 Sex

A descriptive feature emphasized in the school board reports in section A.1 is that boys stay in school only until they have reached puberty and their labour becomes more valuable. To test if this is quantitatively meaningful I examine differences in school attendance for boys versus girls before and after the age of 13, when puberty typically occurs. I first estimate the difference in school attendance between girls and boys with the following specification:

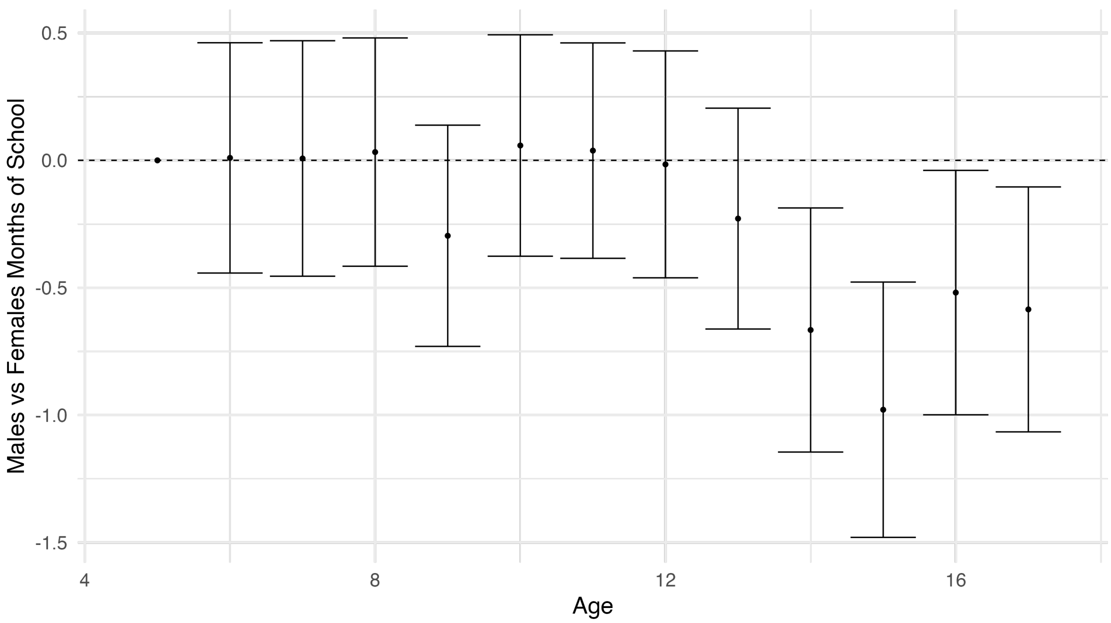
$$y_{ig} = \beta sex_i \times age_i + sex_i + age_i + X_i\alpha_g + \epsilon_{ig} \tag{24}$$

where sex_i is a binary variable with 0 for female and 1 for male, age_i is an age fixed effect for all ages between 6 and 17, and X_i is a series of other controls including the distance to the nearest railway line in 1911, the distance to the nearest school, birthplace, birth order, and whether or not they family lives on homestead land, and α_g is the enumeration subdistrict. As in the previous section I use both the whether a child attended school in 1911 and the number of months they attended school in 1911 as outcome variables.

Figure A12 shows the coefficients of the interaction terms between sex_i and age_i from ages 6 to 17. As described in the school board reports there is a clear decline in male school attendance relative to females starting at age 13. The difference becomes significant and large by the late teens, amounting to approximately 0.75 months of school in a year, or three weeks. The average school attendance for children between the ages of 13 to 17 is 4.2 months, so a decline of 0.75 months represents a 17% decline in the amount of schooling attended.

Figure A13 shows that this result is primarily driven on the intensive margin. Boys decrease their likelihood of going to school only slightly relative to girls, but conditional on going to school decrease the amount that they go. This is consistent with boys remaining attached to the school, but being more selective about which days of school are attended as the value of their farm labour increases.

Figure A12: Males vs Females



Coefficients of the interaction term of sex and age regressed on total months at school. Controls for sex, age, birth order, birthplace, enumeration subdistrict (geography), and whether the family is living on a homestead. The decline in the coefficients represents females attending more school relative to males.

Table A17: Months of School vs Age by Sex

Dependent Variable:	Months School 1911				
Model:	(1)	(2)	(3)	(4)	(5)
Male	0.0895	0.0852	0.0974	0.1004*	0.1008*
	(0.0602)	(0.0601)	(0.0597)	(0.0598)	(0.0602)
Older 12	-0.7744***	-0.8683***	-0.9518***	-0.9531***	-1.345***
	(0.0933)	(0.0937)	(0.0930)	(0.0932)	(0.0994)
Log RR Distance	-0.0818***	-0.0823***	-0.0905***	-0.0910***	
	(0.0290)	(0.0293)	(0.0290)	(0.0291)	
Log School Distance	-0.3392***	-0.3330***	-0.3235***	-0.3183***	
	(0.0621)	(0.0619)	(0.0614)	(0.0619)	
Log Farm Area	0.0501	0.0457	0.0005	-0.0006	
	(0.0376)	(0.0377)	(0.0383)	(0.0384)	
Link Prob.	0.7864***	0.7305***	0.6549***	0.6454***	0.3252
	(0.1774)	(0.1778)	(0.1763)	(0.1759)	(0.2204)
Male × Older 12	-0.6921***	-0.6893***	-0.6916***	-0.6914***	-0.7329***
	(0.1073)	(0.1072)	(0.1073)	(0.1074)	(0.1208)
Birth Order	Yes	Yes	Yes	Yes	Yes
Enumeration Subdistrict	Yes	Yes	Yes	Yes	
Birthplace		Yes	Yes	Yes	Yes
Family					Yes
Homestead			Yes	Yes	
Livestock			Yes	Yes	
Ag. Productive.				Yes	
Observations	26,878	26,878	26,878	26,878	26,878
R^2	0.24081	0.24483	0.24966	0.25051	0.68260

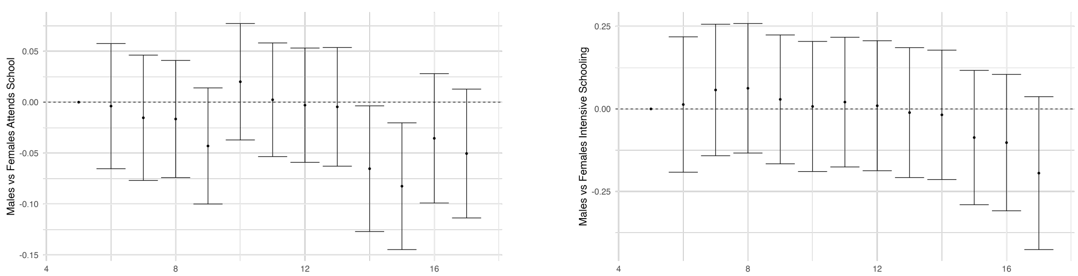
Clustered (Enumeration Subdistrict) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Months of school in 1911 versus sex interacted with age. Column 1 reports the regression without any family characteristics. Column 2 include whether the family lives on a homestead parcel. Column 3 includes enumeration subdistrict fixed effects. Column 4 includes family fixed effects. In each case the coefficient on sex interacted with being older than 12 is nearly identical, indicating that boys decrease their schooling after the age of 12.

I then repeat the estimation in equation 24, but use an indicator for being over the age of 12 (13-17) to capture the effect on boys of puberty. The main result is in column 3, of table A17 where I find a coefficient of -0.6580 on the months of school for boys when over the age of 12, in line with the results in figure A12. I then vary the fixed effects used in the estimation. Column 2 drops the enumeration subdistrict fixed effect, and column 1 drops the homestead fixed effect. In each case I find negligible changes in the interaction effect between sex and being over 12. Column 4 does the opposite, and adds a family fixed effect. The interpretation of this regression is that any changes in male school attendance over the age of 12 are relative to their siblings. Comfortingly I find coefficients that are nearly identical.

These results are unsurprising, and mirror what is report in the Iowa state census in 1915 as reported in Goldin and Katz (2000), that the total stock of education for women was higher than men. A related result from Goldin and Katz (2000) is that the returns to education for women were only significant until the point they were qualified to become a teacher or clerk as adults, and essentially zero afterwords. This suggests that the causal impact of education access may be to drive women towards those occupations, but this remains an area to be further researched.

Figure A13: Intensive and Extensive Margin of School Attendance for Boys vs Girls



(a) School Attendance (Extensive Margin) (b) Months of Schooling (Intensive Margin)

Coefficients of the interaction term of sex and age regressed on whether or not a child attended school in 1911 and the months of school attended in 1911. Controls for sex, age, birth order, birthplace, enumeration subdistrict (geography), and whether the family is living on a homestead. The decline in the coefficients represents females attending more school relative to males.

A.9 Difference-in-Differences Robustness

Table A18: Splitting Farm work by Distance to Railway

Dependent Variable:	Farm Work		
Model:	(1)	(2)	(3)
	Full Sample	Within 5km of RR	Beyond 5km of RR
School Constr. × Cohort	-0.0295*	0.0375	-0.0529**
	(0.0171)	(0.0380)	(0.0209)
School Constr.	0.0029	-0.0088	0.0114
	(0.0177)	(0.0398)	(0.0197)
Cohort	0.0051	-0.0134	0.0142
	(0.0099)	(0.0146)	(0.0139)
Enumeration Subdistrict (1906)	Yes	Yes	Yes
Fam. Bpl. + Homestead	Yes	Yes	Yes
Livestock + Ag.	Yes	Yes	Yes
Observations	14,701	5,299	9,402
R^2	0.07841	0.14284	0.08890
Dependent variable mean	0.62009	0.58464	0.64008

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Likelihood of working on a farm as an adult, split by if there is a railway within 5km. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905 in the baseline specification in column 3. All other columns alternate this value. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

A.10 Farming Income

The 1931 census asked enumerators to record the income of employees only, and did not ask enumerators to record the income of those listed as employers or working on their own account. Some records for employers and own account workers do contain income, likely

Table A19: Distance Robustness for Farm Work in 1931

Dependent Variable:	Farm Work				
Model:	(1)	(2)	(3)	(4)	(5)
	3km	4km	5km	6km	7km
School Constr. × Cohort	-0.0034	-0.0142	-0.0297*	-0.0243	-0.0236
	(0.0180)	(0.0168)	(0.0171)	(0.0172)	(0.0186)
School Constr.	-0.0061	-0.0078	0.0033	0.0101	0.0134
	(0.0164)	(0.0168)	(0.0175)	(0.0170)	(0.0183)
Cohort	-0.0017	0.0008	0.0043	0.0029	0.0026
	(0.0099)	(0.0100)	(0.0099)	(0.0101)	(0.0099)
Log RR Distance (1906)	0.0199***	0.0200***	0.0200***	0.0201***	0.0201***
	(0.0048)	(0.0048)	(0.0048)	(0.0048)	(0.0048)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes	Yes
Fam. Bpl. + Homestead	Yes	Yes	Yes	Yes	Yes
Livestock + Ag.	Yes	Yes	Yes	Yes	Yes
Observations	14,701	14,701	14,701	14,701	14,701
R^2	0.07970	0.07981	0.07989	0.07978	0.07976
Dependent variable mean	0.62009	0.62009	0.62009	0.62009	0.62009

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Likelihood of being having at least one family member in agriculture as an adult in 1931. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905 in the baseline specification in column 3. All other columns alternate this value. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

Table A20: Distance Robustness for Income in 1931

Dependent Variable:	Log Income				
Model:	(1)	(2)	(3)	(4)	(5)
	3km	4km	5km	6km	7km
School Constr. × Cohort	0.0915	0.0749	0.1173*	0.1296**	0.1329**
	(0.0599)	(0.0563)	(0.0598)	(0.0591)	(0.0632)
School Constr.	0.0236	0.0448	0.0086	-0.0211	-0.0080
	(0.0498)	(0.0519)	(0.0553)	(0.0569)	(0.0594)
Cohort	-0.2453***	-0.2440***	-0.2541***	-0.2571***	-0.2559***
	(0.0307)	(0.0313)	(0.0308)	(0.0308)	(0.0308)
Log RR Distance (1906)	-0.0326***	-0.0328***	-0.0330***	-0.0334***	-0.0335***
	(0.0125)	(0.0125)	(0.0126)	(0.0126)	(0.0126)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes	Yes
Fam. Bpl. + Homestead	Yes	Yes	Yes	Yes	Yes
Livestock + Ag.	Yes	Yes	Yes	Yes	Yes
Observations	5,651	5,651	5,651	5,651	5,651
R^2	0.17301	0.17307	0.17320	0.17301	0.17314
Dependent variable mean	6.5234	6.5234	6.5234	6.5234	6.5234

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Income as an adult in 1931. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905 in the baseline specification in column 3. All other columns alternate this value. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

Table A21: Family in Agriculture in 1931

Dependent Variable:	Family in Ag				
Model:	(1)	(2)	(3)	(4)	(5)
School Constr. × Cohort	-0.0385**	-0.0286*	-0.0300*	-0.0312**	-0.0314**
	(0.0154)	(0.0156)	(0.0156)	(0.0156)	(0.0155)
School Constr.	0.0113	0.0057	0.0060	0.0035	0.0039
	(0.0166)	(0.0168)	(0.0167)	(0.0169)	(0.0167)
Cohort	0.0399***	0.0412***	0.0603***	0.0602***	0.0593***
	(0.0093)	(0.0096)	(0.0094)	(0.0094)	(0.0094)
Log RR Distance (1906)					0.0208***
					(0.0046)
Enumeration Subdistrict (1906)	Yes	Yes	Yes	Yes	Yes
Fam. Bpl.		Yes	Yes	Yes	Yes
Homestead			Yes	Yes	Yes
Livestock			Yes	Yes	Yes
Ag. Productive.				Yes	Yes
Observations	14,701	14,701	14,701	14,701	14,701
R^2	0.05878	0.06693	0.07833	0.08073	0.08254
Dependent variable mean	0.69322	0.69322	0.69322	0.69322	0.69322

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Likelihood of being having at least one family member in agriculture as an adult in 1931. School construction is defined as 1 if the distance to school decreased by more than 5km between 1900 and 1905. The post cohort is those who are 0-5 in 1906, as compared to those who are 12-17. Railway distance is the distance in 1906. Homestead is a control for the type of land parcel a family is on in 1906. Family birthplace are controls for own, father's and mother's birthplace. Livestock controls for family holdings of cattle, hogs, horses, milk cows, and sheep in 1906 as taken from the census. Agricultural conditions are for the agricultural productivity in wheat, oats, barley, flax, and reed canary grass.

reflecting some enumerators asking everyone regardless of employment status. We can use this information to have a rough approximation of farmer incomes for non-employee farmers.

Table A22 breaks out farmer income by employment status. There are two possible ways that a farmer could be coded in the data, as either a farmer or as a labourer working on a farm. All six possibilities are listed. Incomes are highest for both Farmer's and Labourer's listed as employers at over $800 per year, with declines to approximately $500 per year if listed as on their own account. Employees are the lowest paid, with an income of only approximately $350.

It is worth noting that these figures are sampled from a very small percentage of farmers. Of the total 1906-1931 sample only 4% of own account farmers reported any income. In addition I drop all farmers reporting exactly $1000 of income. There is a significant excess mass at this value, consistent with farmers whose income is unknown reporting exactly $1000.

Table A22: Estimated Farming Income by Type in 1931

Type	income_1931	has_income_1931	n
Farmer - Employer	$866.43	0.04	3812
Farmer - Own Account	$585.97	0.04	8363
Farmer - Employee	$381.53	0.68	1445
Labour - Employer	$927.46	0.39	33
Labour - Own Account	$318.75	0.18	44
Labour - Employee	$337.02	0.82	612

A.11 Growing up Rural vs Urban Managers

A.12 Teacher Effects

To assign students to individual teachers in the 1911 census I rely on a distance measure based on the distance in pages and lines between two people in the census, as in Card, Domnisoru, and Taylor (2022). I first locate each teacher based on their HISCO code. For each student, I then count the number of lines in the census to the nearest teacher, and

Table A23: Managers in 1931

Industry	n
Unknown	166
Bank	32
Lumber Mill	18
General Store	17
General Farm	14
Grain Elevator	12
Garage	10
Grocery Store	9
Insurance	9
General	8
Other	78

Grew up Rural

Industry	n
Unknown	198
General Store	23
Bank	20
Insurance	10
General	9
Grain Elevator	9
Steam Railway	9
Dairy Farm	8
Lumber Mill	8
Wholesale Grocery	8
Other	107

Grew up Urban

Table A24: Teachers in 1911

Birthplace	N	Percentage
ON	2353	36.38
MB	1248	19.29
United Kingdom	737	11.39
United States	551	8.52
Maritimes	447	6.91
QC	311	4.81
Canada + BC	296	4.58
Rest of Europe	178	2.75
Ireland	122	1.89
Nordic	108	1.67
	90	1.39
ROW	22	0.34
Caribbean	5	0.08

Birthplaces of all teachers location in Alberta, Manitoba, and Saskatchewan in 1911. Teachers local to the prairies consist of those born in Manitoba and Canada + BC for a total of 23.87%. Due to the rapid growth of the schooling system it was impossible to educate sufficient teacher locally, a large portion of which came from Ontario.

assign them to that teacher. If no teacher can be found within an enumeration subdistrict, the child is not assigned to a teacher. I restrict the sample to children who have at least one parent working in agriculture and can therefore be assumed to be in rural areas.

This distance measure is based on the assumption that enumerators walked sequentially between households, so that proximity in the pages of the census corresponds to geographic proximity in rural areas. A visual inspection of the geolocated rural households in figure A7 shows that this is correct. Enumerators walked in paths between households, completing one area before moving on to the next. As a result the distance in the census is likely to be correlated with geographic distance.

To validate this measure further, I repeat the estimation of Card, Domnisoru, and Taylor (2022) and test whether female students perform better when they have a female teacher. Table A25 shows the results. I find that female students are 1.28pp (2.27%) more likely to attend between the ages of 6-11 if they have a female teacher and 2.72pp (5.83%) between the ages of 12-17. This is very similar to the estimated effect of 3.25pp found by Card, Domnisoru, and Taylor (2022) looking at enrollment for children aged 15-18 in 1940. The similar results I find here validates the teacher assignment rule.

I then estimate the effects of teacher birthplace on students outcomes in 1911. The results can be found in table A26. I set teachers from Ontario as the reference group as they are the largest single group, as seen in table A24. There are no differences in school attendance by teacher birthplace. On the intensive margin, children with a teacher from Europe (largely Eastern Europe) attend 8% less relative to having a teacher from Ontario. Column 3 shows that the ability to read is higher if students have a teacher from the United Kingdom, and lower for a teacher from the Caribbean. I interpret these results as there being little effect from teachers on contemporaneous schooling outcomes.

Table A27 repeats the same estimation, but with a linked sample between 1911 and 1931. There are no clear effects of teacher birthplace on income or farm work. Similar effects are true for migration distance with the exception of teachers from the United Kingdom, who

Table A25: School Attendance by Sex and Teacher Sex

Dependent Variable:	Attends School			
	Full Sample	0-5	6-11	12-17
Model:	(1)	(2)	(3)	(4)
Sex = Female	-0.0075***	-0.0014	-0.0166***	-0.0057
	(0.0025)	(0.0012)	(0.0048)	(0.0057)
Teacher = Female	-0.0044	-0.0005	-0.0059	-0.0062
	(0.0044)	(0.0018)	(0.0078)	(0.0085)
Teacher = Female and Child = Female	0.0117***	0.0006	0.0128**	0.0272***
	(0.0031)	(0.0017)	(0.0062)	(0.0069)
Age	Yes	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes	Yes
Religion	Yes	Yes	Yes	Yes
Enumeration Subdistrict (1911)	Yes	Yes	Yes	Yes
Observations	258,011	101,807	84,541	71,663
R^2	0.44294	0.06315	0.30749	0.32061
Dependent variable mean	0.32239	0.02131	0.56297	0.46631

Clustered (Enumeration Subdistrict (1911)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

School Attendance in 1911 by student sex and teacher sex. Sex and Teacher are binary indicators if the child or teacher is female. Controls for age, birthplace, religion and enumeration subdistrict.

Table A26: 1911 Schooling outcomes by Teacher Birthplace

Dependent Variables:	Attends School	Log Months School	Read
Model:	(1)	(2)	(3)
Teacher Birthplace = UnitedStates	-0.0094	0.0003	-0.0054
	(0.0120)	(0.0149)	(0.0091)
Teacher Birthplace = Canada+BC	-2.12×10^{-5}	0.0015	-0.0118
	(0.0145)	(0.0157)	(0.0146)
Teacher Birthplace = Lives in MB	0.0086	0.0014	0.0051
	(0.0113)	(0.0103)	(0.0080)
Teacher Birthplace = Lives in Maritimes	-0.0120	0.0228	-0.0042
	(0.0103)	(0.0161)	(0.0073)
Teacher Birthplace = Lives in QC	0.0033	-0.0049	0.0102
	(0.0216)	(0.0268)	(0.0145)
Teacher Birthplace = Caribbean	0.0054	-0.1023*	-0.1428***
	(0.1345)	(0.0566)	(0.0421)
Teacher Birthplace = Nordic	-0.0107	0.0524*	-0.0019
	(0.0346)	(0.0281)	(0.0203)
Teacher Birthplace = UnitedKingdom	0.0185	0.0039	0.0172**
	(0.0116)	(0.0140)	(0.0086)
Teacher Birthplace = Ireland	-0.0116	-0.0014	-0.0186
	(0.0210)	(0.0250)	(0.0225)
Teacher Birthplace = RestofEurope	-0.0417	-0.0831**	-0.0322
	(0.0265)	(0.0330)	(0.0202)
Teacher Birthplace = ROW	0.0398	-0.0229	0.0612***
	(0.0469)	(0.0457)	(0.0183)
Sex/Age/BO Interactions	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes
Religion	Yes	Yes	Yes
Enumeration Subdistrict (1911)	Yes	Yes	Yes
Observations	97,201	56,462	94,262
R^2	0.31023	0.28285	0.37711
Dependent variable mean	0.58088	1.9499	0.76432

Clustered (Enumeration Subdistrict (1911)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Outcomes for students in 1911 by the birthplace of their teacher. All outcomes expressed relative to having a teacher from Ontario. Includes controls for all interactions of sex, age and birth order, birthplace, religion, and enumeration subdistrict. Standard error are clustered at the enumeration subdistrict level.

are associated with a 9.22% decline in migration distance. If anything this goes against the hypothesis that having a teacher from a farther away location inspires children to move farther away from home as adults.

B Mathematical Appendix

This model is an extension of Hsiao (2025) to two sectors. What follows is largely consistent with his work, while elabourating on some of the longer derivations like choice probabilities from Fréchet distributions, which are standard in the literature but nearly never expressed in full detail[19].

B.1 Market Access and Choice Probabilities

I want to show that the choice probabilities by location are

$$\bar{m}_{jkg\ell} = \frac{\tilde{v}^{\theta}_{jkg\ell}}{\text{MA}_{jkg}}$$

where,

$$\tilde{v}_{jkg\ell} = \frac{a_\ell r_{jg} s_{jk\ell}}{\tau^m_{jkg\ell}}, \quad \text{MA}_{jkg} = \sum_{\ell} \tilde{v}^{\theta}_{jkg\ell}.$$

The utility before realizing the location shock is:

$$v_{jkg\ell}(e, \varepsilon_\ell) = \frac{a_\ell r_{jg} e^{\eta} s_{jk\ell} z_g}{\tau^m_{jkg\ell}} \varepsilon_\ell$$

The probability of choosing location ℓ is the probability that the utility is greater for ℓ than any other location ℓ'. The probability of choosing location ℓ given origin j, cohort k, and

19. I am reflecting some frustration at trying to find these derivations in the early years of my PhD.

Table A27

Dependent Variables:	Log Income	Farm Work	Log Migration Dist.
Model:	(1)	(2)	(3)
Teacher Birthplace = UnitedStates	0.0953	-0.0215	0.0072
	(0.0588)	(0.0197)	(0.0460)
Teacher Birthplace = Canada+BC	-0.0198	-0.0033	0.0068
	(0.0645)	(0.0283)	(0.0662)
Teacher Birthplace = Lives in MB	-0.0242	0.0156	0.0310
	(0.0549)	(0.0180)	(0.0439)
Teacher Birthplace = Lives in Maritimes	-0.0430	-0.0057	-0.0082
	(0.0620)	(0.0198)	(0.0465)
Teacher Birthplace = Lives in QC	0.1043	0.0032	0.0024
	(0.0864)	(0.0374)	(0.0748)
Teacher Birthplace = Caribbean	0.1311	0.1203	0.4311
	(0.1963)	(0.0987)	(0.3825)
Teacher Birthplace = Nordic	0.0564	0.0428	0.0853
	(0.1683)	(0.0516)	(0.1188)
Teacher Birthplace = UnitedKingdom	-0.0089	0.0142	-0.0922**
	(0.0518)	(0.0197)	(0.0453)
Teacher Birthplace = Ireland	-0.1591*	0.0532	0.0069
	(0.0872)	(0.0363)	(0.0754)
Teacher Birthplace = RestofEurope	-0.1625	-0.0495	-0.1507*
	(0.1201)	(0.0336)	(0.0800)
Teacher Birthplace = ROW	-0.6533	0.0891	-0.2904
	(0.4678)	(0.0817)	(0.2079)
Link Prob.	-0.0246	0.3678***	-2.640***
	(0.0575)	(0.0221)	(0.0578)
Enumeration Subdistrict (1911)	Yes	Yes	Yes
Birthplace	Yes	Yes	Yes
Father Birthplace	Yes	Yes	Yes
Mother Birthplace	Yes	Yes	Yes
Religion	Yes	Yes	Yes
Observations	7,144	17,424	16,259
R^2	0.24974	0.15655	0.34813
Dependent variable mean	6.4586	0.62299	4.6434

Clustered (Enumeration Subdistrict (1911)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Outcomes for adults in 1931 by the birthplace of their teacher in 1911. All outcomes expressed relative to having a teacher from Ontario. Includes controls for own birthplace, mothers birthplace, fathers birthplace, religion, and enumeration subdistrict. All regressions are weighted by the probability of a link being correct. Standard error are clustered at the enumeration subdistrict level.

sector g is therefore:

$$P_{jkg\ell} = P(\frac{a_\ell r_{jg} e^\eta s_{jk\ell} z_g}{\tau^m_{jkg\ell}}\varepsilon_\ell > \frac{a_{\ell'} r_{jg} e^\eta s_{jk\ell'} z_g}{\tau^m_{jk\ell'}}\varepsilon_{\ell'} \qquad \forall \ell' \neq \ell)$$

$$P_{jkg\ell} = m_{jkg\ell} = P(\frac{a_\ell r_{jg} s_{jk\ell}}{\tau^m_{jkg\ell}}\varepsilon_\ell > \frac{a_{\ell'} r_{jg} s_{jk\ell'}}{\tau^m_{jk\ell'}}\varepsilon_{\ell'} \qquad \forall \ell' \neq \ell)$$

$$m_{jkg\ell} = P(\tilde{v}_{jkg\ell}\varepsilon_\ell > \tilde{v}_{jkg\ell'}\varepsilon_{\ell'} \qquad \forall \ell' \neq \ell)$$

where $\tilde{v}_{jkg\ell} = \frac{a_\ell r_{jg} s_{jk\ell}}{\tau^m_{jkg\ell}}$, $\quad \text{MA}_{jkg} = \sum_\ell \tilde{v}^\theta_{jkg\ell}$ as above. Recall that the CDF for a Fréchet distribution with dispersion parameter θ is $F(\varepsilon_\ell) = exp(-e^{-\theta})$. The PDF is therefore:

$$\begin{aligned} F(\varepsilon_\ell) =& exp(-\varepsilon^{-\theta}) \\ dF(\varepsilon_\ell) =& -\theta(-\varepsilon_\ell^{-\theta-1}) exp(-\varepsilon_\ell^{-\theta}) d\varepsilon_\ell \\ dF(\varepsilon_\ell) =& \theta\varepsilon_\ell^{-\theta-1} exp(-\varepsilon_\ell^{-\theta}) d\varepsilon_\ell. \end{aligned}$$

We can recast our choice probabilities as a multiplication. The probability that the utility from a given location is larger than any other location is the probability that that location has a higher utility than any of the other locations individually, then multiplied together.

$$
\begin{aligned}
m_{jkg\ell} &= P(\tilde{v}_{jkg\ell'}\varepsilon_{\ell'} < \tilde{v}_{jkg\ell}\varepsilon_{\ell} \qquad \forall \ell' \neq \ell) \\
m_{jkg\ell} &= \prod_{\ell' \neq \ell} P(\tilde{v}_{jkg\ell'}\varepsilon_{\ell'} < \tilde{v}_{jkg\ell}\varepsilon_{\ell}) \\
m_{jkg\ell} &= \prod_{\ell' \neq \ell} P(\varepsilon_{\ell'} < \frac{\tilde{v}_{jkg\ell}}{\tilde{v}_{jkg\ell'}}\varepsilon_{\ell}) \\
m_{jkg\ell} &= \prod_{\ell' \neq \ell} \int_0^\infty exp(-\frac{\tilde{v}_{jkg\ell}}{\tilde{v}_{jkg\ell'}}\varepsilon_{\ell})^{\theta} dF\varepsilon_{\ell} \\
m_{jkg\ell} &= \prod_{\ell' \neq \ell} \int_0^\infty exp(-\tilde{v}_{jkg\ell}^{-\theta}\tilde{v}_{jkg\ell'}^{\theta}\varepsilon_{\ell}^{-\theta}) dF\varepsilon_{\ell} \\
m_{jkg\ell} &= \int_0^\infty exp(-\tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell' \neq \ell}\tilde{v}_{jkg\ell'}^{\theta}\varepsilon_{\ell}^{-\theta}) dF\varepsilon_{\ell} \\
m_{jkg\ell} &= \int_0^\infty exp(-\tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell' \neq \ell}\tilde{v}_{jkg\ell'}^{\theta}\varepsilon_{\ell}^{-\theta})\theta\varepsilon_{\ell}^{-\theta-1} exp(-\varepsilon_{\ell}^{-\theta}) d\varepsilon_{\ell} \\
m_{jkg\ell} &= \theta \int_0^\infty exp(-\varepsilon_{\ell}^{-\theta}(1 + \tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell' \neq \ell}\tilde{v}_{jkg\ell'}^{\theta}))\varepsilon_{\ell}^{-\theta-1} d\varepsilon_{\ell} \\
m_{jkg\ell} &= \theta \int_0^\infty exp(-\varepsilon_{\ell}^{-\theta}\tilde{v}_{jkg\ell}^{-\theta}(\tilde{v}_{jkg\ell}^{\theta} + \sum_{\ell' \neq \ell}\tilde{v}_{jkg\ell'}^{\theta}))\varepsilon_{\ell}^{-\theta-1} d\varepsilon_{\ell} \\
m_{jkg\ell} &= \theta \int_0^\infty exp(-\varepsilon_{\ell}^{-\theta}\tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell'}\tilde{v}_{jkg\ell'}^{\theta})\varepsilon_{\ell}^{-\theta-1} d\varepsilon_{\ell}
\end{aligned}
$$

To solve the integral make a subsitution. Let $t = \varepsilon_{\ell}^{-\theta}$. Then $dt = -\theta\varepsilon_{\ell}^{-\theta-1}d\varepsilon_{\ell}$, and $\varepsilon_{\ell} = \frac{1}{t^{\frac{1}{\theta}}}$. Computing the limits of integration gives that as $\varepsilon_{\ell} \to \infty$ then $t \to 0$ and as $\varepsilon_{\ell} \to 0$ then

$t \to \infty$.

$$m_{jkg\ell} = \theta \int_{\infty}^{0} exp(-t^{-\theta}\tilde{v}_{jkg\ell}^{-\theta} \sum_{\ell'} \tilde{v}_{jkg\ell'}^{\theta})\varepsilon_{\ell}^{-\theta-1} \frac{dt}{(-1)\theta\varepsilon_{\ell}^{-\theta-1}}$$
$$m_{jkg\ell} = \int_{0}^{\infty} exp(-t^{-\theta}\tilde{v}_{jkg\ell}^{-\theta} \sum_{\ell'} \tilde{v}_{jkg\ell'}^{\theta})dt$$
$$m_{jkg\ell} = \frac{\tilde{v}_{jkg\ell}^{\theta}}{\sum_{\ell'} \tilde{v}_{jkg\ell'}^{\theta}} exp(-t^{-\theta}\tilde{v}_{jkg\ell}^{-\theta} \sum_{\ell'} \tilde{v}_{jkg\ell'}^{\theta})\Big|_{0}^{\infty}$$
$$m_{jkg\ell} = \frac{\tilde{v}_{jkg\ell}^{\theta}}{\sum_{\ell} \tilde{v}_{jkg\ell}^{\theta}}(-1)(0-1)$$
$$m_{jkg\ell} = \frac{\tilde{v}_{jkg\ell}^{\theta}}{\sum_{\ell} \tilde{v}_{jkg\ell}^{\theta}}$$

B.2 Expected Utility and Expected Wage

I want to show that $\bar{v}_{jkg}(e) = z_g e^{\eta}\gamma_{\ell}\mathrm{MA}_{jkg}^{\frac{1}{\theta}}$ where $\gamma_{\ell} = \Gamma(1-\frac{1}{\theta})$.

$$\begin{aligned}\bar{v}_{jkg}(e) &= \mathbb{E}[\max_{\ell} v_{jkg\ell}(e,\varepsilon_{\ell})|e] \\ &= \mathbb{E}[\max_{\ell} \tilde{v}_{jkg\ell}e^{\eta}z_g\varepsilon_{\ell}|e] \\ &= e^{\eta}z_g\mathbb{E}[\max_{\ell} \tilde{v}_{jkg\ell}\varepsilon_{\ell}]\end{aligned}$$

By the law of iterated expectations, the expectation of of the maximum is the expected utility if a location ℓ is chosen, expected given the likelihood of choosing each location.

$$\begin{aligned}\bar{v}_{jkg}(e) &= e^{\eta}z_g \sum_{\ell} \mathbb{E}[\tilde{v}_{jkg\ell}\varepsilon_{\ell}|\ell]m_{jkg\ell} \\ &= e^{\eta}z_g \sum_{\ell} \tilde{v}_{jkg\ell}\mathbb{E}[\varepsilon_{\ell}|\ell]m_{jkg\ell}\end{aligned}$$

The final equality holds because conditional on choosing ℓ, $\tilde{v}_{jkg\ell}$ is a constant. I then need to show that $\mathbb{E}[\varepsilon_{\ell}|\ell] = m_{jkg\ell}^{-1/\theta}\Gamma(1-\frac{1}{\theta})$. I can compute the expected value of ε_{ℓ} conditional on choosing ℓ. Note that I need to weight inversely by the probability that ℓ is chosen since I am only are evaluating over the region where ℓ is chosen, which only happens some of the time. I include the CDF of the unchosen option to capture the fact that we only take cases where ℓ is actually chosen[20].

$$\begin{aligned}
\mathbb{E}[\varepsilon_\ell|\ell] &= P(\text{choose } \ell)^{-1}\int_0^\infty F(\varepsilon_\ell)_{\forall \ell' \neq \ell}\varepsilon_\ell f(\varepsilon_\ell)d\varepsilon_\ell \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\sum_{\ell' \neq \ell}\varepsilon_{\ell'}^{-\theta})\varepsilon_\ell \theta \varepsilon_\ell^{-\theta-1} exp(-\tilde{v}_{jkg\ell})d\varepsilon_\ell \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\sum_{\ell'}\varepsilon_{\ell'}^{-\theta})\theta \varepsilon_\ell^{-\theta} d\varepsilon_\ell
\end{aligned}$$

Recall that $\varepsilon_{\ell'} = \frac{\tilde{v}_{jkg\ell}\varepsilon_\ell}{\tilde{v}_{jkg\ell'}}$. Then:

$$\begin{aligned}
\mathbb{E}[\varepsilon_\ell|\ell] &= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\sum_{\ell'}(\frac{\tilde{v}_{jkg\ell}\varepsilon_\ell}{\tilde{v}_{jkg\ell'}})^{-\theta})\theta \varepsilon_\ell^{-\theta} d\varepsilon_\ell \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\varepsilon_\ell^{-\theta}\tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell'}(\frac{1}{\tilde{v}_{jkg\ell'}})^{-\theta})\theta \varepsilon_\ell^{-\theta} d\varepsilon_\ell \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\varepsilon_\ell^{-\theta}\tilde{v}_{jkg\ell}^{-\theta}\sum_{\ell'}(\tilde{v}_{jkg\ell'})^{\theta})\theta \varepsilon_\ell^{-\theta} d\varepsilon_\ell \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\frac{\varepsilon_\ell^{-\theta}}{m_{jkg\ell}})\theta \varepsilon_\ell^{-\theta} d\varepsilon_\ell
\end{aligned}$$

Substitute in $x = \varepsilon_\ell^{-\theta}$. This implies $\varepsilon_\ell = x^{\frac{1}{\theta}}$, $dx = -\theta\varepsilon_\ell^{-\theta-1}d\varepsilon_\ell$, and $d\varepsilon_\ell = -\frac{dx}{\theta x^{1+\frac{1}{\theta}}}$. The limits of integration change from $\varepsilon_\ell = 0 \rightarrow x = \infty$ and $\varepsilon_\ell = \infty \rightarrow x = 0$. Then:

$$\begin{aligned}
\mathbb{E}[\varepsilon_\ell|\ell] &= \frac{1}{m_{jkg\ell}}\int_\infty^0 exp(-\frac{x}{m_{jkg\ell}})x^{-\frac{1}{\theta}}(-1)dx \\
&= \frac{1}{m_{jkg\ell}}\int_0^\infty exp(-\frac{x}{m_{jkg\ell}})x^{-\frac{1}{\theta}}dx
\end{aligned}$$

20. For a full discussion of this seen Matt Turner's excellent note here: https://matthewturner.org/ec2410/lectures/5_Discrete_v4.pdf

I apply a rule for integrals that $\int_0^\infty x^n e^{-ax} dx = \frac{\Gamma(n+1)}{a^{n+1}}$.

$$\mathbb{E}[\varepsilon_\ell|\ell] = \frac{1}{m_{jkg\ell}} \frac{\Gamma(1-\frac{1}{\theta})}{(\frac{1}{m_{jkg\ell}})^{-\frac{1}{\theta}+1}} \tag{25}$$

$$= \Gamma(1-\frac{1}{\theta}) m_{jkg\ell}^{-\frac{1}{\theta}} \tag{26}$$

Plugging back into the above expression for $\bar{v}_{jkg}(e)$ gives:

$$\begin{aligned}
\bar{v}_{jkg}(e) &= e^\eta z_g \sum_\ell \tilde{v}_{jkg\ell} \mathbb{E}[\varepsilon_\ell|\ell] m_{jkg\ell} \\
&= e^\eta z_g \sum_\ell \tilde{v}_{jkg\ell} \Gamma(1-\frac{1}{\theta}) m_{jkg\ell}^{1-\frac{1}{\theta}}
\end{aligned}$$

Recall that $m_{jkg\ell} = \frac{\tilde{v}_{jkg\ell}^\theta}{\sum_\ell \tilde{v}_{jkg\ell}^\theta}$, so $\tilde{v}_{jkg\ell} = m_{jkg\ell}^{\frac{1}{\theta}} \mathrm{MA}_{jkg}^{\frac{1}{\theta}}$ Then:

$$\begin{aligned}
\bar{v}_{jkg}(e) &= e^\eta z_g \sum_\ell \Gamma(1-\frac{1}{\theta}) m_{jkg\ell} \mathrm{MA}_{jkg}^{\frac{1}{\theta}} \\
&= e^\eta z_g \Gamma(1-\frac{1}{\theta}) \mathrm{MA}_{jkg}^{\frac{1}{\theta}} \sum_\ell m_{jkg\ell} \\
&= e^\eta z_g \gamma_\ell \mathrm{MA}_{jkg}^{\frac{1}{\theta}}
\end{aligned}$$

The expected wage follows similarly as the expected utility.

$$\begin{aligned}
\bar{w}_{jkg}(e) &= \mathbb{E}[\max_{\ell} w_{jkg\ell}(e, \varepsilon_\ell)|e] \\
&= \mathbb{E}[\max_{\ell} r_{g\ell} h_{jkg\ell}(e, \epsilon_\ell)|e] \\
&= \mathbb{E}[\max_{\ell} r_{g\ell} e^{\eta} s_{jkl} z_g \varepsilon_\ell|e] \\
&= e^{\eta} z_g \sum_{\ell} r_{g\ell} s_{jkl} m_{jkg\ell} \mathbb{E}[\varepsilon_\ell] \\
&= e^{\eta} z_g \sum_{\ell} r_{g\ell} s_{jkl} m_{jkg\ell} \bar{m}_{jkg\ell}^{-\frac{1}{\theta}} \gamma_\ell \\
&= e^{\eta} z_g \gamma_\ell \sum_{\ell} r_{g\ell} s_{jkl} m_{jkg\ell}^{1-\frac{1}{\theta}}
\end{aligned}$$

Substituting $m_{jkg\ell} = \frac{\tilde{v}_{jkg\ell}^{\theta}}{\sum_{\ell} \tilde{v}_{jkg\ell}^{\theta}}$ gives:

$$\begin{aligned}
\bar{w}_{jkg}(e) &= e^{\eta} z_g \gamma_\ell \sum_{\ell} r_{g\ell} s_{jkl} \left(\frac{\tilde{v}_{jkg\ell}^{\theta}}{\sum_{\ell} \tilde{v}_{jkg\ell}^{\theta}}\right)^{1-\frac{1}{\theta}} \\
&= e^{\eta} z_g \gamma_\ell \sum_{\ell} r_{g\ell} s_{jkl} \frac{\tilde{v}_{jkg\ell}^{\theta-1}}{\mathrm{MA}_{jkg}^{1-\frac{1}{\theta}}}
\end{aligned}$$

Spilling up the market access term and using $\tilde{v}_{jkg\ell} = \frac{a_\ell r_{g\ell} s_{jk\ell}}{\tau_{jkg\ell}^{m}}$ gives:

$$\bar{w}_{jkg}(e) = e^{\eta} z_g \gamma_\ell \mathrm{MA}_{jkg}^{\frac{1}{\theta}} \frac{\sum_{\ell} (\frac{\tau_{jkg\ell}^{m}}{a_\ell}) \tilde{v}_{jkg\ell}^{\theta}}{\mathrm{MA}_{jkg}}$$

B.3 Expected Utility and Expected Productivity

We want to find the expected utility given a choice of industry g. This is the same as finding the expected utility in section B.2, but with a change of variables. First recall that:

$$\bar{z}_{jkg}(e) = \mathbb{E}[\max_{g} z_g|e] = \gamma_g \bar{n}_{jkg}^{-\frac{1}{\phi}}$$

as in equation 25. Then I can compute the expected utility:

$$\begin{aligned}
\bar{u}_{jk}(e) &= \mathbb{E}[\max_g z_g e^{\eta}\gamma_\ell \mathrm{MA}_{jkg}^{\frac{1}{\theta}}|e] \\
&= e^{\eta}\gamma_\ell \sum_g \mathrm{MA}_{jkg}^{\frac{1}{\theta}}\mathbb{E}[z_g]n_{jkg} \\
&= e^{\eta}\gamma_\ell \sum_g \mathrm{MA}_{jkg}^{\frac{1}{\theta}}\mathbb{E}[z_g]n_{jkg} \\
&= e^{\eta}\gamma_\ell \sum_g \mathrm{MA}_{jkg}^{\frac{1}{\theta}}\gamma_g n_{jkg}^{-\frac{1}{\phi}}n_{jkg} \\
&= e^{\eta}\gamma_\ell\gamma_g \sum_g \mathrm{MA}_{jkg}^{\frac{1}{\theta}} n_{jkg}^{1-\frac{1}{\phi}}
\end{aligned}$$

Now recall from equation 10 that $\bar{n}_{jkg}^{\frac{1}{\phi}}MA_{jk}^{\frac{1}{\phi}} = \mathrm{MA}_{jkg}^{\frac{1}{\theta}}$. Then we can recover the result:

$$\begin{aligned}
\bar{u}_{jk}(e) &= e^{\eta}\gamma_\ell\gamma_g \sum_g \bar{n}_{jkg}^{\frac{1}{\phi}}MA_{jk}^{\frac{1}{\phi}}n_{jkg}^{1-\frac{1}{\phi}} \\
&= e^{\eta}\gamma_\ell\gamma_g MA_{jk}^{\frac{1}{\phi}} \sum_g n_{jkg} \\
&= \gamma_\ell\gamma_g e^{\eta} MA_{jk}^{\frac{1}{\phi}}.
\end{aligned}$$

B.4 Estimating Equations

I want to recover the two estimating equations. Starting with equation 13 and aggregating over individuals at the origin j and cohort k gives:

$$\bar{e}_{jk} = \left(\frac{\gamma_g\gamma_\ell\eta\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\tau_{jk}^{e}}\right)^{\frac{1}{1-\eta}}\bar{\epsilon}, \quad \bar{\epsilon} = \mathbb{E}[\epsilon^{-\frac{1}{1-\eta}}] \tag{27}$$

Taking logs gives:

$$(1-\eta)log(\bar{e}_{jk}) = log(\gamma_g) + \log(\gamma_\ell) + (1-\eta)log(\bar{\epsilon}) + log(\eta) + \frac{1}{\phi}log(\mathrm{MA}_{jk}) - log(\tau_{jk}^{e}). \tag{28}$$

I can compute wages in every given sector and location. Wages are given by $w_{jkg\ell}(e) = r_{g\ell} e^{\eta} s_{jkg\ell} z_g \varepsilon_\ell$. Taking the expectation given a choice of ℓ gives:

$$\bar{w}_{jkg\ell} = \mathbb{E}[w_{jkg\ell}] = r_{g\ell}\mathbb{E}[e^{\eta}] s_{jkg\ell}\mathbb{E}[z_g]\mathbb{E}[\varepsilon_\ell]$$

Solving each expectation separately gives:

$$\begin{aligned}\mathbb{E}[e^{\eta}] &= \left(\frac{\gamma_g \gamma_\ell \eta \mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\tau_{jk}^{e}}\right)^{\frac{\eta}{1-\eta}} \mathbb{E}[\epsilon^{-\frac{\eta}{1-\eta}}] \\ &= \left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^{\eta} \tilde{\epsilon}\end{aligned}$$

where the first equality comes from the solution to e^{η}, and the second equality comes from the definitions $\bar{\epsilon} = \mathbb{E}[\epsilon^{-\frac{1}{1-\eta}}]$ and $\tilde{\epsilon} = \mathbb{E}[\epsilon^{-\frac{\eta}{1-\eta}}]$. Next we have:

$$\begin{aligned}\mathbb{E}[z_g] &= \gamma_g \bar{n}_{jkg}^{-\frac{1}{\phi}} \\ &= \gamma_g \left(\frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}}}{\mathrm{MA}_{jk}}\right)^{-\frac{1}{\phi}} \\ &= \gamma_g \frac{\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\mathrm{MA}_{jkg}^{\frac{1}{\theta}}}.\end{aligned}$$

The final expectation is given by:

$$\begin{aligned}\mathbb{E}[\varepsilon_\ell] &= \gamma_\ell \bar{m}_{jkg\ell}^{-\frac{1}{\theta}} \\ &= \gamma_\ell \left(\frac{\tilde{v}_{jkg\ell}^{\theta}}{\mathrm{MA}_{jkg}}\right)^{-\frac{1}{\theta}} \\ &= \gamma_\ell \tilde{v}_{jkg\ell}^{-1} \mathrm{MA}_{jkg}^{\frac{1}{\theta}}\end{aligned}$$

Bringing all of the components together and substituting for $\tilde{v}_{jkg\ell} = \frac{a_\ell r_{g\ell} s_{jk\ell}}{\tau^m_{jkg\ell}}$ gives:

$$\begin{aligned}
\bar{w}_{jkg\ell} &= r_{g\ell}\mathbb{E}[e^\eta] s_{jkg\ell}\mathbb{E}[z_g]\mathbb{E}[\varepsilon_\ell] \\
&= r_{g\ell}\left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^\eta \tilde{\epsilon} s_{jkg\ell}\gamma_g \frac{\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\mathrm{MA}_{jkg}^{\frac{1}{\theta}}}\gamma_\ell \tilde{v}^{-1}_{jkg\ell}\mathrm{MA}_{jkg}^{\frac{1}{\theta}} \\
&= r_{g\ell}\left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^\eta \tilde{\epsilon} s_{jkg\ell}\gamma_g \frac{\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\mathrm{MA}_{jkg}^{\frac{1}{\theta}}}\gamma_\ell \frac{\tau^m_{jkg\ell}}{a_\ell r_{g\ell} s_{jk\ell}}\mathrm{MA}_{jkg}^{\frac{1}{\theta}} \\
&= \left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^\eta \tilde{\epsilon}\gamma_g\gamma_\ell \mathrm{MA}_{jk}^{\frac{1}{\phi}}\frac{\tau^m_{jkg\ell}}{a_\ell}
\end{aligned} \tag{29}$$

Taking logs gives:

$$log(\bar{w}_{jkg\ell}) = \eta log(\bar{e}_{jk}) - \eta log(\bar{\epsilon}) + log(\tilde{\epsilon}) + log(\gamma_g) + log(\gamma_\ell) + \frac{1}{\phi}log(\mathrm{MA}_{jk}) + log(\tau^m_{jkg\ell}) - log(a_\ell) \tag{30}$$

I take equation 7 for location choice and express it in logs. This gives:

$$\begin{aligned}
log(\bar{m}_{jkg\ell}) &= \frac{log(\tilde{v}^\theta_{jkg\ell})}{log(\mathrm{MA}_{jkg})} \\
&= \theta log(\tilde{v}_{jkg\ell}) - log(\mathrm{MA}_{jkg}) \\
&= \theta log(a_\ell) + \theta log(r_{g\ell}) + \theta log(s_{jk\ell}) - \theta log(\tau^m_{jkg\ell}) - log(\mathrm{MA}_{jkg})
\end{aligned}$$

The term MA_{jkg} is the market access conditional on choosing occupation g. We can express this in terms of the market without having chosen g by using the agricultural choice share from equation 10. Rearranging equation 7 gives $\mathrm{MA}_{jkg} = (\bar{n}_{jkg}MA_{jk})^{\frac{\theta}{\phi}}$. Substituting into the equation above and solving for the market access term gives:

$$\begin{aligned}\frac{\theta}{\phi}log(\bar{n}_{jkg}MA_{jk}) &= \theta log(a_\ell) + \theta log(r_{g\ell}) + \theta log(s_{jk\ell}) - \theta log(\tau^m_{jkg\ell}) - log(\bar{m}_{jkg\ell}) \\ \frac{1}{\phi}log(MA_{jk}) &= log(a_\ell) + log(r_{g\ell}) + log(s_{jk\ell}) - log(\tau^m_{jkg\ell}) - \frac{1}{\theta}log(\bar{m}_{jkg\ell}) - \frac{1}{\phi}log(\bar{n}_{jkg})\end{aligned} \tag{31}$$

With these equations I can now derive two estimating equations. Subtracting equation 28 from equation 30 gives:

$$\begin{aligned}log(\bar{w}_{jkg\ell}) - log(\bar{e}_{jk}) &= log(\tilde{\epsilon}) - log(\bar{\epsilon}) - log(\eta) - log(a_\ell) + log(\tau^m_{jkg\ell}) + log(\tau^e_{jk}) \\ log(\bar{w}_{jkg\ell}) - log(\bar{e}_{jk}) &= log(\frac{\tilde{\epsilon}}{\bar{\epsilon}\eta}) - log(a_\ell) + log(\tau^e_{jk}) + log(\tau^m_{jkg\ell})\end{aligned} \tag{32}$$

This equation can be estimated with with OLS. The second estimating equation can be obtained by substituting equation 31 into equation 30, giving:

$$\begin{aligned}log(\bar{w}_{jkg\ell}) &= \eta log(\bar{e}_{jk}) - \eta log(\bar{\epsilon}) + log(\tilde{\epsilon}) + log(\gamma_g) + log(\gamma_\ell) + log(r_{g\ell}) + log(s_{jkg\ell}) \\ &\quad - \frac{1}{\theta}log(m_{jkg\ell}) - \frac{1}{\phi}log(n_{jkg}) \\ &= log(\frac{\tilde{\epsilon}\gamma_g\gamma_\ell}{\bar{\epsilon}^\eta}) + \eta log(\bar{e}_{jk}) + log(r_{g\ell}) - \frac{1}{\theta}log(m_{jkg\ell}) - \frac{1}{\phi}log(n_{jkg}) + log(s_{jkg\ell})\end{aligned} \tag{33}$$

B.5 Hat Algebra

Changes in labour demand are given by $\hat{H}^D_{g\ell} = \frac{{H'}^D_{g\ell}}{H^D_{g\ell}}$. For the non agricultural sector this gives:

$$\begin{aligned}
\hat{H}^D_{N\ell}(r_{N\ell}) &= \frac{{\hat{H}'}^D_{N\ell}(r_{N\ell})}{\hat{H}^D_{N\ell}(r_{N\ell})} \\
&= \frac{\left(\frac{\kappa A_{N\ell}}{{r'}_{N\ell}}\right)^{\frac{1}{1-\kappa}}}{\left(\frac{\kappa A_{N\ell}}{r_{N\ell}}\right)^{\frac{1}{1-\kappa}}} \\
&= \hat{r}_{N\ell}^{-\frac{1}{1-\kappa}}
\end{aligned}$$

The same logic applies for the agricultural sector:

$$\hat{H}^D_{F\ell}(r_{N\ell}) = \hat{r}_{N\ell}^{-\frac{1}{1+\delta-\kappa}}.$$

Changes in labour supply can also be derived in changes. Recall that the following holds for both sectors:

$$H^S_{g\ell}(r_{g\ell}) = \frac{W_{g\ell}}{r_{g\ell}}$$

where $W_{g\ell}$ is the total wage payments in industry g and location ℓ. Changes in labour supply are given by:

$$\begin{aligned}
H^S_{g\ell}(r_{g\ell}) &= \frac{{H'}^S_{g\ell}({r'}_{g\ell})}{H^S_{g\ell}(r_{g\ell})} \\
&= \frac{\frac{{W'}_{g\ell}}{{r'}_{g\ell}}}{\frac{W_{g\ell}}{r_{g\ell}}} \\
&= \frac{\hat{W}_{g\ell}}{\hat{r}_{g\ell}}.
\end{aligned}$$

$\hat{r}_{g\ell}$ is determined in equilibrium, but we can solve for $\hat{W}_{g\ell}$. Recall that $W_{g\ell} = \sum_{jk} N_{jk}\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell}$. This implies that:

$$\hat{W}_{g\ell} = \frac{W'_{g\ell}}{W_{g\ell}} \tag{34}$$

$$= \frac{\sum_{jk} N_{jk}\bar{n'}_{jkg}\bar{m'}_{jkg\ell}\bar{w'}_{jkg\ell}}{\sum_{jk} N_{jk}\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell}} \tag{35}$$

Where $\bar{n'}_{jkg}\bar{m'}_{jkg\ell}\bar{w'}_{jkg\ell}$ are unknown. To recover these variables I need to express them in terms of the base level and changes in variables that I know. By the definition of changes:

$$\bar{n'}_{jkg}\bar{m'}_{jkg\ell}\bar{w'}_{jkg\ell} = \bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell}\hat{n}_{jkg}\hat{m}_{jkg\ell}\hat{w}_{jkg\ell}.$$

From this equation we can see that if $\hat{n}_{jkg}\hat{m}_{jkg\ell}\hat{w}_{jkg\ell}$ is known, and $\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell}$ can come from the data, then $\bar{n'}_{jkg}\bar{m'}_{jkg\ell}\bar{w'}_{jkg\ell}$ can be recovered. To do this I rely on the definitions of $\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell}$, and then compute what happens if there are changes to the underlying parameters. Multiplying together equations 7, 10, and 29:

$$\begin{aligned}
\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell} &= \left(\frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}}}{MA_{jk}}\right)\left(\frac{\tilde{v}_{jkg\ell}^{\theta}}{\mathrm{MA}_{jkg}}\right)\left(\left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^{\eta}\tilde{\epsilon}\gamma_g\gamma_\ell \mathrm{MA}_{jk}^{\frac{1}{\phi}}\frac{\tau_{jkg\ell}^{m}}{a_\ell}\right) \\
&= \frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}-1}}{\mathrm{MA}_{jk}^{1-\frac{1}{\phi}}}\tilde{v}_{jkg\ell}^{\theta}\left(\frac{\bar{e}_{jk}}{\bar{\epsilon}}\right)^{\eta}\tilde{\epsilon}\gamma_g\gamma_\ell\frac{\tau_{jkg\ell}^{m}}{a_\ell}
\end{aligned}$$

Then substituting in equations 5 and 27 to remove $\tilde{v}_{jkg\ell}$ and $\bar{e}_{jk}$:

$$\begin{aligned}
\bar{n}_{jkg}\bar{m}_{jkg\ell}\bar{w}_{jkg\ell} &= \frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}-1}}{\mathrm{MA}_{jk}^{1-\frac{1}{\phi}}} \left(\frac{a_\ell r_{g\ell} s_{jkg\ell}}{\tau^m_{jkg\ell}}\right)^\theta \left(\frac{\left(\frac{\gamma_g\gamma_\ell\eta\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\tau^e_{jk}}\right)^{\frac{1}{1-\eta}}\bar{\epsilon}}{\bar{\epsilon}}\right)^\eta \tilde{\epsilon}\gamma_g\gamma_\ell\frac{\tau^m_{jkg\ell}}{a_\ell} \\
&= \frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}-1}}{\mathrm{MA}_{jk}^{1-\frac{1}{\phi}}} \left(\frac{a_\ell r_{g\ell} s_{jkg\ell}}{\tau^m_{jkg\ell}}\right)^\theta \left(\frac{\gamma_g\gamma_\ell\eta\mathrm{MA}_{jk}^{\frac{1}{\phi}}}{\tau^e_{jk}}\right)^{\frac{\eta}{1-\eta}} \tilde{\epsilon}\gamma_g\gamma_\ell\frac{\tau^m_{jkg\ell}}{a_\ell} \\
&= \frac{\mathrm{MA}_{jkg}^{\frac{\phi}{\theta}-1}}{\mathrm{MA}_{jk}^{1-\frac{1}{\phi}\frac{1}{1-\eta}}} \frac{a_\ell^{\theta-1} r_{g\ell}^\theta s_{jkg\ell}^\theta}{(\tau^m_{jkg\ell})^{\theta-1}(\tau^e_{jk})^{\frac{\eta}{1-\eta}}} \eta^{\frac{\eta}{1-\eta}}\gamma_g^{\frac{1}{1-\eta}}\gamma_\ell^{\frac{1}{1-\eta}}\tilde{\epsilon}
\end{aligned}$$

We now want to express this in changes. All constant terms cancel, leaving only terms which change:

$$\hat{n}_{jkg}\hat{m}_{jkg\ell}\hat{w}_{jkg\ell} = \frac{\hat{\mathrm{MA}}_{jkg}^{\frac{\phi}{\theta}-1}}{\hat{\mathrm{MA}}_{jk}^{1-\frac{1}{\phi}\frac{1}{1-\eta}}} \frac{\hat{r}_{g\ell}^\theta}{(\hat{\tau}^m_{jkg\ell})^{\theta-1}(\hat{\tau}^e_{jk})^{\frac{\eta}{1-\eta}}}$$

Of these terms, $\hat{r}_{g\ell}$ comes from finding a value that clears the labour market, and $\hat{\tau}^m_{jkg\ell}$ and $\hat{\tau}^e_{jk}$ come from assumed changes. This leaves the market access terms $\hat{\mathrm{MA}}_{jkg}$ and $\hat{\mathrm{MA}}_{jk}$ that need to be solved for. $\hat{\mathrm{MA}}_{jk}$ comes from the summation over industries:

$$\begin{aligned}
\hat{\mathrm{MA}}_{jk} &= \frac{\mathrm{MA}'_{jk}}{\mathrm{MA}_{jk}} \\
&= \frac{\sum_g (\mathrm{MA'}_{jkg}^{\frac{1}{\theta}})^\phi}{\sum_g (\mathrm{MA}_{jkg}^{\frac{1}{\theta}})^\phi}.
\end{aligned}$$

$\hat{\text{MA}}_{jkg}$ comes from a summation over locations:

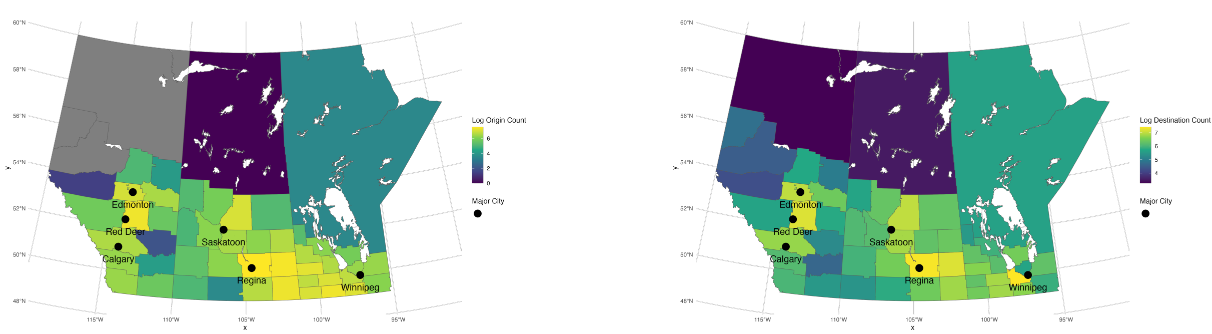
$$\begin{aligned}
\hat{\text{MA}}_{jkg} &= \frac{\text{MA}'_{jkg}}{\text{MA}_{jkg}} \\
&= \frac{\sum_\ell \left(\frac{a_\ell {r'}_{g\ell} s_{jkg\ell}}{{\tau'}^m_{jkg\ell}}\right)^\theta}{\sum_\ell \left(\frac{a_\ell r_{g\ell} s_{jkg\ell}}{\tau^m_{jkg\ell}}\right)^\theta} \\
&= \frac{\sum_\ell \left(\frac{a_\ell \hat{r}_{g\ell} r_{g\ell} s_{jkg\ell}}{\hat{\tau}^m_{jkg\ell} \tau^m_{jkg\ell}}\right)^\theta}{\sum_\ell \left(\frac{a_\ell r_{g\ell} s_{jkg\ell}}{\tau^m_{jkg\ell}}\right)^\theta}
\end{aligned}$$

C Model Data and Calibration

C.1 Census Divisions

For each person in the linked 1906-1931 sample, I use their geolocations to determine their 1931 Census Division in both 1906 and 1931. The migration rate between 1906 and 1921 is significant, 52% of people in 1906 are in a different Census Division in 1931. Figure C14 shows the distribution of origins in 1906 and destinations in 1931. Given that the sample is restricted to the children of farmers in 1906, the highest concentration of origins comes from the fertile regions of southern Manitoba, south-eastern Saskatchewan, and the parkland belt linking to western Alberta. The set of destinations is also striking, with the CDs containing Winnipeg, Regina, Saskatoon, Edmonton, and Calgary all clearly visible.

Figure C14: Origin and Destination Counts from 1906-1931



(a) Log origin counts in 1906

(b) Log destination counts in 1931

C.2 Estimating Wages

As noted in table A22, few farmers reported income in this time period. To know the potential benefit of choosing an agricultural occupation I therefore estimate farm incomes, relying on the fact that a small number of farmers did declare some income.

I utilize the full set of farmers in Alberta, Saskatchewan, and Manitoba in 1931, and estimate the factors determining farmer wages. I rely on the fact that some farmers did declare income. As in table A22 I restrict to the set of farmers who did not earn exactly $1000. Table C28 shows the results. I then use this model to estimate wages for all farmers in the 1906-1931 linked sample.

Table C28: Farmer Wage Model

Dependent Variable:	Log Income	
Model:	(1)	(2)
Employee	-0.1507***	-0.1213***
	(0.0171)	(0.0144)
Employer	0.1587***	0.1511***
	(0.0273)	(0.0240)
Log Farm Area	0.0206***	
	(0.0045)	
Log RR Distance	-0.0103***	
	(0.0033)	
Head of HH	0.2014***	0.2618***
	(0.0124)	(0.0094)
Age	Yes	Yes
Enumeration Subdistrict (1931)	Yes	Yes
Livestock + Ag.	Yes	
Observations	47,063	75,924
R^2	0.29623	0.34526

Clustered (Enumeration Subdistrict (1931)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Factors predicting income of farmers in 1931. I utilize two alternate specifications, one for households with a precise geolocation where I have an estimate of agricultural productivity and farm size, and a second without these factors.

C.3 Estimating Education

To estimate the returns to schooling I first need to estimate the amount of school a person receives over their lifetime. I don't observe the total amount of school received directly, but I observe the amount of school attended in the 1906-1911 census link. I construct a linear probability model for the factors that determine the amount of school attended in 1911. Table C29 shows the results.

Table C29: Factors Determining 1911 Months of School

Dependent Variable:	Months of School	
Model:	(1)	(2)
Log School Dist.	-0.2225***	-0.2955***
	(0.0686)	(0.0683)
Log RR Distance	-0.1080***	-0.1573***
	(0.0372)	(0.0403)
Log Farm Area	-0.0273	-0.0110
	(0.0445)	(0.0446)
Enumeration Subdistrict	Yes	
Enumeration District		Yes
Age	Yes	Yes
Birth Order	Yes	Yes
Fam. Bpl. + Homestead	Yes	Yes
Livestock + Ag.	Yes	Yes
Observations	15,462	15,462
R^2	0.32643	0.25162

Clustered (Enumeration Subdistrict (1906)) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

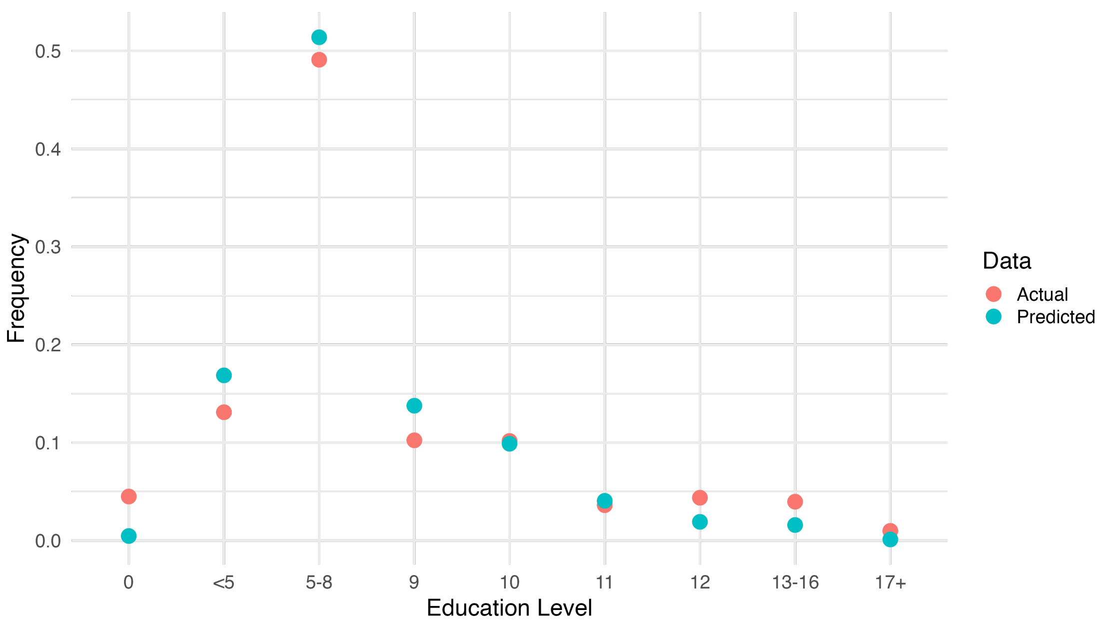
Factors predicting months of school in 1911. I utilize two alternate specifications, one with enumeration subdistrict fixed effects, and another with enumeration district fixed effects.

I then take the estimated models to the data for the 1906-1931 link. For each person in the sample, I predict the number of months of school they would have attended for each year of their childhood. Recall that I do not observe the months of schooling in 1906. Two variables, school distance and railway distance, are time varying. I use school distances from

1900 - 1912, and railway distances from 1895 - 1925. After predicting the number of months of school between the ages of 5 and 17 (Unesco Institute for Statistics 2025), I sum them to get an estimate of the total amount of education obtained.

I can validate this strategy by comparing my predicted amount of schooling in 1931 with the observed amount of education obtained and recorded in the aggregate 1936 census tables. Figure C15 shows the results. In general, I find my predictions are close to the aggregate values reported, with the predicted values being too high for middle bins of education, and too low in the tails.

Figure C15: Predicted vs Actual Education Levels



Comparison between education level obtained in 1936 census, and predicted from 1906 characteristics. I restrict to those approximately in the young and old cohorts used in the difference-in-differences to align with aggregate statistics published in 1936. This consists of males 30-34 and 40-44 in 1931 and males 35-39 and 45-49 in 1936.

With estimates of human capital, I can perform a naive estimation of the returns to schooling. Regressing my estimated level of human capital on income, I find that the estimated return to human capital is 12.8% as shown in table C30. The returns to human capital are higher in the non-agricultural sector, with an estimated return of 13.1% versus

8.4%. The magnitude of these results is significantly higher than found by Goldin and Katz (2000), who finds a return of 4.3% for a year of common schooling (equivalent to one of the rural schools studied here). Although the magnitudes are different, they also find lower returns in agricultural occupations relative to non-agricultural occupations.

The difference in magnitude between my estimates of the returns to education and Goldin and Katz (2000) likely reflect omitted higher levels of education. I don't observe attendance in high school or college, any returns from these forms of higher education will be attributed to rural schools. A second possibility is that the returns to education were significantly higher in the more remote Canadian prairies relative to Iowa. This is plausible given the delayed settlement of western Canada relative to the United States and the delayed development of the education system, but is not yet testable.

Table C30: Returns to Education

Dependent Variable:	Log Income		
Farm Work	Full sample	0	1
Model:	(1)	(2)	(3)
Constant	6.044***	6.257***	5.511***
	(0.0927)	(0.0988)	(0.1557)
Log Predicted Schooling	0.1227***	0.1248***	0.0831**
	(0.0229)	(0.0244)	(0.0385)
Observations	9,907	7,605	2,302
R^2	0.00289	0.00343	0.00202

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table C31: Estimated returns to education in 1931, using predicted schooling. As in table, I drop all farmers who report an income of $1000.

C.4 Estimating Land Share in Agriculture

I need an estimate of the land share in agricultural production δ. I estimate this parameter from the aggregate tables in the 1931 census (Dominion Bureau of Statistics Canada 1936).

Computing the wage rate per unit of human capital and rental rate per unit of land and rearranging gives:

$$\kappa - \delta = \frac{H_F}{Y_F} r_F$$
$$\delta = \frac{T}{Y_F} \rho_F$$

Where ρ_F is the rental rate of land. From the summary tables in the 1931 census I take H_F as the total number of male farm workers including family members, permanent farm workers, and temporary farm workers, multiplied by the average number of weeks of hired farm work the typical farm hires. This averages to approximately 26 weeks per year. I take Y_F as the measured aggregate farm output. r_F I take as the average wage rate for a week of labour as reported in the aggregate tables. T I take as the number of acres of farm land. ρ_F I take as the average rental rate per acre of farm land. I do all computations at the province level, then do a weighted average across Alberta, Saskatchewan, and Manitoba weighted by the number of acres devoted to farming in each province.

From these equations I recover two estimates of δ. The first estimate using the returns to labour gives a value of 0.791. Given the value of κ of 0.925 this implies $\delta = 0.134$. The second estimate using the returns to land gives an estimated value of $\delta = 0.092$. I therefore take the average of these two values of 0.113.